Sobolev Spaces and Applications

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Contents

Notations v

1 Theory of Distributions 1

1.1 History . . . 1
1.2 Motivation . . . 2
1.2.1 Case 1 . . . 2
1.2.2 Case 2 . . . 3
1.3 Space of Test Functions . . . 4
1.3.1 Smooth Non-Analytic (Bump) Functions . . . 6
1.3.2 Mollifiers . . . 10
1.3.3 Multi-index Notation . . . 11
1.3.4 Topology on $C(\Omega)$, $C^\infty(\Omega)$ and $C_c^\infty(\Omega)$ . . . 12
1.3.5 Complete Topology on $C_c^\infty(\Omega)$ . . . 14
1.3.6 Regularization and Cut-off Technique . . . 17
1.4 Space of Distributions . . . 21
1.4.1 Functions as Distributions . . . 23
1.4.2 Measures as Distributions . . . 27
1.4.3 Multipole Distributions . . . 32
1.4.4 Infinite Order Distributions . . . 34
1.4.5 Topology on Distributions . . . 35
1.4.6 Principal Value Distribution . . . 39
1.4.7 Functions, but not Distributions . . . 43
1.5 Operations With Distributions . . . 44
1.5.1 Differentiation . . . 44
1.5.2 Product . . . 54
1.5.3 Support . . . 58
1.5.4 Singular Support . . . 61
1.5.5 Shifting and Scaling . . . 62

1.5.6 Convolution 64
1.6 Tempered Distributions 67
1.7 Fundamental Solution 70
1.8 Further Reading 71

2 Sobolev Spaces **73**
2.1 Space of Continuous Functions 73
2.2 Hölder Spaces 79
2.3 $W^{k,p}(\Omega)$ Spaces 87
2.4 Smooth Approximations 90
2.5 Extension Operators 95

Appendices **101**

A Cantor Set and Cantor Function **103**

B Partition of Unity **109**
B.1 Zero Sets 109
B.2 Continuous Partition of Unity 110
B.3 Smooth Partition of Unity 112

Bibliography **115**

Index **117**

Notations

Symbols

∇ $\left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \ldots, \frac{\partial}{\partial x_n}\right)$

Ω denotes an open subset of $\mathbb{R}^n$, not necessarily bounded

$\mathbb{R}$ denotes the real line

$\mathbb{R}^n$ denotes the n-dimensional Euclidean space

D^α $\frac{\partial^{\alpha_1}}{\partial {x_1}^{\alpha_1}} \ldots \frac{\partial^{\alpha_n}}{\partial {x_n}^{\alpha_n}}$ and $\alpha = (\alpha_1, \ldots, \alpha_n)$

Function Spaces

$C^{k,\gamma}(\Omega)$ all functions in $C^k(\Omega)$ whose k-th partial derivatives ($k \geq 0$) are Hölder continuous with exponent γ

$\mathrm{Lip}(E)$ denotes the space of all Lipschitz functions on E

$C(\Omega)$ is the class of all continuous functions on Ω

$C(X)$ is the class of all continuous functions on a topological space X

$C^k(\Omega)$ is the class of all k-times ($k \geq 1$) continuously differentiable functions on Ω

$C^\infty(\overline{\Omega})$ denotes class of $C^\infty(\Omega)$ functions such that all its derivatives can be extended continuously to $\overline{\Omega}$

$C_c^\infty(\Omega)$ is the class of all infinitely differentiable functions on Ω with compact support

$C_0(\Omega)$ is the class of all continuous functions on Ω that vanishes at boundary (for bounded sets) or ∞

$H^k(\Omega)$ is the space $W^{k,2}(\Omega)$

$H_0^k(\Omega)$ is the space $W_0^{k,2}(\Omega)$

$\mathcal{E}(\Omega)$ is the class of all infinitely differentiable functions on Ω, also denoted as $C^\infty(\Omega)$, endowed with topology of uniform convergence on compact subsets

$\mathcal{D}(\Omega)$ is the class of all infinitely differentiable functions on Ω with compact support endowed with inductive limit topology

$W^{k,p}(\Omega)$ is the class of all functions in $L^p(\Omega)$ whose distributional derivative upto order k are also in $L^p(\Omega)$

Chapter 1

Theory of Distributions

1.1 History

In late 1920's P. A. M. Dirac (cf. [Dir49]) derived the equation

$$\frac{d}{dx}\ln(x) = \frac{1}{x} - i\pi\delta(x) \tag{1.1.1}$$

in the study of quantum theory of collision processes, where δ is a "function" defined and continuous in real line $\mathbb{R}$ satisfying the following properties:

1. $\delta(x) = 0$ for $x \neq 0$.
2. $\int_{-\infty}^{\infty} \delta(x)\,dx = 1$.
3. For any continuous function defined on $\mathbb{R}$, $f(a) = \int_{-\infty}^{\infty} f(x)\delta(a-x)\,dx$ for all $a \in \mathbb{R}$.
4. δ is infinitely differentiable and for any k-times continuously differentiable function f on $\mathbb{R}$, $f^{(k)}(a) = \int_{-\infty}^{\infty} f(x)\delta^{(k)}(a-x)\,dx$ for all $a \in \mathbb{R}$.
5. Given the Heaviside function

$$H(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0, \end{cases} \tag{1.1.2}$$

then the delta function is the "derivative" of H, i.e., $\delta(x) = H'(x)$.

The δ became known as *Dirac's delta function*. It had caused unrest among the mathematicians of the era, because δ did not adhere to the classical notion of function.

H. Lebesgue had laid the foundations of measure theory in his doctoral thesis in 1902. With this knowledge, δ can be viewed as a set function on the σ-algebra (a collection of subsets of $\mathbb{R}$). δ can be viewed as a positive measure, called *Dirac measure*, defined on subsets E from the σ-algebra of $\mathbb{R}$ as,

$$\delta(E) = \begin{cases} 1 & \text{if } 0 \in E \\ 0 & \text{if } 0 \notin E. \end{cases}$$

Thus, mathematically one has given sense to properties in 1, 2 and 3. However, the differentiability of δ and H was yet to be made precise, since classically, every differentiable function is continuous.

In 1944, G. Choquet and J. Deny published a work on polyharmonic functions in two dimensions (cf. [CD44]). Laurent Schwartz, in an attempt to generalise the work of Choquet and Deny in higher dimensions, published an article in 1945 (cf. [Sch45]). With this article, the theory of distributions was discovered and also settled the issue raised in properties 4 and 5, in addition to 1, 2 and 3.

Dirac won the physics Nobel prize (1933) and Schwartz won the Fields medal (1950) for their respective work.

1.2 Motivation

The theory of distribution is a concept that generalises the notion of function, hence is also called *generalised functions*. But the beauty of the theory lies in the fact that this new notion admits the concept of differentiation and every distribution is infinitely differentiable. The need for such a generalisation was felt at various instances in the history of mathematics.

1.2.1 Case 1

Recall that the wave equation $u_{tt}(x,t) = c^2 u_{xx}(x,t)$ on $\mathbb{R} \times (0,\infty)$, describing the vibration of an infinite string, has the general solution $u(x,t) = F(x+ct) + G(x-ct)$. In reality, it happens that even if F and G are *not* twice differentiable, they are "solution" to the wave equation. For instance,

consider F to be the Heaviside function H (cf. (1.1.2)) and $G \equiv 0$ or vice-versa. Now, how do we mathematically describe this nature of "solution"?

1.2.2 Case 2

Consider the Burger's equation

$$\begin{cases} u_t(x,t) + u(x,t)u_x(x,t) &= 0 \qquad x \in \mathbb{R} \text{ and } t \in (0,\infty) \\ u(x,0) &= \phi(x) \quad x \in \mathbb{R} \end{cases}$$

and the corresponding characteristic equations:

$$\frac{dx(r,s)}{ds} = z, \quad \frac{dt(r,s)}{ds} = 1, \text{ and } \frac{dz(r,s)}{ds} = 0$$

with initial conditions,

$$x(r,0) = r, \quad t(r,0) = 0, \text{ and } z(r,0) = \phi(r).$$

Solving the ODE corresponding to z, we get $z(r,s) = c_3(r)$ with initial conditions $z(r,0) = c_3(r) = \phi(r)$. Thus, $z(r,s) = \phi(r)$. Using this in the ODE of x, we get

$$\frac{dx(r,s)}{ds} = \phi(r).$$

Solving the ODE's, we get

$$x(r,s) = \phi(r)s + c_1(r), \quad t(r,s) = s + c_2(r)$$

with initial conditions

$$x(r,0) = c_1(r) = r$$

and

$$t(r,0) = c_2(r) = 0.$$

Therefore,

$$x(r,s) = \phi(r)s + r, \text{ and } t(r,s) = s.$$

Eliminating s, we get the characteristic curves on the $x-t$ plane on which we know u is constant. Thus, the characteristic curves are $x = \phi(x_0)t + x_0$ and value of u on this curve is $\phi(x_0)$. Hence $u(\phi(x_0)t + x_0, t) = \phi(x_0)$.

Suppose we choose ϕ to be the function

$$\phi(x) = \begin{cases} 1 & x < 0 \\ 1 - x & 0 \leq x \leq 1 \\ 0 & 1 \leq x. \end{cases}$$

Then the characteristic curves are

$$x = \begin{cases} t + x_0 & x_0 < 0 \\ (1 - x_0)t & 0 \leq x_0 \leq 1 \\ x_0 & 1 \leq x_0. \end{cases}$$

Note that the characteristic curves are intersecting. This is no issue, as long as, the solution u which is constant on these curves takes the same constant in each characteristic curve. Unfortunately, that is not the case here:

$$u(x,t) = \begin{cases} 1 & x < t \\ \frac{x}{t+1} & 0 \leq 1 - \frac{x}{t} \leq 1 \\ 0 & 1 \leq x. \end{cases}$$

This situation is, usually, referred to as *shock waves*.

1.3 Space of Test Functions

The space of test functions is, in a sense, the argument for the generalised function or distributions (a notion we shall introduce soon). Let $(a, b) \subset \mathbb{R}$ be a non-empty open subset, not necessarily bounded. Let $C_c^\infty(a, b)$ denote the class of all infinitely differentiable functions on Ω with compact support in (a, b). The genius of L. Schwartz is the choice of this function space. The motivation of this choice is the following: If $f \in C^1(\Omega)$ and $\phi \in C_c^\infty(a, b)$, then by classical integration by parts we have

$$\int_a^b \phi(x) f'(x)\, dx = -\int_a^b f(x)\phi'(x)\, dx. \tag{1.3.1}$$

There are no boundary integrals above because ϕ has compact support in (a, b) and vanishes at the end-points a and b. This explains the choice of

compact support. Suppose f was chosen from $C^k(a,b)$, then above integration by parts could be repeated k times to get

$$\int_a^b \phi(x) f^{(k)}(x)\, dx = (-1)^k \int_a^b f(x)\phi^{(k)}\, dx.$$

Observe that the maps $\phi \mapsto \int_a^b f\phi\, dx$ and $\phi \mapsto \int_a^b f\phi'\, dx$ are linear on $C_c^\infty(a,b)$. If there exists a suitable complete topology on $C_c^\infty(a,b)$ such that these linear maps are continuous, then f can be identified with a continuous linear functional on $C_c^\infty(a,b)$ given by

$$\phi \mapsto \int_a^b f\phi\, dx \tag{1.3.2}$$

and the derivative of f, f', can be identified with the mapping ϕ mapped to the right hand side of (1.3.1), even if $f \notin C^1(\Omega)$, as long as the integrals make sense.

Let $\Omega \subset \mathbb{R}^n$ be a non-empty open subset. Let $C_c^\infty(\Omega)$ denote the class of all infinitely differentiable functions on Ω with compact support. Observe that $C_c^\infty(\Omega)$ is a vector space under usual addition and scalar multiplication of real-valued functions. Of course, the zero function is in $C_c^\infty(\Omega)$. We know that polynomials, trigonometric functions, exponential are all C^∞ functions. Any function in $C_c^\infty(\Omega)$ falls to zero within a compact set.

Exercise 1. Show that any non-zero function in $C_c^\infty(\Omega)$, for $\Omega \subset \mathbb{R}^n$, cannot be an analytic function.

Proof. If f were analytic then, for all $a \in \Omega$, we have the Taylor expansion $\phi(x) = \sum_{k=0}^\infty \frac{\phi^{(k)}(a)}{k!}(x-a)^k$. Analyticity of f implies that the radius of convergence of the Taylor series at a is infinite. For $\phi \in C_c^\infty(\Omega)$, choose $a \notin \text{supp}(\phi)$. The Taylor's expansion around a imply that ϕ is zero function, which is a contradiction. □

Therefore, no non-zero analytic function can sit in $C_c^\infty(\Omega)$ because zero function is the only analytic function with compact support. The function $f(x) = e^{ax}$ is a smooth analytic function on $\mathbb{R}$ for all $a \in \mathbb{R}$. Do we have non-zero smooth (non-analytic) functions in $C_c^\infty(\Omega)$?

1.3.1 Smooth Non-Analytic (Bump) Functions

We are looking for a smooth non-analytic functions in $C_c^\infty(\Omega)$. This can be viewed as gluing of the zero function, along the boundary of a compact set, with smooth function inside the compact set. The question is : can they be glued such that the smoothness is preserved in all of domain.

Two continuous maps can be glued together to result in a continuous map. In a different view point: consider a connected subset (interval) $E = (a, b)$ of $\mathbb{R}$ and the function f on E^c which takes zero on the connected component, $x \leq a$ and 1 on the connected component $x \geq b$. This is a step function. A continuous extension is trivial: Consider the linear map $x \mapsto \frac{x-a}{b-a}$ from $[a, b]$ to $[0, 1]$ which glues. However, the function is not differentiable at a and b. Similarly, gluing of the smooth maps $x \mapsto -x$ on $(-\infty, 0]$ and $x \mapsto x$ on $[0, \infty)$ yields the non-differentiable map $|x|$ on $\mathbb{R}$. Thus the smoothness property need not be preserved when glued. A tool to accomplish the gluing of smooth functions in a smooth way is the partition of unity (cf. Appendix B).

Recall that we wish to define f on E such that the extended function is smooth or C^∞ on $\mathbb{R}$. We wish to glue a function in (a, b) such that on $\mathbb{R}$ it is smooth. Note that the smoothness can break only at the end-points a and b. Thus, if we can find a smooth function whose zero derivatives are at a point, then we hope to transit smoothly from 0 to 1 in (a, b).

Recall that the positive function $f(x) = e^{1/x}$ behaves badly at $x = 0$. From the right side it approaches $+\infty$ and from left side it approaches zero. However, for $x \neq 0$, $e^{1/x}$ is infinitely differentiable (smooth). In fact, the k-th derivative of $e^{1/x}$ is $P_k(1/x)e^{1/x}$. The proof is by induction. For $k = 0$, $P_0(t) \equiv 1$, the constant function 1. Let us assume $f^{(k)}(x) = P_k(1/x)e^{1/x}$ for $x \neq 0$. Then,

$$f^{(k+1)}(x) = P_k(1/x)\left(\frac{-1}{x^2}\right)e^{1/x} + e^{1/x}P_k'(1/x)\left(\frac{-1}{x^2}\right) = P_{k+1}(1/x)e^{1/x},$$

where $P_{k+1}(t) = -t^2(P_k(t) + P_k'(t))$.

Exercise 2. Show that the k-th derivative of $e^{1/x}$ for $x \neq 0$ is

$$\frac{d^k}{dx^k}(e^{1/x}) = e^{1/x}\left[(-1)^k \sum_{i=1}^{k} \binom{k}{i}\binom{k-1}{i-1}(k-i)!x^{-k-i}\right].$$

The coefficients of the polynomial appearing in the derivative are called the *Lah* numbers, after Ivo Lah who encountered these numbers in actuarial science.

Example 1.1. Consider the non-negative function $f_0 : (-\infty, 0] \to [0, 1]$ defined as

$$f_0(x) = \begin{cases} \exp(1/x) & \text{if } x < 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Example 1.2. A variant of the above example is

$$f_1(x) = \begin{cases} \exp(-1/|x|) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Example 1.3. Another variant is the function $f_2 : \mathbb{R} \to \mathbb{R}$ defined as

$$f_2(x) = \begin{cases} \exp(-1/x) & \text{if } x > 0 \\ 0 & \text{if } x \leq 0. \end{cases}$$

It is clear that $0 \leq f_i(x) < 1$ and f_i is infinitely differentiable for all $x \neq 0$ and $i = 0, 1, 2$. We need to check differentiability only at $x = 0$. We shall do it only for $f = f_2$ as the proof is similar for other cases. Of course, the left side limit of f and its derivative is zero at $x = 0$. Thus, we consider

$$f'(0) = \lim_{h \to 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \to 0^+} \frac{\exp(-1/h)}{h}.$$

For all $h > 0$ and any fixed positive integer $m > 0$, we have

$$\frac{1}{h} = h^m \frac{1}{h^{m+1}} \leq h^m (m+1)! \sum_{i=0}^{\infty} \frac{1}{i!} \left(\frac{1}{h}\right)^i = h^m (m+1)! \exp(1/h).$$

Thus,

$$\frac{\exp(-1/h)}{h} \leq h^m (m+1)!$$

and the larger the m, the sharper the estimate. Hence,

$$f'(0) = \lim_{h \to 0^+} \frac{\exp(-1/h)}{h} = 0.$$

The same argument follows for any $k + 1$ derivative of f because

$$f^{(k+1)}(0) = \lim_{h \to 0^+} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} = \lim_{h \to 0^+} \frac{P_{k-1}(h) \exp(-1/h)}{h^{2k+1}} = 0.$$

Therefore, $f \in C^\infty(\mathbb{R})$. We shall now observe that f is not analytic at $x = 0$. The Taylor series of f at $x = 0$,

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = 0,$$

converges to the zero function for all $x \in \mathbb{R}$. But for $x > 0$, we know that $f(x) > 0$ and hence do not converge to the Taylor series at $x = 0$. Thus, f is not analytic.

Example 1.4. The function $f : \mathbb{R} \to \mathbb{R}$ defined as

$$f_3(x) = \begin{cases} \exp(-1/x^2) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is infinitely differentiable but not analytic.

Example 1.5. The function $f : \mathbb{R} \to \mathbb{R}$ defined as

$$f_4(x) = \begin{cases} \exp(-1/x^2) & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

is infinitely differentiable but not analytic.

One can, in fact, construct bump functions where the function takes the value 1 on a given subset of the support of f.

Lemma 1.3.1 (Smooth Cut-off Function)**.** *For any interval (a, b) of $\mathbb{R}$, there is a decreasing smooth function $g : \mathbb{R} \to \mathbb{R}$ such that $0 < g(x) < 1$ for all $x \in (a, b)$ and*

$$g(x) = \begin{cases} 1 & x \leq a \\ 0 & x \geq b. \end{cases}$$

Proof. Choose $f = f_i$, for $i = 2$ or $i = 4$ in the above examples. This is precisely the choice of function whose zero derivatives are all at a point. Note that the function $f(b-x) = 0$ for all $x \geq b$ and the function $f(x-a) = 0$ for all $x \leq a$. Consider the non-zero positive function

$$h(x) = f(b - x) + f(x - a)$$

on $\mathbb{R}$ which is

$$h(x) = \begin{cases} f(b-x) & x \leq a \\ f(x-a) & x \geq b. \end{cases}$$

Since $f(b-x) < f(b-x) + f(x-a)$, addition of positive function. Moreover, h is smooth. Define the function

$$g(x) = \frac{f(b-x)}{f(b-x) + f(x-a)}$$

and it satisfies all the desired properties. □

In particular, when $f = f_2$, we have

$$g(x) = \frac{e^{-1/(b-x)}}{e^{-1/(b-x)} + e^{-1/(x-a)}}.$$

Note that the increasing function

$$1 - g(x) = \frac{f(x-a)}{f(b-x) + f(x-a)}$$

is identically zero for $x \leq a$ and one on $x \geq b$ satisfying the other properties of above lemma.

Lemma 1.3.2 (Smooth Bump Function)**.** *For any positive $a, b \in \mathbb{R}$ such that $a < b$, there is a $h \in C_c^\infty(\mathbb{R})$ such that $0 \leq h(x) \leq 1$, for all $x \in \mathbb{R}$, $h \equiv 1$ on $[-a, a]$ and supp(h)$= [-b, b]$.*

Proof. Recall the function g obtained in above lemma. Then the function $x \mapsto g(|x|)$ is identically one in $[-a, a]$ and zero outside $(-b, b)$. The desired function h is obtained by setting $h(x) = g(|x|)$. The function h is smooth because it is composed with the function $|x|$, which is smooth except at $x = 0$. But $h \equiv 1$ around the origin. □

Lemma 1.3.3 (Smooth Bump Function on $\mathbb{R}^n$)**.** *For any $0 < a < b$, there is a $h \in C_c^\infty(\mathbb{R}^n)$ with range in $[0, 1]$ such that $0 \leq h(x) \leq 1$, for all $x \in \mathbb{R}$, $h \equiv 1$ on the closed ball $B_a(0)$ and supp(h) is the closure of $B_b(0)$.*

Proof. The same function $h(x) = g(|x|)$ works. □

Theorem 1.3.4. *For any compact subset K of $\mathbb{R}^n$, then there exists a $\phi \in C_c^\infty(\mathbb{R}^n)$ such that $\phi \equiv 1$ on K.*

Proof. One can chose an open set $U \supset K$ such that $\overline{U}$ is compact. Then, U and K^c form an open cover of $\mathbb{R}^n$. By Theorem B.3.5, there is a partition of unity subordinate to the given cover of $\mathbb{R}^n$. Thus, there are smooth non-negative functions ϕ and ψ on $\mathbb{R}^n$ such that $\text{supp}(\phi)$ is in U, $\text{supp}(\psi)$ is in K^c and $\phi + \psi = 1$ on $\mathbb{R}^n$. On K, $\psi = 0$ and hence $\phi = 1$. Also, $\text{supp}(\phi)$ is a closed subset of a compact subset $\overline{U}$, hence ϕ has compact set. Thus, ϕ is the desired function. □

1.3.2 Mollifiers

The functions f_2 and f_4 in the above examples can be tweaked to construct a function in $C_c^\infty(\mathbb{R})$ such that $\text{supp}(f) = [-b, b]$ for any positive $b \in \mathbb{R}$. Note that this time we do not demand that the function takes constant value 1 in a subset of the support. Consider the transformation $g_2^b(x) = f_2(1 - |x|/b)$. Thus,

$$g_2^b(x) = \begin{cases} \exp\left(\frac{-b}{b-|x|}\right) & \text{if } |x| < b \\ 0 & \text{if } |x| \geq b. \end{cases}$$

Also, using the function

$$f_5(x) = \begin{cases} \exp\left(\frac{-1}{1-x^2}\right) & \text{if } |x| < 1 \\ 0 & \text{if } |x| \geq 1 \end{cases}$$

one defines the function

$$g_5^b(x) = \begin{cases} \exp\left(\frac{-b^2}{b^2-|x|^2}\right) & \text{if } |x| < b \\ 0 & \text{if } |x| \geq b. \end{cases}$$

In fact, the functions g_2^b, g_5^b can be extended to n dimensions and are in $C_c^\infty(\mathbb{R}^n)$ with support in $\overline{B}(0; b)$, the disk with centre at origin and radius b.

We shall now introduce an important sequence of functions in $C_c^\infty(\mathbb{R}^n)$, called *mollifiers*. Recall the $C_c^\infty(\mathbb{R}^n)$ functions g_2^b and g_5^b introduced in the previous section. For $\varepsilon > 0$, we set $b = \varepsilon$ in g_5^b (or g_2^b) and

$$\begin{aligned} \int_{\mathbb{R}^n} g_5^\varepsilon(x)\, dx &= \int_{|x|<\varepsilon} \exp\left(\frac{-\varepsilon^2}{\varepsilon^2 - |x|^2}\right) dx \\ &= \varepsilon^n \int_{|y|<1} \exp\left(\frac{-1}{1-|y|^2}\right) dy \quad \text{(by setting } y = x/\varepsilon) \\ &= \varepsilon^n c^{-1}, \end{aligned}$$

where

$$c^{-1} = \int_{|y|\leq 1} \exp\left(\frac{-1}{1-|y|^2}\right)\, dy.$$

Now, set $\rho_\varepsilon(x) = c\varepsilon^{-n} g_5^\varepsilon(x)$, equivalently,

$$\rho_\varepsilon(x) = \begin{cases} c\varepsilon^{-n}\exp\left(\frac{-\varepsilon^2}{\varepsilon^2-|x|^2}\right) & \text{if } |x| < \varepsilon \\ 0 & \text{if } |x| \geq \varepsilon. \end{cases} \tag{1.3.3}$$

Note that $\rho_\varepsilon \geq 0$ and is in $C_c^\infty(\mathbb{R}^n)$ with support in $B(0;\varepsilon)$. The sequence $\{\rho_\varepsilon\}$ is an example of mollifiers, a particular case of the *Dirac Sequence*.

Definition 1.3.5. *A sequence of functions $\{\rho_k\}$, say on $\mathbb{R}^n$, is said to be a* Dirac Sequence *if*

(i) $\rho_k \geq 0$ *for all* k.

(ii) $\int_{\mathbb{R}^n} \rho_k(x)\, dx = 1$ *for all* k.

(iii) For every given $r > 0$ *and* $\varepsilon > 0$, *there exists a* $N_0 \in \mathbb{N}$ *such that*

$$\int_{\mathbb{R}^n \setminus B(0;r)} \rho_k(x)\, dx < \varepsilon, \quad \forall k > N_0.$$

The connection between the sequence of mollifiers and Dirac delta function will become evident in the sequel. The notion of mollifiers is also an example for the *approximation of identity* concept in functional analysis and ring theory.

Definition 1.3.6. *An* approximate identity *is a sequence (or net) $\{\rho_k$ in a Banach algebra or ring (possible with no identity), $(X, \star)$ such that for any element a in the algebra or ring, the limit of $a \star \rho_k$ (or $\rho_k \star a$) is a.*

1.3.3 Multi-index Notation

We introduce the Schwartz's multi-index notation for derivative. If $\alpha = (\alpha_1, \ldots, \alpha_n)$ is an n-tuple of non-negative integers, i.e., $\alpha_i \geq 0$, $i = 1, 2, \ldots, n$, we denote $|\alpha| = \alpha_1 + \ldots + \alpha_n$ and the partial differential operator of *order* α,

$$D^\alpha = \frac{\partial^{\alpha_1}}{\partial {x_1}^{\alpha_1}} \cdots \frac{\partial^{\alpha_n}}{\partial {x_n}^{\alpha_n}} = \frac{\partial^{|\alpha|}}{\partial {x_1}^{\alpha_1} \ldots \partial {x_n}^{\alpha_n}}.$$

If $|\alpha| = 0$, then $D^\alpha f = f$. If α and β are two multi-indices, then $\alpha \leq \beta$ means $\alpha_i \leq \beta_i$ for $1 \leq i \leq n$. Also, $\alpha \pm \beta = (\alpha_1 \pm \beta_1, \ldots, \alpha_n \pm \beta_n)$, $\alpha! = \alpha_1! \ldots \alpha_n!$ and $x^\alpha = x_1^{\alpha_1} \ldots x_n^{\alpha_n}$. For $k \geq 1$, we define $D^k u(x) := \{D^\alpha u(x) \mid |\alpha| = k\}$. Thus, for $k = 1$, we regard Du as being arranged in a vector,

$$\nabla = \left(D^{(1,0,\ldots,0)}, D^{(0,1,0,\ldots,0)}, \ldots, D^{(0,0,\ldots,0,1)}\right) = \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \ldots, \frac{\partial}{\partial x_n}\right).$$

We call this the *gradient* vector. Similarly, for $k = 2$, we regard D^2 as being arranged in a matrix form (called the *Hessian* matrix),

$$D^2 = \begin{pmatrix} \frac{\partial^2}{\partial x_1^2} & \cdots & \frac{\partial^2}{\partial x_1 \partial x_n} \\ \frac{\partial^2}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2}{\partial x_2 \partial x_n} \\ & \ddots & \\ \frac{\partial^2}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2}{\partial x_n^2} \end{pmatrix}_{n \times n}.$$

The trace of the Hessian matrix is called the Laplace operator, denoted as $\Delta := \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$. Note that under some prescribed order on multi-indices α, $D^k u(x)$ can be regarded as a vector in R^{n^k}.

1.3.4 Topology on $C(\Omega)$, $C^\infty(\Omega)$ and $C_c^\infty(\Omega)$

Recall that our aim is to define a suitable topology in $C_c^\infty(\Omega)$ such that the operations described in (1.3.2) are continuous. Recall that for any compact set K, the norm on $C(K)$, space of continuous functions on K, is defined as

$$\|\phi\|_{\infty,K} := \max_{x \in K} |\phi(x)|$$

is complete and the space is a Banach space. Moreover, for the space of k-times differentiable functions on K, $C^k(K)$, we restructure the norm as follows to preserve completeness

$$\|\phi\|_{k,K} := \sum_{|\alpha|=0}^{k} \|D^\alpha \phi\|_{\infty,K}$$

and the space is a Banach space. Note that $\|\cdot\|_0 = \|\cdot\|_\infty$.[1] For any open subset Ω of $\mathbb{R}^n$, the spaces $C(\Omega)$ and $C^k(\Omega)$ are not normable but are

[1] Please note the abuse of notation. The infinity symbol (∞) is not denoting any derivative and is inherited as a limit of the p-norms.

metrizable[2]. They form a locally convex complete metric space called *Fréchet space*. We describe this briefly in the following paragraph.

For any open subset Ω of $\mathbb{R}^n$, there is a sequence K_j of non-empty compact subsets of Ω such that $\Omega = \cup_{j=0}^{\infty} K_j$ and $K_j \subset \text{Int}(K_{j+1})$, for all j (exhaustion of an open set by compact sets, cf. Lemma B.2.2). We define a countable family of semi-norms on $C(\Omega)$ as

$$p_j(\phi) = \|\phi\|_{\infty, K_j}.$$

The metric induced by the family of semi-norms on $C(\Omega)$ is

$$d(\phi, \psi) = \max_{j \in \mathbb{N} \cup \{0\}} \frac{1}{2^j} \frac{p_j(\phi - \psi)}{1 + p_j(\phi - \psi)}$$

and the metric is complete and $C(\Omega)$ is a Fréchet space. This is precisely the topology of compact convergence (uniform convergence on compact sets) or the compact-open topology in this case. The metric defined above is called the *Fréchet* metric and is equivalent to the metric

$$d(\phi, \psi) = \sum_{j=0}^{\infty} \frac{1}{2^j} \frac{p_j(\phi - \psi)}{1 + p_j(\phi - \psi)}.$$

Now that we have constructed a topology on the space $C(\Omega)$, we shall construct one for $C^\infty(\Omega)$. We define a countable family of semi-norms on $C^\infty(\Omega)$ as

$$p_j(\phi) = \sum_{|\alpha|=0}^{j} \|D^\alpha \phi\|_{\infty, K_j} = \|\phi\|_{j, K_j}.$$

Again, as before, the family of semi-norms induces a locally convex complete metric on $C^\infty(\Omega)$ making it a Fréchet space.

Exercise 3. Show that the topology given in $C(\Omega)$ and $C^\infty(\Omega)$ are independent of the choice the exhaustion compact sets $\{K_j\}$ of Ω.

The space $C_c^\infty(\Omega)$ is a subset of $C^\infty(\Omega)$ and the semi-norms defined in $C^\infty(\Omega)$ restricted to $C_c^\infty(\Omega)$ becomes a norm. For any $\phi \in C_c^\infty(\Omega)$, the family

[2]Any norm space is a metric space but the converse is not true always because $d(x, 0)$ may fail to satisfy the properties of norm. A metric d induces a norm if $d(x + z, y + z) = d(x, y)$ and $d(\alpha x, \alpha y) = |\alpha| d(x, y)$.

of norms ($j = 0, 1, 2, \ldots$)

$$\|\phi\|_j = \sum_{|\alpha|=0}^{j} \|D^\alpha \phi\|_{\infty,\Omega} = \|\phi\|_{j,\Omega}$$

induces the same topology as the one inherited from $C^\infty(\Omega)$. However, this norm induced topology on $C_c^\infty(\Omega)$ is not complete and its completion is $C^\infty(\Omega)$. For more details and proof of the semi-norm induced locally convex topology refer to [Rud91].

Exercise 4. Let $\phi \in C_c^\infty(\mathbb{R})$ with supp(ϕ)= $[0, 1]$ and $\phi > 0$ in $(0, 1)$. Then the sequence

$$\psi_m(x) = \sum_{i=1}^{m} \frac{1}{i}\phi(x - i) = \phi(x - 1) + \frac{1}{2}\phi(x - 2) + \ldots + \frac{1}{m}\phi(x - m)$$

is Cauchy in the topology induced by the norms, but $\lim \psi_m \notin C_c^\infty(\mathbb{R})$.

1.3.5 Complete Topology on $C_c^\infty(\Omega)$

We shall construct a complete (non-metrizable) topology on $C_c^\infty(\Omega)$ different from the one inherited from $C^\infty(\Omega)$. For every compact subset $K \subset \Omega$, let $C^\infty(K)$ denote the class of all functions in $C^\infty(\Omega)$ such that their support is in K. The space $C^\infty(K)$ is given the topology inherited from $C^\infty(\Omega)$, the same induced by the family of norms defined at the end of previous section,

$$\|\phi\|_{j,K} = \sum_{|\alpha|=0}^{j} \|D^\alpha \phi\|_{\infty,K} \quad \forall j \geq 0.$$

Exercise 5. $C^\infty(K)$ is a closed subspace of $C^\infty(\Omega)$ under the inherited topology of $C^\infty(\Omega)$.

Proof. For each $x \in \Omega$, define the functional $T_x : C^\infty(\Omega) \to \mathbb{R}$ defined as $T_x(\phi) = \phi(x)$. For each $x \in \Omega$, there is a j_0 such that $x \in K_j$ for all $j \geq j_0$. Then,

$$|T_x(\phi)| = |\phi(x)| \leq p_j(\phi) \quad \forall j \geq j_0.$$

The functional T_x is continuous[3] because uniform convergence implies pointwise convergence. The topology on $C^\infty(\Omega)$ is uniform convergence on compact subsets. Therefore the kernel of T_x,

$$\ker(T_x) := \{\phi \in C^\infty(\Omega) \mid T_x(\phi) = 0\},$$

is a closed subspace of $C^\infty(\Omega)$. Note that $\ker(T_x)$ is precisely those $\phi \in C^\infty(\Omega)$ such that $\phi(x) = 0$. We claim that

$$C^\infty(K) = \cap_{x\in K^c}\ker(T_x).$$

If $\phi \in C^\infty(K)$ then ϕ is in the intersection because $\phi(x) = 0$ for all $x \in K^c$. Conversely, If $\phi(x) = 0$ for all $x \in K^c$, then $\text{supp}(\phi)\subseteq K$. Thus, for any compact subset K of Ω, we have our claim that $C^\infty(K)$ is an arbitrary intersection of closed sets. Thus, $C^\infty(K)$ is closed in $C^\infty(\Omega)$. □

Recall that K_j is a sequence of non-empty compact subsets of Ω such that $\Omega = \cup_{j=1}^\infty K_j$ and $K_j \subset \text{Int}(K_{j+1})$, for all j. Note that

$$C_c^\infty(\Omega) = \cup_{j=1}^\infty C^\infty(K_j)$$

and $C^\infty(K_\ell) \subset C^\infty(K_m)$ for all $\ell < m$. With the given topology on these spaces, inherited from $C^\infty(\Omega)$, the inclusion map $I_{k\ell} : C^\infty(K_\ell) \to C^\infty(K_m)$ is continuous. This is because the local base in $C^\infty(K_m)$ is

$$\left\{\phi \in C^\infty(K_m) \mid \|\phi\|_m < \frac{1}{m}\right\}.$$

For any such ϕ in the local base, we have $\|\phi\|_\ell < 1/\ell$ and is in the local base of $C^\infty(K_\ell)$. Thus, we endow C_c^∞ with the finest[4] topology that makes the inclusion maps $I_j : C^\infty(K_j) \to C_c^\infty(\Omega)$ continuous, for all j. In other words, a set U in $C_c^\infty(\Omega)$ is said to be open if and only if $I_j^{-1}(U)$ is open in $C^\infty(K_j)$ for all $j \geq 1$. Such a topology is called the *inductive limit topology* with respect to $C^\infty(K_j)$ and the maps $I_{k\ell}$. The space $C_c^\infty(\Omega)$ is complete with respect to the inductive limit topology because any Cauchy sequence is Cauchy in $C^\infty(K_j)$, for some j. Since $C^\infty(K_j)$ is closed, the space $C_c^\infty(\Omega)$ is complete w.r.t the inductive limit topology. Though each $C^\infty(K)$ is metrizable, the space $C_c^\infty(\Omega)$ is not metrizable (cf. Exercise 7).

[3] Compare the functional T_x with Dirac distribution to be introduced later

[4] strongest or largest topology, the one with more open sets

Exercise 6. Every proper subspace of a topological vector space has empty interior.

Proof. Let X be a topological vector space and $V \subsetneq X$ be a vector subspace of X. We need to show that for $x \in V$ there is an open set U containing x such that $U \subset V$. It is enough to show the claim for $x = 0$ $(0 \in V)$ because if U contains 0, then $U + \{x\} \subset V$ is an open set (due to continuity of addition) containing x. Due to vector space structure $U + \{x\}$ is in V. For every $x \in X$, we can define a function $f_x : \mathbb{R} \to X$ as $f_x(\lambda) = \lambda x$. The function f_x is continuous because the scalar multiplication map from $\mathbb{R} \times X$ to X is continuous. Suppose that there is an open set V containing 0 such that $V \subset X$. Then $f_x^{-1}(V)$ will be an open set containing $0 \in \mathbb{R}$. Thus, for any $\lambda \in f_x^{-1}(V)$, we have $\lambda x \in V$. Thus, $x \in V$. This argument is true for all $x \in X$, thus $X \subset V$. This implies $X = V$, a contradiction. □

Exercise 7. The inductive limit topology on $C_c^\infty(\Omega)$ is not metrizable.

Proof. Recall that $C_c^\infty(\Omega) = \cup_{j=1}^\infty C^\infty(K_j)$, where each closed set $C^\infty(K_j)$ has empty interior (cf. Exercise 6). Therefore, the complete space $C_c^\infty(\Omega)$ is a countable union of no-where dense sets. If $C_c^\infty(\Omega)$ was metrizable then it would contradict the Baire's category theorem. □

Definition 1.3.7. *The space $C_c^\infty(\Omega)$ endowed with the inductive limit topology, and denoted as $\mathcal{D}(\Omega)$, is called the space of* test functions.

Exercise 8 (cf. Exercise 3). Show that the topology defined on $\mathcal{D}(\Omega)$ is independent of the choice of (exhaustion sets) the sequence of compact sets K_j of Ω.

The sequential characterisation of the inductive limit topology on $\mathcal{D}(\Omega)$ is given below. We define only for the zero converging sequence due to continuity of the addition operation.

Proposition 1.3.8. *A sequence of functions $\{\phi_m\} \subset \mathcal{D}(\Omega)$ converges to zero iff there exists a compact set $K \subset \Omega$ such that supp$(\phi_m) \subset K$, for all m, and ϕ_m and all its derivatives converge uniformly to zero on K.*

1.3.6 Regularization and Cut-off Technique

In this section, we show that that space of test functions is densely contained in L^p space. This observation is very useful since most conjectures can be checked for test functions and then carried on to the required function in L^p. As usual, let Ω be an open subset of $\mathbb{R}^n$. We already know the following results from the theory of Lebesgue measure on $\mathbb{R}^n$.

Theorem 1.3.9 (cf. Theorem **??**)**.** *For $1 \leq p < \infty$, the class of all simple*[5] *functions $S(\Omega)$ is dense in $L^p(\Omega)$.*

Theorem 1.3.10 (cf. Theorem **??**)**.** *For $1 \leq p < \infty$, $C_c(\Omega)$ is dense in $L^p(\Omega)$.*

Further, a step ahead, we show that the space of test functions, $C_c^\infty(\Omega)$ is densely contained in $L^p(\Omega)$. This result is established by using the *technique of regularization by convolution* introduced by Leray and Friedrichs.

Definition 1.3.11. *Let $f, g \in L^1(\mathbb{R}^n)$. The convolution $f * g$ is defined as,*

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x-y)g(y)\,dy \quad \forall x \in \mathbb{R}^n.$$

The integral on RHS is well-defined, since by Fubini's Theorem and the translation invariance of the Lebesgue measure, we have

$$\int_{\mathbb{R}^n \times \mathbb{R}^n} |f(x-y)g(y)|\,dx\,dy = \int_{\mathbb{R}^n} |g(y)|\,dy \int_{\mathbb{R}^n} |f(x-y)|\,dx = \|g\|_1 \|f\|_1.$$

Thus, for a fixed x, $f(x-y)g(y) \in L^1(\mathbb{R}^n)$.

Theorem 1.3.12. *The convolution operation on $L^1(\mathbb{R}^n)$ is both commutative and associative.*

[5] By our definition, simple function is non-zero on a finite measure. A simple function ϕ is a non-zero function on $\mathbb{R}^n$ having the (canonical) form

$$\phi(x) = \sum_{i=1}^{k} a_i 1_{E_i}$$

with disjoint measurable subsets $E_i \subset \mathbb{R}^n$ with $\mu(E_i) < +\infty$ and $a_i \neq 0$, for all i, and $a_i \neq a_j$ for $i \neq j$.

Theorem 1.3.13 (Young's inequality)**.** *Let $1 \leq p, q, r < \infty$ such that $(1/p)+(1/q) = 1 + (1/r)$. If $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$, then the convolution $f * g \in L^r(\mathbb{R}^n)$ and*

$$\|f * g\|_r \leq \|f\|_p \|g\|_q.$$

In particular, for $1 \leq p < \infty$, if $f \in L^1(\mathbb{R}^n)$ and $g \in L^p(\mathbb{R}^n)$, then the convolution $f * g \in L^p(\mathbb{R}^n)$ and

$$\|f * g\|_p \leq \|f\|_1 \|g\|_p.$$

Proposition 1.3.14. [6] *Let $f \in L^1(\mathbb{R}^n)$ and $g \in L^p(\mathbb{R}^n)$, for $1 \leq p \leq \infty$. Then*

$$supp(f * g) \subset \overline{supp(f) + supp(g)}$$

If both f and g have compact support, then support of $f * g$ is also compact. The convolution operation preserves smoothness.

Proposition 1.3.15. [7] *Let $f \in C_c^k(\mathbb{R}^n)$ $(k \geq 1)$ and let $g \in L^1_{loc}(\mathbb{R}^n)$. Then $f * g \in C^k(\mathbb{R}^n)$ and for all $|\alpha| \leq k$*

$$D^\alpha(f * g) = D^\alpha f * g = f * D^\alpha g.$$

Theorem 1.3.16 (Regularization technique)**.** *$C^\infty(\mathbb{R}^n)$ is dense in $C(\mathbb{R}^n)$ under the uniform convergence on compact sets topology.*

Proof. Let $g \in C(\mathbb{R}^n)$ and $K \subset \mathbb{R}^n$ be a compact subset. Note that g is uniformly continuous on K. Hence, for every $\eta > 0$, there exist a $\delta > 0$ (independent of x and dependent on K and η) such that $|g(x-y) - g(x)| < \eta$ whenever $|y| < \delta$ for all $x \in K$. For each $m \in \mathbb{N}$, set $\rho_m := \rho_{1/m}$, the sequence of mollifiers. Define $g_m := \rho_m * g$. Note that $g_m \in C^\infty(\mathbb{R}^n)$ $(D^\alpha g_m = D^\alpha \rho_m * g)$. Now, for all $x \in \mathbb{R}^n$,

$$\begin{aligned} |g_m(x) - g(x)| &= \left| \int_{|y| \leq 1/m} g(x-y)\rho_m(y)\, dy - g(x) \int_{|y| \leq 1/m} \rho_m(y)\, dy \right| \\ &\leq \int_{|y| \leq 1/m} |g(x-y) - g(x)| \rho_m(y)\, dy \end{aligned}$$

[6] Refer Brezis for proof
[7] Refer Brezis for proof

Hence, for all $x \in K$ and $m > 1/\delta$, we have

$$\begin{aligned} |g_m(x) - g(x)| &\leq \int_{|y|<\delta} |g(x-y) - g(x)| \rho_m(y)\, dy \\ &\leq \eta \int_{|y|<\delta} \rho_m(y)\, dy = \eta \end{aligned}$$

Since the δ is independent of $x \in K$, we have $\|g_m - g\|_\infty < \eta$ for all $m > 1/\delta$. Hence, $g_m \to g$ uniformly on K. □

Theorem 1.3.17. *For any $\Omega \subseteq \mathbb{R}^n$, $C_c^\infty(\Omega)$ is dense in $C_c(\Omega)$ under the uniform topology.*

Proof. Let $g \in C_c(\Omega)$ and $K := \operatorname{supp}(g)$. One can view $C_c(\Omega)$ as a subset of $C_c(\mathbb{R}^n)$ under the following identification: Each $g \in C_c(\Omega)$ is extended to $\mathbb{R}^n$ as $\tilde{g}$

$$\tilde{g}(x) = \begin{cases} g(x) & x \in K \\ 0 & x \in \mathbb{R}^n \setminus K. \end{cases}$$

By Theorem 1.3.16, the sequence $g_m := \rho_m * \tilde{g}$ in $C^\infty(\mathbb{R}^n)$ converges to $\tilde{g}$ uniformly on every compact subsets of $\mathbb{R}^n$. Note that $\operatorname{supp}(g_m) \subset K + B(0; 1/m)$ is compact because K is compact. Since we want $g_m \in C_c^\infty(\Omega)$, we choose $m_0 \in \mathbb{N}$ such that $1/m_0 < \operatorname{dist}(K, \Omega^c)$. Thus, $\operatorname{supp}(g_m) \subset \Omega$ and $g_m \in C_c^\infty(\Omega)$, for all $m \geq m_0$. The proof of the uniform convergence of g_m to g on Ω is same as in Theorem 1.3.16. □

Corollary 1.3.18. *For any $\Omega \subseteq \mathbb{R}^n$, $C_c^\infty(\Omega)$ is dense in $C(\Omega)$ under the uniform convergence on compact sets topology.*

Theorem 1.3.19 (Regularization technique)**.** *The space $C^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$, for $1 \leq p < \infty$, under the p-norm.*

Proof. Let $f \in L^p(\mathbb{R}^n)$. For each $m \in \mathbb{N}$, set $\rho_m := \rho_{1/m}$, the sequence of mollifiers. Then the sequence $f_m := \rho_m * f$ is in $C^\infty(\mathbb{R}^n)$. Since $\rho_m \in L^1(\mathbb{R}^n)$, by Young's inequality, $f_m \in L^p(\mathbb{R}^n)$. We shall prove that f_m converges to f in p-norm. For any given $\varepsilon > 0$, by Theorem **??**, we choose a $g \in C_c(\mathbb{R}^n)$ such that $\|g - f\|_p < \varepsilon/3$. Therefore, by Theorem 1.3.17, there is a compact subset $F \subset \mathbb{R}^n$ such that $\|\rho_m * g - g\|_\infty < \varepsilon/3(\mu(F))^{1/p}$. Hence, $\|\rho_m * g - g\|_p < \varepsilon/3$.

Thus, for sufficiently large m, we have

$$\begin{aligned}\|f_m - f\|_p &\leq \|\rho_m * f - \rho_m * g\|_p + \|\rho_m * g - g\|_p + \|g - f\|_p \\ &< \|\rho_m * (f-g)\|_p + \frac{2\varepsilon}{3} \leq \|f-g\|_p \|\rho_m\|_1 + \frac{2\varepsilon}{3} \\ &< \frac{\varepsilon}{3} + \frac{2\varepsilon}{3} = \varepsilon.\end{aligned}$$

The first term has been handled using Young's inequality. □

Theorem 1.3.20 (Cut-Off Technique)**.** *For $1 \leq p < \infty$ and $\Omega \subseteq \mathbb{R}^n$, $C_c^\infty(\Omega)$ is dense in $L^p(\Omega)$.*

Proof. Any $f \in L^p(\Omega)$ can be viewed as an element in $L^p(\mathbb{R}^n)$ under the extension

$$\tilde{f}(x) = \begin{cases} f(x) & x \in \Omega \\ 0 & x \in \Omega^c. \end{cases}$$

By Theorem 1.3.19, the sequence $f_m := \rho_m * \tilde{f}$ converges to $\tilde{f}$ in p-norm. The sequence $f_m\}$ may fail to have compact support in Ω because support of $\tilde{f}$ is not necessarily compact in Ω. To fix this issue, we shall multiply the sequence with suitable choice of test functions in $C_c^\infty(\Omega)$. Choose the sequence of exhaustion compact sets $\{K_m\}$ in Ω (cf. Lemma B.2.2). In particular, for $\Omega = \mathbb{R}^n$, we can choose $K_m = B(0;m)$. Note that $\Omega = \cup_m K_m$. Consider[8] $\{\phi_m\} \subset C_c^\infty(\Omega)$ such that $\phi_m \equiv 1$ on K_m and $0 \leq \phi_m \leq 1$, for all m. We extend ϕ_m by zero on Ω^c. Define $F_m := \phi_m f_m$ and, hence, $F_m \in C_c^\infty(\Omega)$. Also, $F_m = f_m$ on K_m and $|F_m| \leq |f_m|$ in $\mathbb{R}^n$. Thus,

$$\begin{aligned}\|F_m - f\|_{p,\Omega} &= \|F_m - \tilde{f}\|_{p,\mathbb{R}^n} \leq \|\phi_m f_m - \phi_m \tilde{f}\|_{p,\mathbb{R}^n} + \|\phi_m \tilde{f} - \tilde{f}\|_{p,\mathbb{R}^n} \\ &\leq \|f_m - \tilde{f}\|_{p,\mathbb{R}^n} + \|\phi_m \tilde{f} - \tilde{f}\|_{p,\mathbb{R}^n}.\end{aligned}$$

The first term converges to zero by Theorem 1.3.19 and the second term converges to zero by Dominated convergence theorem. □

The case $p = \infty$ is ignored in the above results, because the L^∞-limit of $\rho_m * f$ is continuous and we do have discontinuous functions in $L^\infty(\Omega)$.

[8] The type of functions, ϕ_k, are called cut-off functions

1.4 Space of Distributions

Definition 1.4.1. *A linear functional T on $\mathcal{D}(\Omega)$ is said to be* continuous *if inverse image of open sets of $\mathbb{R}$ are open in $\mathcal{D}(\Omega)$. A linear functional T on $\mathcal{D}(\Omega)$ is said to be* sequential continuous *if $T\phi_m \to 0$ in $\mathbb{R}$ whenever $\phi_m \to 0$ in $\mathcal{D}(\Omega)$.*

Exercise 9. For any compact set $K \subset \Omega$, the restriction to $C^\infty(K)$ of any continuous map on $C_c^\infty(\Omega)$ is also continuous on $C^\infty(K)$. Similarly, for any compact set $K \subset \Omega$, the restriction to $C^\infty(K)$ of any sequentially continuous map on $C_c^\infty(\Omega)$ is also sequentially continuous on $C^\infty(K)$.

Hint. For each compact set K there is an positive integer i_0 such that $K \subset K_i$, for all $i \geq i_0$. □

It is enough to define for zero convergent sequences because addition operation is continuous. For a first countable space the notion of continuity and sequential continuity are equivalent. A Hausdorff topological vector space is metrizable iff it is first countable. We know that $\mathcal{D}(\Omega)$ is not metrizable (cf. Exercise 7) and hence cannot be first countable.

Theorem 1.4.2. *Let $T : \mathcal{D}(\Omega) \to \mathbb{R}$ be a linear map. Then the following are equivalent:*

(i) T is continuous, i.e., inverse image of open set in $\mathbb{R}$, under T, is open in $\mathcal{D}(\Omega)$.

(ii) For every compact subset $K \subset \Omega$, there exists a constant $C_K > 0$ and an integer $N_K \geq 0$ (both depending on K) such that

$$|T(\phi)| \leq C_K \|\phi\|_{N_K}, \quad \forall \phi \in C^\infty(K).$$

(iii) T is sequentially continuous.

Proof. ((i) $\implies$ (ii)): Let T be continuous on $\mathcal{D}(\Omega)$. Then, for any compact subset $K \subset \Omega$, the restriction of T to $C^\infty(K)$ is continuous (cf. Exercise 9). The inverse image of $(-c, c)$ under T is open set in $C^\infty(K)$ containing origin. Since $C^\infty(K)$ is first countable (normed space), there is a local base at 0. Thus, there is a N_K and for all $\phi \in C^\infty(K)$ such that $\|\phi\|_{N_K} \leq 1/N_K$, we have $|T(\phi)| < c$. Thus, for $\phi \in C^\infty(K)$,

$$\left| T\left(\frac{\phi}{N_K \|\phi\|_{N_K}} \right) \right| < c$$

and hence $|T(\phi)| < N_K c\|\phi\|_{N_K}$.

((ii) $\implies$ (iii)): Let $\phi_m \to 0$ in $\mathcal{D}(\Omega)$. By definition of the test function convergence, there is a compact set K such that $\|\phi_m\|_j \to 0$ for all j. Using the fact that $|T(\phi)| \leq C_K\|\phi\|_{N_K}$, we get $|T(\phi_m)| \to 0$. Thus $T(\phi_m) \to 0$.

((iii) $\implies$ (i)): Let T be sequentially continuous. Then, by Exercise 9, the restriction of T on $C^\infty(K)$, for every compact subset $K \subset \Omega$, is also sequentially continuous. But $C^\infty(K)$ is metrizable and hence T on $C^\infty(K)$ is continuous, for all compact K of Ω. Thus, T is continuous on $\mathcal{D}(\Omega)$. □

Definition 1.4.3. *A linear functional T on $\mathcal{D}(\Omega)$ is said to be a* distribution *on Ω, if for every compact subset $K \subset \Omega$, there exists a constant $C_K > 0$ and an integer $N_K \geq 0$ (both depending on K) such that*

$$|T(\phi)| \leq C_K\|\phi\|_{N_K}, \quad \forall \phi \in C^\infty(K).$$

By Theorem 1.4.2, the above definition is saying that any continuous linear functional on $\mathcal{D}(\Omega)$ is a distribution. The space of all distributions in Ω is denoted by $\mathcal{D}'(\Omega)$.

Definition 1.4.4. *If the N_K is independent of K, i.e., the same N is enough for all compact sets K, then the smallest such N is called the* order *of T. If there exist no such N, we say T is of* infinite *order.*

Exercise 10. Which of the following are distributions? If your answer is affirmation, give the order of the distribution. If your answer is in negation, give reasons. For $\phi \in \mathcal{D}(\mathbb{R})$, $T(\phi)$ is defined as:

(i) $\phi'(1) - \phi''(-2)$.

(ii) $\sum_{k=0}^{\infty} \phi^{(k)}(\pi)$.

(iii) $\sum_{k=0}^{\infty} \phi^{(k)}(k)$.

(iv) $\sum_{k=1}^{\infty} \frac{1}{k}\phi^{(k)}(k)$.

(v) $\int_{\mathbb{R}} \phi^2(x)\,dx$.

Proof. (i) This is a distribution of order 2.

(ii) This is not a distribution because it is not defined for all $\phi \in \mathcal{D}(\mathbb{R})$. For instance, choose $\phi(x) = e^{x-\pi}\psi(x)$, where $\psi \in \mathcal{D}(\mathbb{R})$ such that $\psi(x) = 1$ in a neighbourhood of π. Note that $\phi^{(k)}(\pi) = 1$, for all k and the sum is infinite.

(iii) This is a distribution of infinite order.

(iv) This is a distribution of infinite order.

(v) This is not a distribution, because it is nonlinear.

□

Exercise 11. Show that if $T \in \mathcal{D}'(\Omega)$ and ω is an open subset of Ω, then $T \in \mathcal{D}'(\omega)$.

Definition 1.4.5. *Two distributions $S, T \in \mathcal{D}'(\Omega)$ are said to be* equal *if $S(\phi) = T(\phi)$ for all $\phi \in \mathcal{D}(\Omega)$.*

1.4.1 Functions as Distributions

Definition 1.4.6. *We say a function f is* locally integrable *in Ω, denoted as $f \in L^1_{loc}(\Omega)$, if f is measurable and $\int_K |f(x)|\,dx < +\infty$, for every compact set $K \subset \Omega$.*

In other words, $f \in L^1_{\text{loc}}(\Omega)$ if $f \in L^1(K)$ for all compact subsets K of Ω. In particular, $L^1(\Omega) \subset L^1_{\text{loc}}(\Omega)$. This inclusion is strict because, if Ω is not of finite measure, then constant functions do not belong to $L^1(\Omega)$ but they are in $L^1_{\text{loc}}(\Omega)$.

Example 1.6. The function $\ln|x| \in L^1_{\text{loc}}(\mathbb{R})$. To see this it is enough to show that $\lim_{\varepsilon\to 0^+} \int_\varepsilon^1 |\ln x|\,dx$ exists and is finite. Consider

$$\lim_{\varepsilon\to 0^+} \int_\varepsilon^1 |\ln x|\,dx = \lim_{\varepsilon\to 0^+} -(x\ln x - x)\,|_\varepsilon^1 = -[-1 - \lim_{\varepsilon\to 0^+}(\varepsilon\ln(\varepsilon) - \varepsilon)] = 1.$$

Because $\varepsilon\ln(\varepsilon) \to 0$ as $\varepsilon \to 0^+$ (use L'Hospital's rule after proper substitution).

The function $1/x \notin L^1_{\text{loc}}(\mathbb{R})$ (after assigning a real value at $x = 0$). Similarly, $e^{1/x} \notin L^1_{\text{loc}}(\mathbb{R})$. In both these cases the integral is not finite on a compact set containing origin. For any $n \geq 2$, the function $|x|^{-\alpha} \in L^1_{\text{loc}}(\mathbb{R}^n)$ for all $\alpha < n$. For instance, when $n = 2$ and $\alpha = 1$, it is enough to show for every compact set containing origin. Working in polar coordinates, we have for every ball of radius ρ centred at origin of $\mathbb{R}^2$

$$\int_{B(0;\rho)} |x|^{-1}\,dx = \int_0^\rho \int_0^{2\pi} r^{-1} r\,dr\,d\theta = 2\pi\rho < +\infty.$$

The notion of locally integrable functions can be extended to L^p spaces, for all $p \geq 1$.

Definition 1.4.7. *For $1 \leq p < \infty$, we say a function f is* locally p-integrable *in Ω, denoted as $f \in L^p_{loc}(\Omega)$, if f is measurable and $\int_K |f(x)|^p \, dx < +\infty$, for every compact set $K \subset \Omega$.*

We, in general, will not treat separately the notion of locally p-integrable functions because any locally p-integrable function is locally integrable. i.e., if $f \in L^p_{\text{loc}}(\Omega)$ for all $1 < p < \infty$, then $f \in L^1_{\text{loc}}(\Omega)$. Because the Hölder's inequality implies that for any compact subset K of Ω,

$$\int_K |f(x)| \, dx \leq \left(\int_K |f|^p \, dx \right)^{1/p} (\mu(K))^{1/q} < +\infty.$$

Thus, $f \in L^1_{\text{loc}}(\Omega)$. Since any $L^p(\Omega)$, for $1 \leq p < \infty$, is in $L^p_{\text{loc}}(\Omega)$ and hence are in $L^1_{\text{loc}}(\Omega)$.

Exercise 12. If f is continuous on Ω, then $f \in L^1_{\text{loc}}(\Omega)$.

We shall now observe that to every locally integrable function one can associate a distribution. For any $f \in L^1_{\text{loc}}(\Omega)$, we define the functional T_f on $\mathcal{D}(\Omega)$ defined as,

$$T_f(\phi) = \int_\Omega f(x)\phi(x) \, dx.$$

The functional T_f is continuous on $\mathcal{D}(\Omega)$ (is in $\mathcal{D}'(\Omega)$) because for every compact set K in Ω,

$$|T_f(\phi)| \leq \left(\int_K |f| \, dx \right) \|\phi\|_0, \quad \forall \phi \in C^\infty(K).$$

The distribution T_f is of zero order. We usually identify the distribution T_f with the function f that induces it. Hence, every continuous function and every L^p function, for $p \geq 1$, induces the distribution described above.

Exercise 13. If $f : \Omega \to \mathbb{R}$ is such that

$$\int_\Omega |f\phi| \, dx < +\infty \quad \forall \phi \in \mathcal{D}(\Omega)$$

then $f \in L^1_{\text{loc}}(\Omega)$.

Proof. Let K be a compact subset of Ω. Then there is a $\phi_K \in \mathcal{D}(\Omega)$ such that $\phi_K \equiv 1$ on K and $0 \leq \phi_K \leq 1$. Hence,

$$\int_K |f(x)|\,dx = \int_\Omega |f(x)| \cdot 1_K\,dx \leq \int_\Omega |f(x)\phi_K(x)|\,dx < +\infty.$$

Thus, $f \in L^1_{\text{loc}}(\Omega)$. The function 1_K denotes the characteristic function which takes 1 on K and zero on the K^c. □

The constant function 0 on Ω induces the zero distribution (the zero functional on $\mathcal{D}(\Omega)$).

Exercise 14. If $f \in C(\Omega)$ such that $T_f \equiv 0$, i.e., for all $\phi \in \mathcal{D}(\Omega)$,

$$\int_\Omega f\phi\,dx = 0$$

then $f \equiv 0$ in Ω.

Proof. Let us assume f is continuous. If $f \neq 0$ in Ω, then there is a $x_0 \in \Omega$ such that $f(x_0) = \lambda \neq 0$. Define the function $g = f/\lambda$ which is also continuous on Ω such that $g(x_0) = 1$. Note the $\int g\phi = 0$ for all $\phi \in \mathcal{D}(\Omega)$. By the continuity of g, there is an $r > 0$ such that, for all $x \in B_r(x_0) \subset \Omega$, $g(x) > 1/2$. Choose a test function $\psi \in \mathcal{D}(\Omega)$ such that $\text{supp}(\psi) \subset B_r(x_0)$ and $\int_\Omega \psi(x)\,dx = 1$. For instance, choose ψ to be the mollifier function $\psi(x) = \rho_r(x - x_0)$. Then, in particular,

$$0 = \int_\Omega g\psi\,dx = \int_{B_r(x_0)} g\psi\,dx > \frac{1}{2}\int_{B_r(x_0)} \psi\,dx = \frac{1}{2} > 0,$$

a contradiction. Thus, $g \equiv 0$ on Ω and hence $f \equiv 0$ on Ω. □

Exercise 15. If $f \in L^1_{\text{loc}}(\Omega)$ is non-negative, $f \geq 0$, such that $T_f \equiv 0$, i.e., for all $\phi \in \mathcal{D}(\Omega)$,

$$\int_\Omega f\phi\,dx = 0$$

then $f = 0$ a.e. in Ω.

Proof. Consider a compact subset K of Ω. By outer regularity of Lebesgue measure, for every $\varepsilon > 0$, there is an open set $U_\varepsilon \supset K$ such that $\mu(U_\varepsilon \backslash K) < \varepsilon$. Let $\{\phi_\varepsilon\}$ be a sequence of functions in $\mathcal{D}(\Omega)$ such that $\phi_\varepsilon \equiv 1$ on K, $\phi_\varepsilon \equiv 0$

on U_ε^c and $0 \le \phi_\varepsilon \le 1$. Note that ϕ_ε converge point-wise to the characteristic function 1_K, as ε goes to 0. Therefore, by dominated convergence theorem,

$$0 = \lim_{\varepsilon \to 0} \int_\Omega f\phi_\varepsilon \, dx = \int_\Omega f \cdot 1_K \, dx = \int_K f \, dx.$$

Since $f \ge 0$, $\int_K f \, dx = \int_K |f| \, dx$. Thus, $\int_K f \, dx = \|f\|_{1,K} = 0$ for all compact subsets K of Ω. Hence $f = 0$ a.e. on all compact K of Ω and hence $f = 0$ a.e. in Ω. □

Exercise 16. If $f \in L^1_{\text{loc}}(\Omega)$, such that $T_f \equiv 0$, i.e., for all $\phi \in \mathcal{D}(\Omega)$,

$$\int_\Omega f\phi \, dx = 0$$

then $f = 0$ a.e. in Ω.

Proof. Note that it is enough to prove that $f = 0$ a.e. on all compact subsets K of Ω. For any compact subset K of Ω, choose $\phi_K \in \mathcal{D}(\Omega)$ such that $\phi_K \equiv 1$. Define, for each $x \in K$,

$$f_\varepsilon(x) := f\phi_K * \rho_\varepsilon(x) = \int_K f(y)\phi_K(y)\rho_\varepsilon(x-y) \, dy.$$

Since $y \mapsto \phi_K(y)\rho_\varepsilon(x-y)$ is in $\mathcal{D}(\Omega)$, we have $f_\varepsilon \equiv 0$ on K. Since $f_\varepsilon \to f$ in $L^1(K)$,

$$0 = \lim_{\varepsilon \to 0} \|f_\varepsilon - f\|_1, K = \lim_{\varepsilon \to 0} \int_K |f_\varepsilon - f| \, dx = \int_K |f| \, dx = \|f\|_{1,K}$$

and hence $f = 0$ a.e. in K. □

A consequence of the above observations is that the association $f \mapsto T_f$ is well-defined because if $f = g$ a.e. in Ω, then the distributions $T_f = T_g$. Thus, we have seen $L^1_{\text{loc}}(\Omega) \subset \mathcal{D}'(\Omega)$. In particular, $\mathcal{D}(\Omega) \subset \mathcal{D}'(\Omega)$. Any distribution that is induced by a locally integrable function is called a *regular* distribution. Otherwise, the distribution is called *singular*.

1.4.2 Measures as Distributions

Recall that a natural σ-algebra compatible with the topological space Ω is the Borel σ-algebra, $\mathcal{B}(\Omega)$. It is the smallest σ-algebra containing all open subsets of Ω.

Definition 1.4.8. *A* Borel measure *is a regular measure defined on the Borel σ-algebra, $\mathcal{B}(\Omega)$, of Ω. A* Radon measure *μ is a Borel measure such that $\mu(K) < +\infty$, for every compact set $K \subset \Omega$.*

The Radon measure is about mass concentration or distribution of finite matter. For instance, if α units of mass are concentrated at a single point $x \in \mathbb{R}^n$, then

$$\mu(E) = \begin{cases} \alpha & x \in E \\ 0 & x \notin E. \end{cases}$$

If α units of mass are uniformly distributed on a curve $\gamma \subset \mathbb{R}^n$ of length one ($|\gamma| = 1$), then $\mu(E) = \alpha \times |\gamma \cap E|$. The Lebesgue measure on $\mathbb{R}^n$ is a Radon measure and the counting measure is not a Radon measure.

To each signed Radon measure μ, we associate a linear functional on $\mathcal{D}(\Omega)$ as follows:

$$T_\mu(\phi) = \int_\Omega \phi \, d\mu, \quad \forall \phi \in \mathcal{D}(\Omega).$$

The functional T_μ is continuous on $\mathcal{D}(\Omega)$ (and is in $\mathcal{D}'(\Omega)$) because for every compact set K in Ω,

$$|T_\mu(\phi)| \leq \left(\int_K d\mu \right) \|\phi\|_0 = |\mu|(K) \|\phi\|_0, \quad \forall \phi \in C^\infty(K),$$

where $|\mu|$ denotes the total variation[9] of the signed measure. The distribution T_μ is of zero order.

Example 1.7. The Lebesgue measure on Ω restricted to the Borel σ-algebra, $\mathcal{B}(\Omega)$, of Ω is a Radon measure. The distribution induced by the Lebesgue measure is same as the one induced by locally integrable constant function 1 on Ω. Thus, the distribution induced by Lebesgue measure is a *regular* distribution.

[9] If you are not aware of total variation concept just think of the measure as non-negative, instead of signed measure, and replace $|\mu|(K)$ by $\mu(K)$

Example 1.8 (Dirac Measure). The Dirac measure is a Radon measure that assigns a total mass of 1 at a point $a \in \Omega$. Thus, on the Borel measurable subsets of Ω,

$$\delta_a(E) = \begin{cases} 1 & \text{if } a \in E \\ 0 & \text{if } a \notin E. \end{cases}$$

If $a = 0$ then the corresponding measure $\delta_0 = \delta$ is the Dirac measure discussed in §1.1. The distribution induced by Dirac measure δ_a, called *Dirac distribution*, is:

$$\delta_a(\phi) = \int_\Omega \phi(x)\, d\delta_a.$$

Exercise 17. Show that, for any function[10] g on Ω,

$$\int_\Omega g(x)\, d\delta_a = g(a).$$

Proof. We shall give two proofs: The first proof is the usual measure theory technique, If $g = 1_E$, the characteristic function of E, then

$$\int_\Omega 1_E\, d\delta_a = \int_E d\delta_a = \delta_a(E) = 1_E(a).$$

For any simple function $g = \sum_{i=1}^k \alpha_i 1_{E_i}$, then

$$\int_\Omega g(x)\, d\delta_a = \sum_{i=1}^k \alpha_i \delta_a(E_i) = g(a).$$

For any non-negative g, by choosing an increasing sequence of simple functions g_i converging to g, one can show that

$$\int_\Omega g(x)\, d\delta_a = g(a).$$

Further any $g = g^+ - g^-$, which ends the first proof.

The second proof is quite elegant. Consider the constant function h on Ω defined as $h(x) = g(a)$. Let

$$E := \{x \in \Omega / g(x) \neq h(x)\}.$$

[10] Yes! I mean "any" function, because the class of Dirac measurable sets is the power set and hence any function is Dirac measurable. Though, in the above context we have restricted the Dirac measure to the class of Borel σ-algebra

Hence, $a \notin E$ and therefore $\delta_a(E) = 0$. This implies that $g = h$ almost everywhere w.r.t δ_a and, hence, their integrals coincide

$$\int_\Omega g(x)\, d\delta_a = \int_\Omega h(x) d\delta_a.$$

But

$$\int_\Omega h(x) d\delta_a = g(a)\delta_a(\Omega) = g(a).$$

□

With the above observations, we note that the Dirac distribution is just

$$\delta_a(\phi) = \phi(a) \quad \forall \phi \in \mathcal{D}(\Omega). \tag{1.4.1}$$

Recall that the distribution induced by Lebesgue measure is same as the one induced by the constant function $1 \in L^1_{\text{loc}}(\Omega)$. If all measure induced distributions were also induced by some locally integrable functions, then we have not achieved anything different by considering measures as distribution. But it turns out that not all measure induced distribution are induced by locally integrable functions.

Proposition 1.4.9. *The Dirac distribution is a* singular *distribution, i.e., there is no $f \in L^1_{loc}(\Omega)$ such that $T_f = \delta_a$.*

Proof. Note that for any $a \in \Omega$, since Ω is open, there is a $\varepsilon_0 > 0$ such that $B(a, \varepsilon_0) \subset \Omega$. For each $0 < \varepsilon < \varepsilon_0$, we choose $\phi_\varepsilon \in \mathcal{D}(\Omega)$ with support in $B(a; \varepsilon)$, $0 \leq \phi_\varepsilon \leq 1$ and $\phi_\varepsilon = 1$ in $B(a; \frac{\varepsilon}{2})$. Thus,

$$\delta_a(\phi_\varepsilon) = \phi_\varepsilon(a) = 1 \quad \text{for all } 0 < \varepsilon < \varepsilon_0.$$

Suppose f is a locally integrable function such that $T_f = \delta_a$, then

$$\delta(\phi_\varepsilon) = T_f(\phi_\varepsilon) = \int_\Omega f\phi_\varepsilon\, dx = \int_{B(a;\varepsilon)} f\phi_\varepsilon\, dx \leq \int_{B(a;\varepsilon)} |f|\, dx.$$

Therefore, $1 \leq \int_{B(a;\varepsilon)} |f|\, dx$. Since $f \in L^1_{\text{loc}}(\Omega)$, the quantity on RHS is finite. Hence, as $\varepsilon \to 0$, we get a contradiction $1 \leq 0$. □

There are many more measures that induce singular distribution. In fact, one can identify the class of measures which induce regular distribution. They are, precisely, the absolutely continuous[11] measures w.r.t the Lebesgue measure. We shall not dwell on this topic, but a quick summary is as follows: Note that for each $f \in L^1_{\text{loc}}(\Omega)$, one can define a signed measure

$$\mu_f(E) := \int_E f\, dx \tag{1.4.2}$$

for all measurable subsets E of Ω. Conversely, the Radon-Nikodym theorem states that, for any absolutely continuous measure μ w.r.t the Lebesgue measure, there is a $f \in L^1_{\text{loc}}(\Omega)$ such that

$$\mu(E) = \int_E f\, dx$$

for all μ-measurable subsets of Ω. This is a generalisation of the Fundamental theorem of Calculus (FTC).

Recall the fact that $\mathcal{D}(\Omega) \subsetneq L^1_{\text{loc}}(\Omega) \subsetneq \mathcal{D}'(\Omega)$. Let $\mathcal{M}(\Omega)$ denote the space of Radon measures on the Borel σ-algebra of Ω then, in this section, we have shown the that $\mathcal{D}(\Omega) \subsetneq L^1_{\text{loc}}(\Omega) \subsetneq \mathcal{M}(\Omega) \subset \mathcal{D}'(\Omega)$. We end this section with the result that there is a classical function space $X \supset \mathcal{D}(\Omega)$ such that the dual of X is $\mathcal{M}(\Omega)$, i.e., $X' = \mathcal{M}(\Omega)$.

Theorem 1.4.10 (Riesz)**.** *Let $C_c(\Omega)$ be given the topology of compact convergence and the Banach space $\mathcal{M}(\Omega)$ with the total variation norm. For any positive linear functional $T : C_c(\Omega) \to \mathbb{R}$, there is a unique positive Radon measure μ on Ω such that*

$$T(f) = \int_\Omega f(x)\, d\mu \quad \forall f \in C_c(\Omega).$$

In the statement of the above theorem, by *positive linear functional* T we mean that $T(f) \geq 0$ for all $f \geq 0$. Any positive linear functional on $C_c(\Omega)$, with respect to the inductive limit topology, is continuous (hence bounded) for locally compact Hausdorff Ω.

Let $C_0(\Omega)$ be the class of all continuous function on Ω vanishing at ∞, i.e., for every $\varepsilon > 0$ there is a compact set $K \subset \Omega$ such that $|f(x)| < \varepsilon$ for all

[11] A measure μ is absolutely continuous w.r.t another measure ν if for every element E of the σ-algebra $\mu(E) = 0$ whenever $\nu(A) = 0$, denoted as $\mu \ll \nu$. The Lebesgue measure is absolutely continuous w.r.t counting measure but the converse is not true.

$x \in K^c$. The space $C_0(\Omega)$ is a Banach space with the supremum norm. For non-compact Ω, $C_c(\Omega)$ is densely contained in $C_0(\Omega)$ under supremum norm (cf. Theorem 1.3.18). Thus, we have the following extension:

Theorem 1.4.11 (Riesz-Markov)**.** *For any continuous linear functional $T : C_0(\Omega) \to \mathbb{R}$ there is a unique Radon measure $\mu \in \mathcal{M}(\Omega)$ such that*

$$T(f) = \int_\Omega f(x)\, d\mu \quad \forall f \in C_0(\Omega).$$

We have the following inclusions

$$\mathcal{D}(\Omega) \subsetneq C_c(\Omega) \subseteq C_0(\Omega) \subsetneq L^1_{\text{loc}}(\Omega) \subsetneq \mathcal{M}(\Omega) \subset \mathcal{D}'(\Omega).$$

Note that so far we have only seen examples of distributions which are of zero order. This is because of the following result:

Theorem 1.4.12. *A distribution $T \in \mathcal{D}'(\Omega)$ is of zero order iff T is a distribution induced by a Radon measure.*

Proof. The implication that a Radon measure induced distribution is of zero order is already shown. Conversely, let T be a distribution of zero order. Then, for every compact set $K \subset \Omega$, there is a $C_K > 0$ such that

$$|T(\phi)| \leq C_K \|\phi\|_0 \quad \forall \phi \in C^\infty(K).$$

The idea is to continuously extend T to $C_0(\Omega)$, in a unique way, and then consider the Radon measure obtained from Riesz-Markov result. Let $\phi \in C_0(\Omega)$ and define $\phi_\varepsilon := \phi * \rho_\varepsilon$. By Theorem 1.3.18, ϕ_ε converges uniformly to ϕ. The sequence $\{T(\phi_\varepsilon)\}$ is Cauchy in $\mathbb{R}$ because

$$|T(\phi_\varepsilon) - T(\phi_\delta)| = |T(\phi_\varepsilon - \phi_\delta)| \leq C_K \|\phi_\varepsilon - \phi_\delta\|_0.$$

Hence the sequence $\{T(\phi_\varepsilon)\}$ converges, thus we extend T uniquely on $C_0(\Omega)$ as $T(\phi) = \lim_{\varepsilon \to 0} T(\phi_\varepsilon)$. Thus, by Riesz-Markov, there is a Radon measure associated to the extended T. □

Definition 1.4.13. *A distribution $T \in \mathcal{D}'(\Omega)$ is said to be* positive *if $T(\phi) \geq 0$ for all $\phi \geq 0$ in $\mathcal{D}(\Omega)$.*

Exercise 18. Any positive functional on $\mathcal{D}(\Omega)$ is a distribution of order zero and hence corresponds to a positive Radon measure.

Proof. For any compact set $K \subset \Omega$, consider $\phi \in C^\infty(K)$ and $\psi \in \mathcal{D}(\Omega)$, a non-negative function, such that $\psi = 1$ on K. Set $\lambda = \|\phi\|_0$, then $\lambda - \phi(x) \geq 0$ on K. Thus, $\lambda\psi(x) - \phi(x) \geq 0$ is non-negative on Ω. Therefore,

$$\begin{aligned} 0 &\leq T(\lambda\psi(x) - \phi(x)) \\ T(\phi) &\leq \lambda T(\psi) = T(\psi)\|\phi\|_0. \end{aligned}$$

Hence, T is of order zero. Thus, there is a Radon measure associated to T. □

1.4.3 Multipole Distributions

Recall the formulation of Dirac distribution as given in the equation (1.4.1). This motivates, for a fixed non-negative integer k and $a \in \mathbb{R}$, the linear functional on $\mathcal{D}(\mathbb{R})$

$$\delta_a^{(k)}(\phi) = \phi^{(k)}(a), \quad \forall \phi \in \mathcal{D}(\mathbb{R}),$$

where $\phi^{(k)}$ denotes the k-th derivative of ϕ. This functional is continuous on $\mathcal{D}(\mathbb{R})$ because, for all $\phi \in C^\infty(K)$,

$$|\delta_a^{(k)}(\phi)| = |\phi^{(k)}(a)| \leq \|\phi\|_k.$$

The order of this distribution is, at most, k. The situation $k = 0$ corresponds to the Dirac distribution. The situation $k = 1$ is called the *dipole or doublet distribution.*

Example 1.9. We show in this example that the order of the dipole distribution cannot be zero. Suppose that for all compact set $K \subset \mathbb{R}$ and $\phi \in C^\infty(K)$ we have

$$|\delta_0^{(1)}(\phi)| \leq C\|\phi\|_0.$$

Now, choose $\phi \in C^\infty(K)$ such that $0 \in \text{Int}(K)$ and $\phi'(0) \neq 0$. For each integer $m \geq 1$, set $\phi_m(x) := (1/m)\phi(mx)$. Note that $\phi_m'(x) = \phi'(mx)$ and hence

$$0 \neq |\phi'(0)| = |\phi_m'(0)| = |\delta_0^{(1)}(\phi_m)| \leq C\|\phi_m\|_0 = \frac{C}{m}\|\phi\|_0.$$

This is a contradiction because RHS converges to zero, as m increases, and LHS is strictly positive. □

Example 1.10. The example above can be tweaked to show that any k-th multipole distribution, $\delta_0^{(k)}$ cannot have order less than k and hence is of order k. Suppose that for all compact set $K \subset \mathbb{R}$ and $\phi \in C^\infty(K)$ we have

$$|\delta_0^{(k)}(\phi)| \leq C\|\phi\|_j \quad \text{for some } 0 \leq j < k.$$

Now, choose $\phi \in C^\infty(K)$ such that $0 \in \text{Int}(K)$ and $\phi^{(k)}(0) \neq 0$. For each integer $m \geq 1$, set $\phi_m(x) := m^{-k}\phi(mx)$. Note that $\phi_m^{(j)}(x) = m^{j-k}\phi^{(j)}(mx)$. Thus,

$$\begin{aligned} 0 &\neq |\phi^{(k)}(0)| = |\phi_m^{(k)}(0)| = |\delta_0^{(k)}(\phi_m)| \leq C\|\phi_m\|_j = C\sum_{i=0}^{j} \|\phi_m^{(i)}\|_0 \\ &\neq C\sum_{i=0}^{j} m^{i-k}\|\phi^{(i)}\|_0 \leq Cm^{j-k}\sum_{i=0}^{j}\|\phi^{(i)}\|_0 = Cm^{j-k}\|\phi\|_j. \end{aligned}$$

This is a contradiction because $j < k$ and RHS converges to zero, as m increases whereas LHS is strictly positive. □

One may generalise these notions appropriately to higher dimensions. For any open subset Ω of $\mathbb{R}^n$, we define the distribution

$$\delta_a^\alpha(\phi) = D^\alpha\phi(a), \quad \forall\phi \in \mathcal{D}(\Omega),$$

where α is the n tuple of non-negative integers. Just as Dirac distribution models charge density at a point charge, the dipole distribution models charge density for an electric dipole. In fact, any dipole behaviour, viz., the magnetic dipole layer on a surface. For instance, given a smooth surface Γ and f is a continuous function on Γ representing the density of the magnetic moment, one can define the distribution corresponding to the magnetic dipole layer on Γ as,

$$T(\phi) = \int_\Gamma f(x)\frac{d\phi(x)}{dn}\,ds(x)$$

where ds denotes the surface element of Γ and d/dn is the derivation in the direction of normal of Γ.

The dipole distribution is singular. In fact, the situation is much worse. The dipole distribution is not induced by any Radon measure.

Proposition 1.4.14. *The dipole distribution $\delta_a^{(1)} \in \mathcal{D}(\mathbb{R})$ is not induced by any Radon measure, i.e., there is no Radon measure μ such that $T_\mu = \delta_a^{(1)}$.*

Proof. Let $\Omega = \mathbb{R}$ and $a = 0$. Choose $\phi \in \mathcal{D}(\mathbb{R})$ with support in $[-1, 1]$ such that $0 \leq \phi \leq 1$ and $\phi \equiv 1$ on $[-1/2, 1/2]$. Set $\phi_m(x) = \sin(mx)\phi(x)$. Thus, $\text{supp}(\phi_m) \subset [-1, 1]$ and $|\phi_m| \leq 1$. Note that the derivative of ϕ_m is $\phi'_m(x) = \sin(mx)\phi'(x) + m\cos(mx)\phi(x)$ and, hence, $\phi'_m(0) = m$ for all positive integer $m > 0$. Therefore, the dipole distribution takes the value,

$$\delta_0^{(1)}(\phi_m) = \phi'_m(0) = m, \quad \forall m \in \mathbb{N}.$$

Suppose there exists a Radon measure μ inducing the dipole distribution $\delta_0^{(1)}$, then

$$m = |\delta_0^{(1)}(\phi_m)| = |T_\mu(\phi_m)| = \left| \int_{-1}^{1} \phi(x) \sin(mx)\, d\mu \right| \leq |\mu|(B(0; 1)).$$

The inequality $|\mu|(B(0; 1)) \geq m$, for all $m \in \mathbb{N}$, implies that $|\mu|(B(0; 1))$ is infinite which contradicts the fact that the Radon measure μ is finite on compact subsets of $\mathbb{R}$. □

We shall see later that the multipole distribution δ_a^α is, in fact, the α-th distributional derivative of the Dirac measure δ_a up to a sign change. This motivation was behind the choice of notation of multipole distribution. In this section, we have noted that the space of Radon measures is properly contained in the space of distributions, i.e., $\mathcal{M}(\mathbb{R}) \subsetneq \mathcal{D}'(\mathbb{R})$.

Exercise 19. Do we have singular distributions which are neither induced by Radon measures nor distributional derivatives of Radon measures? (Hyperfunctions?)

1.4.4 Infinite Order Distributions

We have introduced distributions of finite order thus far. Does there exist distributions of infinite order? In this section, we given an example of an infinite order distribution.

Example 1.11. Define the functional T on $\mathcal{D}(\mathbb{R})$ as $T(\phi) := \sum_{k=0}^{\infty} \phi^{(k)}(k)$. Without loss of generality, consider the compact set $[0, m]$. If $\phi \in C^\infty[0, m]$, then

$$|T(\phi)| \leq \sum_{k=0}^{m} |\phi^{(k)}(k)| = \sum_{k=0}^{m} \|\phi^{(k)}\|_0 = \|\phi\|_m.$$

The larger the compact set becomes, the higher the derivatives of ϕ needs to be taken in. Thus, there is no fixed m for all compact subsets of $\mathbb{R}$.

Example 1.12. Define the functional $T \in \mathcal{D}'(0,\infty)$ as

$$T(\phi) = \sum_{k=1}^{\infty} \phi^{(k)}(1/k).$$

Let $\phi \in \mathcal{D}(0,\infty)$ be such that supp$(\phi) \subset [1/m, m]$ (exhaustion sets), then

$$T(\phi) = \sum_{k=1}^{m} \phi^{(k)}(1/k) \leq \sum_{k=1}^{m} \|\phi^{(k)}\|_0 \leq \sum_{k=0}^{m} \|\phi^{(k)}\|_0 = \|\phi\|_m.$$

Thus, $T \in \mathcal{D}'(0,\infty)$ is a distribution of infinite order.

1.4.5 Topology on Distributions

In this section, we give a suitable topology on the space of distributions. Recall that the space of distributions $\mathcal{D}'(\Omega)$ is the (topological) dual of $\mathcal{D}(\Omega)$. For every $\phi \in \mathcal{D}(\Omega)$, one can define the linear functional $\Lambda_\phi : \mathcal{D}'(\Omega) \to \mathbb{R}$ as follows, $\Lambda_\phi(T) = T(\phi)$. The linear functionals Λ_ϕ are included in the second (algebraic) dual of $\mathcal{D}(\Omega)$. We consider the coarsest[12] topology on $\mathcal{D}'(\Omega)$ such that all the linear maps $\Lambda_\phi : \mathcal{D}'(\Omega) \to \mathbb{R}$, corresponding to each $\phi \in \mathcal{D}(\Omega)$, are continuous. This is called the *weak-** topology on $\mathcal{D}'(\Omega)$. For every open subset of $V \subset \mathbb{R}$, consider the collection of subsets $\{\Lambda_\phi^{-1}(V)\}$ of $\mathcal{D}'(\Omega)$, for all $\phi \in \mathcal{D}(\Omega)$. The weak-* topology is the topology generated by this collection of subsets in $\mathcal{D}'(\Omega)$. The space $\mathcal{D}'(\Omega)$ is sequentially weak-* complete but not weak-* complete.

Definition 1.4.15. *We say a sequence of distributions $\{T_m\}$ is* Cauchy *in $\mathcal{D}'(\Omega)$, if $T_m(\phi)$ is Cauchy in $\mathbb{R}$, for all $\phi \in \mathcal{D}(\Omega)$. A sequence $\{T_m\} \subset \mathcal{D}'(\Omega)$ converges to T in weak-* topology of $\mathcal{D}'(\Omega)$ or in the distribution sense if $T_m(\phi) \to T(\phi)$ for all $\phi \in \mathcal{D}(\Omega)$, denoted as $T_m \rightharpoonup T$.*

Exercise 20. If S_m and T_m are sequences of distributions converging to S and T, respectively, then show that $\lambda S_m + \mu T_m$ converges to $\lambda S + \mu T$, for all $\lambda, \mu \in \mathbb{R}$.

Proposition 1.4.16 (Sequential Completeness)**.** *Let $\{T_m\}$ be a Cauchy sequence (of distributions) in $\mathcal{D}'(\Omega)$ and let $T(\phi) := \lim_{m\to\infty} T_m(\phi)$ (T is well-defined because $\mathbb{R}$ is complete). Then $T \in \mathcal{D}'(\Omega)$.*

[12] weakest or smallest topology, one with fewer open sets

Proof. To show $T \in \mathcal{D}'(\Omega)$, it is enough to show that $T : C^\infty(K) \to \mathbb{R}$ is continuous for all compact sets K in Ω. Fix a compact set K in Ω and $\phi \in \Omega$. Since $\{T_m(\phi)\}$ is convergent in $\mathbb{R}$ it is bounded and hence there is a real constant $C_\phi > 0$ (may depend on ϕ) such that $\sup_m |T_m(\phi)| \leq C_\phi$. The family $\{T_m\}$ is point-wise bounded on $C^\infty(K)$. Then, by Banach-Steinhaus theorem (uniform boundeness principle) of $C^\infty(K)$, T is uniformly bounded, i.e., there is a constant $C > 0$ (independent of ϕ) such that $\sup_m |T_m(\phi)| \leq C$ for all $\phi \in C^\infty(K)$. Therefore, for all $\phi \in C^\infty(K)$,

$$|T(\phi)| \leq \sup_m |T_m(\phi)| \leq C \quad \forall \phi \in C^\infty(K).$$

and hence T is continuous on $C^\infty(K)$ for all compact subsets K of Ω. Thus, T is continuous on $\mathcal{D}(\Omega)$ and hence $T \in \mathcal{D}'(\Omega)$. □

A natural question is the relation between distributional convergence and poin-wise, uniform or L^p convergence of regular distributions. The weak-* convergence in $\mathcal{D}'(\Omega)$ is weaker than any other topology that preserves the continuity of the functionals Λ_ϕ. For instance, this topology is weaker that L^p-topology or uniform norm topology etc., i.e., convergence in p-norm or uniform norm implies convergence in distribution sense of classical functions.

Example 1.13. Consider the sequence $f_m(x) = e^{imx}$ on $\mathbb{R}$ which converges point-wise to 0 iff $x \in 2\pi\mathbb{Z}$. The sequence of distributions corresponding to f_m converges to zero in the distributional sense because

$$\int e^{imx}\phi(x)\,dx = -\frac{1}{im}\int e^{imx}\phi'(x)\,dx.$$

We have used integration by parts to get the second integral and since the integrand is bounded in the second integral, it converges to zero as $m \to \infty$.

Example 1.14. Let $\rho \in C_c(\mathbb{R}^n)$ such that $\text{supp}(\rho) \subset \overline{B(0;1)}$ and $\int_{\mathbb{R}^n} \rho(y)\,dy = 1$. Now set $\rho_\varepsilon(x) := \varepsilon^{-n}\rho(\frac{x}{\varepsilon})$. This sequence is called the *Dirac Sequence*. In particular, the sequence of mollifiers (cf. (1.3.3)) is one example. Another trivial example is

$$\rho(x) = \begin{cases} 2 & |x| < 1 \\ 0 & |x| \geq 1. \end{cases}$$

We claim that the sequence ρ_ε converges to the Dirac distribution δ_0, in the distribution sense. Let T_ε denote the distribution corresponding to ρ_ε. For

any $\phi \in \mathcal{D}(\mathbb{R}^n)$, consider

$$T_\varepsilon(\phi) = \int_{\mathbb{R}^n} \rho_\varepsilon(x)\phi(x)\,dx = \int_{\mathbb{R}^n} \varepsilon^{-n}\rho\left(\frac{x}{\varepsilon}\right)\phi(x)\,dx = \int_{\mathbb{R}^n} \rho(y)\phi(\varepsilon y)\,dy.$$

Taking limit $\varepsilon \to 0$ both sides we get

$$\lim_{\varepsilon\to 0} T_\varepsilon(\phi) = \int_{\mathbb{R}^n} \rho(y)\phi(0)\,dy = \phi(0) = \delta_0(\phi).$$

The interchange of limit is possible due to uniform continuity of ϕ which induces the uniform convergence of $\phi(\varepsilon y) \to \phi(0)$. Thus, $T_\varepsilon \to \delta_0$ in the distribution sense, whereas the point-wise limit at $x = 0$ did not exist for ρ_ε.

Example 1.15. Consider the sequence of functions

$$f_m x = \begin{cases} m^2 x & 0 \le x < \frac{1}{m} \\ m^2\left(\frac{2}{n} - x\right) & \frac{1}{m} < x \le \frac{2}{m} \\ 0 & \text{otherwise.} \end{cases}$$

Note that they converge point-wise to zero for all $x \in \mathbb{R}$. Let T_m denote the distribution corresponding to f_m and hence

$$T_m(\phi) = m^2 \int_0^{1/m} x\phi(x)\,dx + m^2 \int_{1/m}^{2/m} \left(\frac{2}{n} - x\right)\phi(x)\,dx.$$

Both the integral above converges to $\phi(0)/2$ and hence the sequence of distributions converges to δ_0. Let us give the proof for the first integral. Since

$$m^2 \int_0^{1/m} x\,dx = \frac{1}{2}$$

we consider

$$\begin{aligned} \left| m^2 \int_0^{1/m} x\phi(x)\,dx - \frac{1}{2}\phi(0) \right| &= \left| m^2 \int_0^{1/m} x[\phi(x) - \phi(0)]\,dx \right| \\ &\le m^2 \int_0^{1/m} |x||\phi(x) - \phi(0)|\,dx. \end{aligned}$$

Since ϕ is continuous at $x = 0$, for every $\varepsilon > 0$, there is a $\delta > 0$ such that $|\phi(x) - \phi(0)| < \varepsilon$ whenever $|x| < \delta$. Thus, for all m such that $1/m < \delta$, we have

$$\left| m^2 \int_0^{1/m} x\phi(x)\,dx - \frac{1}{2}\phi(0) \right| \le m^2 \frac{\varepsilon}{m} \int_0^{1/m} dx < \varepsilon.$$

Hence,

$$\lim_{m\to\infty} m^2 \int_0^{1/m} x\phi(x)\,dx = \frac{1}{2}\phi(0).$$

Similarly, one can show that the second integral is also $(1/2)\phi(0)$ and hence $\int f_m(x)\phi(x)\,dx = \phi(0)$.

The following theorem gives the condition that is violated by the above example for the point-wise limit to coincide with the distributional limit.

Theorem 1.4.17 (Weak Dominated Convergence Theorem)**.** *Let $\{f_m\} \subset L^1_{loc}(\Omega)$ such that $f_m(x) \to f(x)$ point-wise for a.e. $x \in \Omega$ and there is a $g \in L^1(\Omega)$ such that for every compact set $K \subset \Omega$ $|f_m| \le g$ for all m. Then $f \in L^1_{loc}(\Omega)$ and $f_m \rightharpoonup f$ in $\mathcal{D}'(\Omega)$.*

Proof. Let T_m denote the distribution corresponding to f_m and fix $\phi \in \mathcal{D}(\Omega)$. Set $h_m(x) = f_m(x)\phi(x)$ and $h(x) = f(x)\phi(x)$. Note that $h_m(x) \to h(x)$ point-wise a.e. in Ω (in fact in the $\mathrm{supp}(\phi)$). Also, $|h_m| \le g|\phi|$ and $g|\phi| \in L^1(\Omega)$. By classical Lebesgue's dominated convergence theorem, $h \in L^1(\Omega)$ and

$$\int_\Omega h(x)\,dx = \lim_{m\to\infty} \int_\Omega h_m(x)\,dx = \lim_{m\to\infty} \int_\Omega f_m(x)\phi(x)\,dx.$$

In particular, choosing $\phi \equiv 1$ on a given compact set K of Ω and since for this ϕ, $h \in L^1(\Omega)$, we have $f \in L^1_{\mathrm{loc}}(\Omega)$ because

$$\infty > \int_\Omega |h|\,dx = \int_\Omega |f\phi|\,dx = \int_K |f|\,dx.$$

Let T_f be the distribution corresponding to f. Moreover,

$$\lim_{m\to\infty} T_m(\phi) = \lim_{m\to\infty} \int_\Omega f_m\phi\,dx = \int_\Omega f\phi\,dx = T_f(\phi).$$

□

Corollary 1.4.18. *Let $\{f_m\} \subset C(\Omega)$ be a sequence of continuous functions that converges uniformly on compact subsets of Ω to f. Then $f_m \rightharpoonup f$ in $\mathcal{D}'(\Omega)$.*

By the vector space structure of $\mathcal{D}'(\Omega)$, we already have the finite sum of distributions. If $\{T_i\}_1^k \subset \mathcal{D}'(\Omega)$ then $T := \sum_{i=1}^k T_i \in \mathcal{D}'(\Omega)$ defined as $T(\phi) = \sum_{i=1}^k T_i(\phi)$. The topology on $\mathcal{D}'(\Omega)$ can be used to give the notion of series of distributions.

Definition 1.4.19. *For any countable collection of distributions* $\{T_i\}_1^\infty \subset \mathcal{D}'(\Omega)$ *the* series $\sum_{i=1}^\infty T_i$ *is said to* converge *to* $S \in \mathcal{D}'(\Omega)$ *if the sequence of partial sums* $T_m := \sum_{i=1}^m T_i$ *converges to* S *in* $\mathcal{D}'(\Omega)$.

1.4.6 Principal Value Distribution

In practice, we encounter functions whose integral diverges. This motivates other ways of studying divergent integrals, like principal part, finite part of divergent integral.

Definition 1.4.20 (Singularity at infinity)**.** *We say that the integral of a function* f *(with singularity at* ∞*) exists in the* generalised sense *if the limit*

$$\lim_{a,b\to+\infty} \int_{-a}^{b} f(x)\,dx$$

exists. We say that the principal value *of the integral exists if the limit (with* $b = a$*)*

$$\lim_{a\to+\infty} \int_{-a}^{b} f(x)\,dx$$

exists.

Definition 1.4.21 (Singularity at point)**.** *We say that the integral of a compact supported function* f *(with singularity at a point* 0*) exists in the* generalised sense *if the limit*

$$\lim_{a,b\to 0^+} \int_{-\infty}^{-a} f(x)\,dx + \int_{b}^{\infty} f(x)\,dx$$

exists. We say that the principal value *of the integral exists if the limit (with* $b = a$*)*

$$\lim_{a\to 0^+} \int_{|x|>a} f(x)\,dx$$

exists.

We have already seen that every locally integrable function can be identified with a distribution. We have already noted, in §1.4.1, that $1/x \notin L^1_{\text{loc}}(\mathbb{R})$. Consider the function $f_\varepsilon : \mathbb{R} \to \mathbb{C}$, for each $\varepsilon > 0$, defined as

$$f_\varepsilon(x) = \frac{1}{x + i\varepsilon}.$$

What is the limit of f_ε, as $\varepsilon \to 0$? Of course, when $x \neq 0$, $f_\varepsilon(x) \to 1/x$. What is the limit when $x = 0$? Classically we cannot make sense of this limit. Similarly, note that the integral $\int_{-1}^{1}(1/x)\,dx$ does not converge. However,

$$\lim_{\varepsilon\to 0}\int_{-1}^{1} f_\varepsilon(x)\,dx = \lim_{\varepsilon\to 0}\ln(x+i\varepsilon)\,|_{-1}^{1} = \ln(1)-\ln(-1) = 0-(i\pi) = -i\pi.$$

Thus, we expect $\int_{-1}^{1}(1/x)\,dx = -i\pi$. This cannot be made sense classically[13]. Using the theory of distributions one can give a meaning to this observation.

We do know that $1/x \in L^1_{\text{loc}}(\mathbb{R}\setminus\{0\})$ and hence induces a distribution in $\mathcal{D}(\mathbb{R}\setminus\{0\})$. One can extend this distribution to yield a distribution corresponding $1/x$ on $\mathbb{R}$. We define the linear functional $\text{PV}\left(\frac{1}{x}\right)$ on $\mathcal{D}(\mathbb{R})$ as

$$\text{PV}\left(\frac{1}{x}\right)(\phi) = \lim_{\varepsilon\to 0}\int_{|x|\geq\varepsilon}\frac{1}{x}\phi(x)\,dx. \tag{1.4.3}$$

As defined before, the limit on RHS is called the *Cauchy's Principal Value* and hence the use of 'PV' in the notation. Note that $PV(1/x)$ is defined as a distributional limit of the sequence of distributions corresponding to the locally integrable functions

$$f_\varepsilon(x) := \begin{cases} \frac{1}{x} & |x| > \varepsilon, \\ 0 & |x| \leq \varepsilon, \end{cases}$$

i.e., $f_\varepsilon \rightharpoonup PV\left(\frac{1}{x}\right)$ in $\mathcal{D}'(\Omega)$.

Lemma 1.4.22. *The principal value of* $1/x$ *(limit on the RHS) exists.*

Proof. Let $\phi \in \mathcal{D}(\mathbb{R})$. Without loss of generality, let $\text{supp}(\phi) \subset [-a,a]$ for some real $a > 0$. For $\varepsilon > 0$ small enough, consider

$$\begin{aligned}\int_{|x|\geq\varepsilon}\frac{\phi(x)}{x}\,dx &= \int_{-a}^{-\varepsilon}\frac{\phi(x)}{x}\,dx + \int_{\varepsilon}^{a}\frac{\phi(x)}{x}\,dx \\ &= \int_{\varepsilon}^{a}\frac{\phi(x)}{x}\,dx - \int_{\varepsilon}^{a}\frac{\phi(-x)}{x}\,dx = \int_{\varepsilon}^{a}\frac{\phi(x)-\phi(-x)}{x}\,dx.\end{aligned}$$

[13] recall (1.1.1)

Therefore, taking limit both sides as $\varepsilon \to 0$,

$$\begin{aligned}
\mathrm{PV}\left(\frac{1}{x}\right)(\phi) &= \int_0^a \frac{\phi(x)-\phi(-x)}{x}\,dx \\
&= \int_0^a \frac{1}{x}\left(\int_{-x}^{x}\phi'(s)\,ds\right)dx \quad \text{(Using FTC)} \\
&= \int_0^a \frac{1}{x}\left(\int_{-1}^{1} x\phi'(xt)\,dt\right)dx.
\end{aligned}$$

Set $\psi(x) = \int_{-1}^{1}\phi'(xt)\,dt$, then

$$\left|\mathrm{PV}\left(\frac{1}{x}\right)(\phi)\right| = \left|\int_0^a \frac{1}{x}x\psi(x)\,dx\right| = \left|\int_0^a \psi(x)\,dx\right| \le \int_0^a |\psi(x)|\,dx.$$

But $|\psi(x)| \le 2\|\phi'\|_0 \le 2\|\phi\|_1$. Hence,

$$\left|\mathrm{PV}\left(\frac{1}{x}\right)(\phi)\right| \le 2a\|\phi\|_1.$$

Since RHS is finite, the limit is finite. □

We have shown that the principal value functional is a distribution, called the *Principal Value (PV) Distribution* corresponding to $1/x$. The $1/x$ in (1.1.1) derived by Dirac will make sense as a principal value distribution. The proof also highlights that, at most, the PV distribution can have order one. We show in the example below that the PV distribution cannot be of order zero.

Example 1.16. The PV distribution cannot be of zero order and hence its order is one. Observe the following characterisation of principal value distribution in the above argument when $\mathrm{supp}(\phi) \subset [-a, a]$,

$$\mathrm{PV}\left(\frac{1}{x}\right)(\phi) = \int_0^a \frac{\phi(x)-\phi(-x)}{x}\,dx.$$

Choose ϕ such that $\mathrm{supp}(\phi) \subset [0, a]$, $0 \le \phi \le 1$ and $\phi \equiv 1$ on $[c, d] \subset [0, a]$

with $c \neq 0$ and $d \neq a$. Then,

$$\begin{aligned} PV(1/x)(\phi) &= \int_0^c \frac{\phi(x)}{x}\,dx + \int_c^d \frac{\phi(x)}{x}\,dx + \int_d^a \frac{\phi(x)}{x}\,dx \\ &> \int_c^d \frac{1}{x}\,dx \quad \text{first and third integral being non-negative} \\ &> \frac{1}{a}\int_c^d dx \quad \text{since } x < a \\ &= \frac{1}{a}(d-c) \geq \frac{d-c}{a}\sup_{x\in[0,a]}|\phi| = \frac{d-c}{a}\|\phi\|_0. \end{aligned}$$

Thus, the distribution cannot be of zero order. □

The principal value distribution is not the only choice for the function $1/x$. Consider the linear functional on $\mathcal{D}(\mathbb{R})$ corresponding to $1/x$,

$$T(\phi) = \lim_{\varepsilon\to 0}\int_{-\infty}^{-\varepsilon}\frac{\phi(x)}{x}\,dx + \int_{2\varepsilon}^{\infty}\frac{\phi(x)}{x}\,dx.$$

Exercise 21. Show that the T defined above is a distribution. Also, compare T and PV$(1/x)$.

Proof. Let $\phi \in \mathcal{D}(\mathbb{R})$. Without loss of generality, let supp$(\phi)\subset [-a,a]$ for some real $a > 0$. For $\varepsilon > 0$ small enough, consider

$$\begin{aligned} \int_{-\infty}^{-\varepsilon}\frac{\phi(x)}{x}\,dx + \int_{2\varepsilon}^{\infty}\frac{\phi(x)}{x}\,dx &= \int_{-a}^{-2\varepsilon}\frac{\phi(x)}{x}\,dx + \int_{-2\varepsilon}^{-\varepsilon}\frac{\phi(x)}{x}\,dx \\ &\quad + \int_{2\varepsilon}^{a}\frac{\phi(x)}{x}\,dx \\ &= \int_{2\varepsilon}^{a}\frac{\phi(x)-\phi(-x)}{x}\,dx + \int_{-2\varepsilon}^{-\varepsilon}\frac{\phi(x)}{x}\,dx \\ &= \int_{2\varepsilon}^{a}\psi(x)\,dx - \int_1^2\frac{\phi(-\varepsilon x)}{x}\,dx. \end{aligned}$$

Therefore,

$$T(\phi) = \int_0^a \psi(x)\,dx - \int_1^2\frac{\phi(0)}{x}\,dx = \int_0^a\psi(x)\,dx - \phi(0)\ln 2.$$

Thus, $|T(\phi)| \leq 2a\|\phi'\|_0 + \|\phi\|_0 \ln 2 \leq C\|\phi\|_1$ where $C = \max(2a, \ln 2)$. T is again a distribution of order one. Now, consider

$$(\mathrm{PV}(1/x) - T)(\phi) = \lim_{\varepsilon \to 0} \int_{\varepsilon}^{2\varepsilon} \frac{\phi(x)}{x}\, dx = \lim_{\varepsilon \to 0} \int_{1}^{2} \frac{\phi(\varepsilon x)}{x}\, dx = \phi(0) \ln 2.$$

□

Remark 1.4.23. Yet another way to defining $PV(1/x)$ is as follows, this time from the complex plane. Consider the distributions T_ε corresponding to

$$f_\varepsilon(x) := \frac{1}{x + i\varepsilon}.$$

We shall show that $T_\varepsilon \rightharpoonup PV(1/x) - i\pi\delta_0$. Similarly, if S_ε are distributions corresponding to

$$f_\varepsilon(x) := \frac{1}{x - i\varepsilon}.$$

We shall show that $S_\varepsilon \rightharpoonup PV(1/x) + i\pi\delta_0$.

1.4.7 Functions, but not Distributions

It is not always possible to extend the notion of distributions to non-locally integrable functions as done above.

Example 1.17. The function $e^{1/x} \notin L^1_{\mathrm{loc}}(\mathbb{R})$ but is in $L^1_{\mathrm{loc}}(\mathbb{R} \setminus \{0\})$. Let $T \in \mathcal{D}'(0, \infty)$ be the distribution corresponding to $e^{1/x}$ on $(0, \infty)$. We shall show that there is no distribution $S \in \mathcal{D}'(\mathbb{R})$ corresponding to $e^{1/x}$ on $\mathbb{R}$ whose restriction coincides with the distribution T on $(0, \infty)$.

Example 1.18. Consider the distribution $T \in \mathcal{D}'(0, \infty)$ defined in Example 1.12, i.e.,

$$T(\phi) = \sum_{k=1}^{\infty} \phi^{(k)}(1/k).$$

We shall show that there is no distribution $S \in \mathcal{D}'(\mathbb{R})$ whose restriction to $(0, \infty)$ coincides with T. Suppose there exists a $S \in \mathcal{D}'(\mathbb{R})$ such that $T = S$ on $(0, \infty)$ then for the compact set $[-1, 1]$ there exists a $C > 0$ and $N \in \mathbb{N} \cup \{0\}$ such that

$$|S(\phi)| \leq C\|\phi\|_N \quad \forall \phi \in C^\infty[-1, 1].$$

Now, choose $m > N$ and ψ such that $\text{supp}(\psi) \subset (1/(m+1), 1/(m-1))$. Then $T(\psi) = \psi^{(m)}(1/m)$. Since T and S coincide for $\psi \in \mathcal{D}(0, \infty)$, we have

$$|\delta^{(m)}_{1/m}(\psi)| = |\psi^{(m)}(1/m)| = |T(\psi)| = |S(\psi)| \leq C\|\psi\|_N.$$

Thus, we have that the order of the multipole distribution $\delta^{(m)}_{1/m}$ is at most N, a quantity smaller than m. This is a contradiction. Hence, we can have no extension S of T to $\mathbb{R}$.

1.5 Operations With Distributions

The space of distributions $\mathcal{D}'(\Omega)$ is a vector space over $\mathbb{R}$ (or $\mathbb{C}$). Thus, we already have the operation of addition of distribution and multiplication by reals induced from the property of linear functionals. Hence, for any two $S, T \in \mathcal{D}'(\Omega)$, $(S+T)(\phi) = S(\phi)+T(\phi)$. Also, for any $\lambda \in \mathbb{R}$ and $T \in \mathcal{D}'(\Omega)$, the distribution λT is defined as $(\lambda T)(\phi) = \lambda T(\phi)$. We begin by introducing the concept of derivative of a distribution.

1.5.1 Differentiation

Recall the discussion leading to (1.3.1). For any $f \in C^\infty(\mathbb{R})$, $\phi \in \mathcal{D}(\mathbb{R})$ and all $k \in \mathbb{N}$, using integration by parts we have

$$\int f^{(k)}\phi \, dx = (-1)^k \int f\phi^{(k)} \, dx.$$

This motivates the following definition of derivative of a distribution.

Definition 1.5.1. *For any $T \in \mathcal{D}'(\Omega)$ and n-tuple α such that $|\alpha| = k$, the derivative $D^\alpha T$ of T is defined as*

$$(D^\alpha T)(\phi) = (-1)^{|\alpha|} T(D^\alpha \phi), \quad \phi \in \mathcal{D}(\Omega).$$

The derivative $D^\alpha T$ is also a distribution because for all compact subsets K of Ω there are $C_K > 0$ and non-negative integer $N_K = N$ such that $|T(\phi)| \leq C_K\|\phi\|_N$ for all $\phi \in C^\infty(K)$. Thus, for any compact subset K of Ω and $\phi \in C^\infty(K)$,

$$|(D^\alpha T)(\phi)| = |(-1)^{|\alpha|} T(D^\alpha \phi)| = |T(D^\alpha \phi)| \leq C_K\|D^\alpha \phi\|_N \leq C_K\|\phi\|_{N+|\alpha|}.$$

Thus, $D^\alpha T \in \mathcal{D}'(\Omega)$. For a finite order distribution T, the derivative $D^\alpha T$ has order more than T, i.e., differentiation operation increases the order of the distribution. Observe that every distribution is infinitely differentiable and the mixed derivatives are equal because the same holds for test functions. The following result highlights some of these properties of the differentiation operation:

Exercise 22. Let $S, T \in \mathcal{D}'(\Omega)$. Show that

(i) $D^\beta(D^\alpha T) = D^\alpha(D^\beta T) = D^{\alpha+\beta}T$ for all multi-indices α, β.

(ii) $D^\alpha(\lambda S + \mu T) = \lambda D^\alpha S + \mu D^\alpha T$, for each $\lambda, \mu \in \mathbb{R}$.

Proof. (i) We see that

$$\begin{aligned} D^\beta(D^\alpha T)(\phi) &= (-1)^{|\beta|} D^\alpha T(D^\beta \phi) = (-1)^{|\beta|+|\alpha|} T(D^\alpha D^\beta \phi) \\ &= (-1)^{|\beta|+|\alpha|} T(D^{\alpha+\beta}\phi) = (D^{\alpha+\beta}T)(\phi). \end{aligned}$$

Similarly for $D^\alpha(D^\beta T)$.

(ii) Consider

$$\begin{aligned} [D^\alpha(\lambda S + \mu T)](\phi) &= (-1)^{|\alpha|}(\lambda S + \mu T)(D^\alpha \phi) \\ &= (-1)^{|\alpha|} \left[\lambda S(D^\alpha \phi) + \mu T(D^\alpha \phi)\right] \\ &= (\lambda D^\alpha S + \mu D^\alpha T)(\phi). \end{aligned}$$

□

Definition 1.5.2. *A function $f \in L^1_{loc}(\Omega)$ is said to be* α-weakly differentiable *if $D^\alpha T_f$ is a regular distribution. In other words, for any given multi-index α, a function $f \in L^1_{loc}(\Omega)$ is said to be* α-weakly differentiable *if there exists a $g_\alpha \in L^1_{loc}(\Omega)$ such that*

$$\int_\Omega g_\alpha \phi \, dx = (-1)^{|\alpha|} \int_\Omega f D^\alpha \phi \, dx \quad \forall \phi \in \mathcal{D}(\Omega).$$

Example 1.19. Consider the continuous function $f : \mathbb{R} \to \mathbb{R}$ defined as $f(x) = |x|$. Classically, its derivative exists a.e. and is the function

$$f'(x) = \begin{cases} -1 & -\infty < x < 0 \\ 1 & 0 < x < \infty. \end{cases}$$

The distribution corresponding to f' is

$$T_{f'}(\phi) = -\int_{-\infty}^{0} \phi(x)\,dx + \int_{0}^{\infty} \phi(x)\,dx$$

However, for $\phi \in \mathcal{D}(\mathbb{R})$, the distributional derivative of T_f is

$$\begin{aligned} DT_f(\phi) &= -T_f(\phi') = -\int_{-\infty}^{0} (-x)\phi'(x)\,dx - \int_{0}^{\infty} x\phi'(x)\,dx \\ &= x\phi(x)\,|_{-\infty}^{0} - \int_{-\infty}^{0} \phi(x)\,dx + \int_{0}^{\infty} \phi(x)\,dx - x\phi(x)\,|_{0}^{\infty} \\ &= \int_{0}^{\infty} \phi(x)\,dx - \int_{-\infty}^{0} \phi(x)\,dx \\ &= T_{f'}(\phi). \end{aligned}$$

Therefore, $DT_f = T_{f'}$. Moreover, $|x|$ is weakly differentiable.

Example 1.20. Consider the everywhere discontinuous function

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q}^c \\ 0 & x \in \mathbb{Q}. \end{cases}$$

The function f is not differentiable at any point. The Lebesgue measure of $\mathbb{Q}$ is zero, hence,

$$DT_f(\phi) = -T_f(\phi') = -\int_{\mathbb{R}} \phi'(x)\,dx = 0.$$

Thus, $DT_f = 0$ and f is weakly differentiable because $DT_f = T_0$.

Example 1.21. Consider the everywhere discontinuous function

$$f(x) = \begin{cases} \sin x & x \in \mathbb{Q}^c \\ 0 & x \in \mathbb{Q}. \end{cases}$$

The function f is not differentiable at any point. The Lebesgue measure of $\mathbb{Q}$ is zero, hence,

$$DT_f(\phi) = -T_f(\phi') = -\int_{\mathbb{R}} \sin x\phi'(x)\,dx = \int_{\mathbb{R}} \cos x\phi(x)\,dx.$$

Thus, $DT_f = T_{\cos x}$ and hence f is weakly differentiable.

Example 1.22. We shall now give an example of a function for which the classical and distributional derivative do not coincide. Consider the locally integrable function H_a, for every $a \in \mathbb{R}$,

$$H_a(x) = \begin{cases} 1 & \text{if } x > a \\ 0 & \text{if } x \leq a. \end{cases}$$

When $a = 0$, the function H_0 is the Heaviside function H as defined in (1.1.2). Classically, the function H_a is differentiable a.e. and $H_a'(x) = 0$ a.e. (except at a). Therefore, $T_{H_a'} = 0$. We shall now compute the distributional derivative of the regular distribution T_{H_a}, induced by H_a. For $\phi \in \mathbb{R}$, the derivative of T_{H_a} is computed as,

$$DT_{H_a}(\phi) = -T_{H_a}(\phi') = -\int_a^\infty \phi'(x)\,dx = \phi(a) = \delta_a(\phi).$$

Thus, the distributional derivative of the function H_a is the Dirac distribution, a singular distribtuion. Therefore, $DT_{H_a} = \delta_a(\phi) \neq 0 = T_{H_a'}$. In fact, H_a is not weakly differentiable.

Observe above that the function H_a had a jump discontinuity and Dirac measure appeared as its derivative at the point of "jump". This is the feature of Dirac measure. It appears as a derivative at points of jump. However, note that in Example 1.20 and 1.21, the jump at every point did not give rise to a Dirac measure because outside the set of jump points (in that case $\mathbb{Q}$) the function was continuous.

Example 1.23. Consider the locally integrable function $f : \mathbb{R} \to \mathbb{R}$ defined as

$$f(x) = \begin{cases} |x| & |x| < a \\ 0 & |x| > a. \end{cases}$$

Classically, its derivative exists a.e. and is the function

$$f'(x) = \begin{cases} -1 & -a < x < 0 \\ 1 & 0 < x < a \\ 0 & |x| > a. \end{cases}$$

The distribution corresponding to f' is

$$T_{f'}(\phi) = -\int_{-a}^0 \phi(x)\,dx + \int_0^a \phi(x)\,dx$$

However, for $\phi \in \mathcal{D}(\mathbb{R})$, the distributional derivative of T_f is

$$\begin{aligned} DT_f(\phi) &= -T_f(\phi') = -\int_{-a}^{0} (-x)\phi'(x)\,dx - \int_{0}^{a} x\phi'(x)\,dx \\ &= x\phi(x)\,|_{-a}^{0} - \int_{-a}^{0} \phi(x)\,dx + \int_{0}^{a} \phi(x)\,dx - x\phi(x)\,|_{0}^{a} \\ &= a\phi(-a) - a\phi(a) + \int_{0}^{a} \phi(x)\,dx - \int_{-a}^{0} \phi(x)\,dx \\ &= a(\delta_{-a}(\phi) - \delta_a(\phi)) + T_{f'}(\phi). \end{aligned}$$

Therefore, $DT_f = a\delta_{-a} - a\delta_a + T_{f'}$. The function f is not weakly differentiable (compare this with Example 1.19).

Example 1.24. Consider the discontinuous function

$$f(x) = \begin{cases} x^2 + x & x < 1 \\ e^{-5x} & x > 1. \end{cases}$$

Classically, its derivative exists a.e. and is the function

$$f'(x) = \begin{cases} 2x + 1 & x < 1 \\ -5e^{-5x} & x > 1. \end{cases}$$

The distribution corresponding to f' is

$$T_{f'}(\phi) = \int_{-\infty}^{1} (2x+1)\phi(x)\,dx - 5\int_{1}^{\infty} e^{-5x}\phi(x)\,dx.$$

The distributional derivative of f (T_f) is

$$\begin{aligned} DT_f(\phi) &= -T_f(\phi') = -\int_{-\infty}^{1} (x^2+x)\phi'(x)\,dx - \int_{1}^{\infty} e^{-5x}\phi'(x)\,dx \\ &= -(x^2+x)\phi(x)\,|_{-\infty}^{1} + \int_{-\infty}^{1} (2x+1)\phi(x)\,dx - 5\int_{1}^{\infty} e^{-5x}\phi(x)\,dx \\ &\quad - e^{-5x}\phi(x)\,|_{1}^{\infty} \\ &= (e^{-5} - 2)\phi(1) + T_{f'}(\phi). \end{aligned}$$

Therefore, $DT_f = (e^{-5} - 2)\delta_1 + T_{f'}$. The function f is not weakly differentiable.

Example 1.25. Consider the Cantor function $f_C : [0,1] \to [0,1]$ (cf. Appendix A) extended continuously to $\mathbb{R}$ by setting

$$f_C(x) = \begin{cases} 0 & x \leq 0 \\ 1 & x \geq 1. \end{cases}$$

Note that $f_C' = 0$ for all $x \leq 0$ and $x \geq 1$. Since C is of Lebesgue measure zero and f_C is constant on each interval removed from $[0,1]$, we have $f_C' = 0$ a.e. in $\mathbb{R}$. Note that

$$\begin{aligned} DT_{f_C}(\phi) &= -T_{f_C}(\phi') = -\int_0^1 f_C\phi'\,dx - \int_1^\infty \phi'\,dx = \phi(1) - \int_{[0,1]\setminus C} f_C\phi'\,dx \\ &= \phi(1) - \int_{\cup_{i=1}^\infty C_i^c} f_C\phi'\,dx \quad \text{(cf. Appendix A)} \\ &= \phi(1) - \sum_{i=1}^\infty \int_{C_i^c} f_C\phi'\,dx \quad \text{(since the union is disjoint)} \\ &= \phi(1) - \sum_{k=1}^\infty c_k \int_{a_k}^{b_k} \phi'\,dx \\ &\qquad (a_k, b_k \text{ are end-points of intervals removed}) \\ &= \phi(1) + \sum_{k=1}^\infty c_k(\phi(a_k) - \phi(b_k)) = \sum_{k=1}^\infty c_k(\delta_{a_k} - \delta_{b_k})(\phi) + \delta_1(\phi). \end{aligned}$$

Note that f_C is not weakly differentiable. Moreover, the classical derivative and distributional derivative do not coincide.

Example 1.26. The k-th distributional derivative of the Dirac distribution is the k-th multipole distribution. For $\phi \in \mathcal{D}(\Omega)$,

$$\begin{aligned} D^\beta \delta_a^\alpha(\phi) &= (-1)^{|\beta|}\delta_a^\alpha(D^\beta\phi) = (-1)^{|\beta|}D^\alpha(D^\beta\phi(a)) \\ &= (-1)^{|\beta|}D^{\alpha+\beta}\phi(a)) = (-1)^{|\beta|}\delta_a^{\alpha+\beta}(\phi). \end{aligned}$$

In particular, the distributional derivative of the Dirac distribution is the dipole distribution, up to a factor of sign.

Example 1.27. Let us compute the distributional derivative of the distribu-

tion induced by $\ln|x|$. For $\text{supp}(\phi) \subset [-a, a]$,

$$\begin{aligned} DT_{\ln|x|}(\phi) &= -T_{\ln|x|}(\phi') = -\int_{\mathbb{R}} \ln|x|\phi'\,dx \\ &= -\lim_{\varepsilon\to 0^+}\left(\int_{-a}^{-\varepsilon} \ln|x|\phi'\,dx + \int_{\varepsilon}^{a} \ln|x|\phi'\,dx\right) \\ &= \lim_{\varepsilon\to 0^+}\left[(\phi(\varepsilon)-\phi(-\varepsilon))\ln\varepsilon + \int_{-a}^{-\varepsilon}\frac{\phi(x)}{x}\,dx + \int_{\varepsilon}^{a}\frac{\phi(x)}{x}\,dx\right] \\ &= \lim_{\varepsilon\to 0^+}\int_{\varepsilon}^{a}\frac{\phi(x)-\phi(-x)}{x}\,dx = \lim_{\varepsilon\to 0^+}\int_{|x|>\varepsilon}\frac{\phi(x)}{x}\,dx \\ &= PV\left(\frac{1}{x}\right)(\phi). \end{aligned}$$

Exercise 23. For any $\alpha \in \mathbb{R}$, we have already noted that the function $f(x) = |x|^{-\alpha}$ is in $L^1_{\text{loc}}(\mathbb{R}^n)$ for $\alpha < n$. However, the function is weakly differentiable for $\alpha + 1 < n$ and its weak partial derivative is $f_{x_i}(x) = -\frac{\alpha}{|x|^{\alpha+1}}\frac{x_i}{|x|}$.

Exercise 24. Let $\phi \in \mathcal{D}(\mathbb{R})$. Show that $\int_{\mathbb{R}} \phi(x)\,dx = 0$ iff there is a $\psi \in \mathcal{D}(\mathbb{R})$ such that $\phi(x) = \psi'(x)$.

Proof. Let there exist a $\psi \in \mathcal{D}(\mathbb{R})$ such that $\phi(x) = \psi'(x)$. Then,

$$\int_{\mathbb{R}} \phi(x)\,dx = \lim_{a\to\infty}\int_{-a}^{a}\psi'(x)\,dx = \lim_{a\to\infty}[\psi(a)-\psi(-a)] = 0.$$

Conversely, let $\int_{\mathbb{R}} \phi(x)\,dx = 0$. Then, set $\psi(x) := \int_{-\infty}^{x}\phi(t)\,dt$ and $\psi' = \phi$. It only remains to show that $\psi \in \mathcal{D}(\mathbb{R})$. $\psi \in C^\infty(\mathbb{R})$ since $\phi \in C^\infty(\mathbb{R})$. Suppose that $\text{supp}(\phi) \subset [-M, M]$. Then for all $x \in (\infty, -M)$, $\psi(x) = 0$ since $\phi(x) = 0$. For all $x \in (M, \infty)$, $\psi(x) = 0$ because $\int_{-M}^{M}\phi = 0$. Thus, $\text{supp}(\psi) \subset [-M, M]$. □

Exercise 25. Let $T \in \mathcal{D}'(\mathbb{R})$ be such that $DT = 0$, then show that T is a regular distribution generated by a constant (a.e.) function.

Proof. Let $T \in \mathcal{D}'(\mathbb{R})$ be such that $DT(\phi) = 0$ for all $\phi \in \mathcal{D}(\mathbb{R})$. Therefore,

$$0 = DT(\phi) = -T(\phi')$$

for all $\phi \in \mathcal{D}(\mathbb{R})$. Consider the subset $E \subset \mathcal{D}(\mathbb{R})$, defined as

$$E := \{\psi \in \mathcal{D}(\mathbb{R}) \mid \psi = \phi' \text{ for some } \phi \in \mathcal{D}(\mathbb{R})\}.$$

We have $T(\psi) = 0$ for all $\psi \in E$. Using Exercise 24, we know that $\psi \in E$ iff $\int_{\mathbb{R}} \psi(x)\,dx = 0$. Choose any $\chi \in \mathcal{D}(\mathbb{R})$ such that $\int_{\mathbb{R}} \chi = 1$ and, for each $\phi \in \mathcal{D}(\mathbb{R})$, set $\psi := \phi - \alpha\chi$ where $\alpha = \int_{\mathbb{R}} \phi$. Then $\psi \in E$ and $T(\psi) = 0$. Therefore,

$$T(\phi) = \alpha T(\chi) = \int_{\mathbb{R}} T(\chi)\phi(x)\,dx.$$

Thus, $T = T_\lambda$ where $\lambda = T(\chi)$, a constant. □

An alternate proof of above exercise is given in Exercise 41. By "regular distribution generated by a constant function" we mean the function is constant except on a set of measure zero. (cf. Example 1.20). The same is not true with weak derivative. If the weak derivative is zero, then the function is constant on each connected component of Ω.

Exercise 26. Solve the equation $DT = S$, for any given distribution $S \in \mathcal{D}'(\mathbb{R})$.

Note that the distributional derivative of a locally integrable functions that are also differentiable in classical sense, admit two notions of differentiability. We have already seen in the examples above that the two notions of differentiability need not coincide. For smooth functions the two notions do coincide. Let $f \in C^\infty(\Omega)$ and T_f be its induced distribution. For any n-tuple α, $g := D^\alpha f \in C^\infty(\Omega)$ and let T_g be its induced distribution. Then

$$\begin{aligned} D^\alpha T_f(\phi) &= (-1)^{|\alpha|} T_f(D^\alpha \phi) = (-1)^{|\alpha|} \int f D^\alpha \phi\, dx \\ &= \int D^\alpha f \phi\, dx = \int g\phi = T_g(\phi). \end{aligned}$$

Thus, the two notions of derivatives coincide for all functions which respect integration by parts.

Proposition 1.5.3. *Let f, g be integrable functions on $[a, b]$ and c, d be some real constants. Set*

$$F(x) := c + \int_a^x f(t)\,dt; \quad \text{and } G(x) := d + \int_a^x g(t)\,dt.$$

Then,

$$\int_a^b \left[F(x)g(x) + G(x)f(x)\right]\,dx = F(b)G(b) - F(a)G(a).$$

Proof. Note that $c = F(a)$ and $d = G(a)$. Now, consider

$$\begin{aligned}\int_a^b [Fg + Gf]\, dt &= \int_a^b g(t)\left(F(a) + \int_a^t f(s)\, ds\right) dt \\ &\quad + \int_a^b f(t)\left(G(a) + \int_a^t g(s)\, ds\right) dt \\ &= F(a)[G(b) - G(a)] + \int_a^b g(t)\left(\int_a^t f(s)\, ds\right) dt \\ &\quad + G(a)[F(b) - F(a)] + \int_a^b f(t)\left(\int_a^t g(s)\, ds\right) dt.\end{aligned}$$

Set $s = x$, $t = y$ in the second integral and $s = y$, $t = x$ in the last integral. Then,

$$\begin{aligned}\int_a^b [Fg + Gf]\, dt &= \int_a^b \int_a^y f(x)g(y)\, dx\, dy + \int_a^b \int_a^x f(x)g(y)\, dy\, dx \\ &\quad + F(a)G(b) + G(a)F(b) - 2F(a)G(a).\end{aligned}$$

Consider the square $[a, b] \times [a, b]$ in $\mathbb{R}^2$ and its bisection by the line $y = x$ in $\mathbb{R}^2$. Then the first integral is evaluated in the region above the bisecting line and the second integral is evaluated below the bisecting line. Thus, combined together the region of computation is the square $[a, b] \times [a, b]$. Hence,

$$\begin{aligned}\int_a^b [Fg + Gf]\, dt &= \int_a^b \int_a^b f(x)g(y)\, dx\, dy + F(a)G(b) + G(a)F(b) \\ &\quad - 2F(a)G(a) \\ &= \left(\int_a^b f(x)\, dx\right)\left(\int_a^b g(y)\, dy\right) + F(a)G(b) + G(a)F(b) \\ &\quad - 2F(a)G(a) \\ &= [F(b) - F(a)][G(b) - G(a)] + F(a)G(b) + G(a)F(b) \\ &\quad - 2F(a)G(a) \\ &= F(b)G(b) - F(a)G(a).\end{aligned}$$

□

We know that F and G are bounded variation and $F' = f$ and $G' = g$. Thus, we have the following consequence:

Corollary 1.5.4. *If $f, g \in AC(\mathbb{R})$ then, for each subinterval $[a, b] \subset \mathbb{R}$,*

$$\int_a^b [f(x)g'(x) + f'(x)g(x)]\, dx = f(b)g(b) - f(a)g(a).$$

In particular, if $f \in AC(\mathbb{R})$ then, for all $\phi \in \mathcal{D}(\mathbb{R})$,

$$\int_{\mathbb{R}} f(x)\phi'(x) = -\int_{\mathbb{R}} f'(x)\phi(x)\, dx.$$

The above proposition shows that, in one dimensional case, the integration by parts formula can be extended, precisely, to the class of absolutely continuous functions. Every absolutely continuous function on $\mathbb{R}$ is of bounded variation and admits a derivative almost everywhere. We have the following inclusions:

$$AC(\mathbb{R}) \subsetneq C(\mathbb{R}) \cap BV(\mathbb{R}) \subsetneq BV(\mathbb{R}) \subsetneq L^1_{\text{loc}}(\mathbb{R}) \subsetneq \mathcal{M}(\mathbb{R}) \subsetneq \mathcal{D}'(\mathbb{R}).$$

We know that $f \in AC(\mathbb{R})$ iff $f(b) - f(a) = \int_a^b f'(t)\, dt$ for all subintervals $[a, b] \subset \mathbb{R}$.

Proposition 1.5.5. *Conversely, if $f : \mathbb{R} \to \mathbb{R}$ is an integrable function such that*

$$\int_{\mathbb{R}} f(x)\phi'(x)\, dx = -\int_{\mathbb{R}} g(x)\phi(x)\, dx,$$

for some integrable function $g : \mathbb{R} \to \mathbb{R}$ and for all $\phi \in \mathcal{D}(\mathbb{R})$, then there is a $h \in AC(\mathbb{R})$ such that $h = f$ a.e. and $h' = g$.

Remark 1.5.6. Every $f \in \mathrm{AC}(\mathbb{R})$ can be associated with a measure μ_f, defined as $\mu_f(a, b) = f(b) - f(a)$, for every subinterval $(a, b) \subset \mathbb{R}$. Note that this measure is different from the measure $\mu_{f'}$ induced by $f' \in L^1_{\text{loc}}(\mathbb{R})$ as in (1.4.2). Observe that $T_f \neq T_{\mu_f}$. In fact, $T_{\mu_f} = T_{f'}$.

We have already seen in Example 1.25 that for the Cantor function f_C, which is not absolutely continuous, the two derivatives do not coincide.

Exercise 27. Compute and compare the classical and distributional derivative of

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x^2} & x \neq 0 \\ 0 & x = 0. \end{cases}$$

Theorem 1.5.7. *The differential operator D^α, for each α, is a continuous map from $\mathcal{D}'(\Omega)$ to $\mathcal{D}'(\Omega)$.*

Proof. It is enough to show that when $T_m \to T$ in $\mathcal{D}'(\Omega)$ then $D^\alpha T_m \to D^\alpha T$ in $\mathcal{D}'(\Omega)$, for all multi-index α. Consider

$$\lim_{m\to\infty} D^\alpha T_m(\phi) = \lim_{m\to\infty} (-1)^\alpha T_m(D^\alpha \phi) = (-1)^\alpha T(D^\alpha \phi) = D^\alpha T(\phi).$$

□

Example 1.28. The above theorem is very special to the space of distributions $\mathcal{D}'(\Omega)$. For instance, the result is not true in $C^\infty(\Omega)$. Consider the sequence $f_m = (1/\sqrt{m}) \sin mx$ that converges uniformly to 0 and hence converges to 0 in the distribution sense too. However, it's derivative $f'_m(x) = \sqrt{m} \cos mx$ does not converge pointwise but converges to 0 in the distribution sense.

Corollary 1.5.8 (Term-by term differentiation of series)**.** *If $S := \sum_{i=1}^\infty T_i$, then $D^\alpha S = \sum_{i=1}^\infty D^\alpha T_i$.*

1.5.2 Product

Now that we have addition of distributions, a natural question is whether one can define product of any two distributions. The answer is in negation. The space of distribution cannot be made an algebra which extends the classical notion of point-wise multiplication. This is called the *Schwartz impossibility result*. L. Schwartz himself showed that it is not possible to define product of Dirac distributions, i.e., $\delta_0 \cdot \delta_0$. However, one may define product of a distribution with a C^∞ function which is a generalisation of the point-wise multiplication of two functions. To motivate this definition, we note that if $f \in L^1_{\text{loc}}(\Omega)$ and $\psi \in C^\infty(\Omega)$ then

$$\int_\Omega [\psi(x) f(x)] \phi(x)\, dx = \int_\Omega f(x) [\psi(x)\phi(x)]\, dx.$$

If T_f is the distribution induced by f, then above equality is same as saying $T_{\psi f}(\phi) = T_f(\psi\phi)$. But $T_f(\psi\phi)$ makes sense only when $\psi\phi \in \mathcal{D}(\Omega)$. This is the reason we demand $\psi \in C^\infty(\Omega)$ because then $\text{supp}(\psi\phi) \subseteq \text{supp}(\phi)$ and $\psi\phi$ has compact support. Also, by Leibniz' rule,

$$D^\alpha(\psi\phi) = \sum_{\beta\le\alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!} D^\beta\psi\, D^{\alpha-\beta}\phi$$

and hence $\psi\phi \in \mathcal{D}(\Omega)$.

Definition 1.5.9. *Let $\psi \in C^\infty(\Omega)$ and $T \in \mathcal{D}'(\Omega)$, then we define the functional $\psi T : \mathcal{D}(\Omega) \to \mathbb{R}$ as $\psi T(\phi) = T(\psi\phi)$.*

Proposition 1.5.10. *For every $\psi \in C^\infty(\Omega)$, $\psi T \in \mathcal{D}'(\Omega)$.*

Proof. We need to show that the functional ψT is continuous on $\mathcal{D}(\Omega)$. For any $\phi \in C^\infty(K)$,

$$\begin{aligned} |\psi T(\phi)| &= |T(\psi\phi)| \le C_K \|\psi\phi\|_{N_K} = C_K \sum_{|\alpha|=0}^{N_K} \|D^\alpha(\psi\phi)\|_0 \\ &\le C_K \sum_{|\alpha|=0}^{N_K} \sum_{\beta\le\alpha} \left| \frac{\alpha!}{\beta!(\alpha-\beta)!} \right| \|D^\beta\psi\|_0 \, \|D^{\alpha-\beta}\phi\|_0 \\ &\le C_K C_0 \|\phi\|_{N_K}. \end{aligned}$$

□

If T is a distribution of order k, then it is enough to demand $\psi \in C^k(\Omega)$.

Example 1.29. Let $\psi(x) = x$ on $\mathbb{R}$ and $T = \delta_a \in \mathcal{D}'(\mathbb{R})$, then $x\delta_a \in \mathcal{D}'(\mathbb{R})$ and $x\delta_a(\phi) = \delta_a(x\phi) = a\phi(a)$.

Theorem 1.5.11 (Leibniz' formula)**.** *If $\psi \in C^\infty(\Omega)$, then*

$$D^\alpha(\psi T) = \sum_{\beta\le\alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!} D^\beta\psi \, D^{\alpha-\beta}T.$$

Exercise 28. Let $\phi \in \mathcal{D}(\mathbb{R})$. Show that $\phi^{(k)}(0) = 0$ for $k = 0, 1, 2, \ldots, m-1$ iff there is a $\psi \in \mathcal{D}(\mathbb{R})$ such that $\phi(x) = x^m\psi(x)$.

Proof. Let $\psi \in \mathcal{D}(\mathbb{R})$ be such that $\phi(x) = x^m\psi(x)$. Then

$$\phi^{(k)}(x) = \sum_{i=0}^{k} \frac{k!}{i!(k-i)!} \frac{d^i(x^m)}{dx^i} \psi^{(k-i)}$$

where $\frac{d^i(x^m)}{dx^i} = \frac{m!}{(m-i)!}x^{m-i}$. Thus, for all $i = 0, 1, \ldots, m-1$, the value of $\frac{d^i(x^m)}{dx^i}$ at $x = 0$ is 0 and $\frac{d^m(x^m)}{dx^m} = m! \neq 0$. Therefore, $\phi^{(k)}(0) = 0$ for all $k = 0, 1, 2, \ldots, m-1$, irrespective of the value of $\psi(0)$. Conversely, let $\phi^{(k)}(0) = 0$ for all $k = 0, 1, 2, \ldots, m-1$. Let $k = 0$, then

$$\phi(x) = \phi(0) + \int_0^x \phi^{(1)}(t)\, dt.$$

Since $\phi(0) = 0$ and setting $t = xs$, we get

$$\phi(x) = \int_0^x \phi^{(1)}(t)\, dt = x \int_0^1 \phi^{(1)}(xs)\, ds = x\psi_0(x),$$

where $\psi_0(x) := \int_0^1 \phi^{(1)}(xs)\, ds$ is in $\mathcal{D}(\mathbb{R})$. Since $\phi^{(1)}(x) = x\psi_0^{(1)}(x) + \psi_0(x)$, we have $\psi(0) = 0$ and hence $\psi_0(x) = x\psi_1(x)$ for some $\psi_1 \in \mathcal{D}(\mathbb{R})$. Thus, $\phi(x) = x^2\psi_1(x)$. Using the fact that $\phi^{(2)}(0) = 0$, we get $\psi_1(0) = 0$ and hence $\phi(x) = x^3\psi_2(x)$ for some $\psi_2 \in \mathcal{D}(\mathbb{R})$. Proceeding this way, using $\phi^{(m-1)}(0) = 0$, we get $\psi_{m-1} \in \mathcal{D}(\mathbb{R})$ such that $\phi(x) = x^m\psi_{m-1}(x)$. □

Exercise 29. (i) Find all real valued functions f on $\mathbb{R}$ which is a solution to the equation $xf(x) = 0$ for all $x \in \mathbb{R}$.

(ii) Find all distributions $T \in \mathcal{D}'(\mathbb{R})$ that solve the equation $xT = 0$ in the distribution sense.

(iii) Find all distributions $T \in \mathcal{D}'(\mathbb{R})$ that solve the equation $x^{2013}T = 0$ in the distribution sense.

Proof. (i) Consider the function $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) = \begin{cases} 0 & x \neq 0 \\ 1 & x = 0. \end{cases}$$

Then f solves $xf(x) = 0$ for all $x \in \mathbb{R}$. Moreover, for any scalar $\lambda \in \mathbb{R}$, the function λf is also a solution of $xf(x) = 0$.

(ii) For any $\phi \in \mathcal{D}(\mathbb{R})$, since $0 = xT(\phi) = T(x\phi)$. By Exercise 28), any solution $T \in \mathcal{D}'(\mathbb{R})$ satisfies $T(\psi) = 0$ for all $\psi \in \mathcal{D}(\mathbb{R})$ such that $\psi(0) = 0$. Obviously, $T \equiv \delta_0$ is a solution. In fact, for any scalar $\lambda\delta_0 \in \mathcal{D}'(\mathbb{R})$ is also a solution. We need to show that any solution is a scalar multiple of δ_0. Let $T \in \mathcal{D}'(\mathbb{R})$ be a solution of $xT = 0$. Hence $T(\psi) = 0$ for all test functions on $\mathbb{R}$ such that $\psi(0) = 0$. Let $\chi \in \mathcal{D}(\mathbb{R})$ such that $\chi(0) = 1$, then for any $\phi \in \mathcal{D}(\mathbb{R})$, $\psi(x) := \phi(x) - \phi(0)\chi(x)$ is a test function such that $\psi(0) = 0$. Therefore,

$$0 = T(\psi) = T(\phi) - T(\chi)\phi(0) = T(\phi) - T(\chi)\delta_0(\phi).$$

Thus, $T(\phi) = \lambda\delta_0(\phi)$ where the choice of λ depends on the choice of χ.

(iii) By Exercise 28, any solution $T \in \mathcal{D}'(\mathbb{R})$ satisfies $T(\psi) = 0$, for all $\psi \in \mathcal{D}(\mathbb{R})$, such that $\psi^{(k)}(0) = 0$ for all $k = 0, 1, 2, \ldots, 2012$. Therefore, $\delta_0^{(k)}$ is a solution for all $k = 0, 1, 2, \ldots, 2012$. Thus,

$$T \equiv \sum_{k=0}^{2012} \lambda_k \delta_0^{(k)}$$

is a solution for every possible choice of $\lambda_k \in \mathbb{R}$.

□

Example 1.30. Let $\psi(x) = x$ on $\mathbb{R}$ and $T = PV(1/x) \in \mathcal{D}'(\mathbb{R})$, then $xPV(1/x) \in \mathcal{D}'(\mathbb{R})$ and

$$\begin{aligned} xPV\left(\frac{1}{x}\right)(\phi) &= PV\left(\frac{1}{x}\right)(x\phi) = \lim_{\varepsilon \to 0^+} \int_{|x| \geq \varepsilon} \frac{1}{x} x\phi(x)\, dx \\ &= \int_{\mathbb{R}} \phi(x)\, dx = T_1(\phi) \end{aligned}$$

where T_1 is the distribution corresponding to the constant function 1.

Exercise 30. Find all distributions $T \in \mathcal{D}'(\mathbb{R})$ that solve the equation $xT = 1$ in the distribution sense.

Proof. The principal value distribution $T \equiv PV(1/x)$ is one particular solution of $xT = 1$. Also any solution S of $xS = 0$ when added to $PV(1/x)$ is also a solution. Thus, the general solution is $PV(1/x) + \lambda\delta_0$ for every $\lambda \in \mathbb{R}$. □

In light of the above example, we come back to the issue of not being able to have a product in the space of distributions. Suppose we give a notion of product of distributions, the product cannot be associative because $PV(1/x) \cdot (x \cdot \delta_0) = 0 \neq \delta_0 = (PV(1/x) \cdot x) \cdot \delta_0$. Further, recall that $f(x) = |x|^{-1/2} \in L^1_{\text{loc}}(\mathbb{R})$, but $g(x) = |x| \notin L^1_{\text{loc}}(\mathbb{R})$. How do we give a notion of product, in a suitable way, such that $(T_f)^2$ coincides with classical pointwise product. The failure to define a suitable notion of product is what makes the theory of distributions unsuitable for nonlinear differential equations.

Exercise 31. For any given $\psi \in C^\infty(\mathbb{R})$, find the solutions of the equation $DT + \psi T = 0$ in $\mathcal{D}'(\mathbb{R})$.

Proof. If $\psi \equiv 0$, we know from Exercise 25 that T is a regular distribution induced by a constant function. Therefore, we assume w.l.o.g that $\psi \not\equiv 0$. We copy the idea of integrating factor from ODE course. We shall now prove the existence of $\Psi \in C^\infty(\mathbb{R})$ (nowhere zero) such that $D(\Psi T) = \Psi(DT + \psi T)$. Multiply Ψ on both sides of the given equation to get $\Psi DT + \Psi\psi T = 0$. If we want Ψ such that $D(\Psi T) = \Psi(DT + \psi T)$, then $\Psi\psi = \Psi'$. Thus, $\Psi(x) = e^{\int_{-\infty}^{x} \psi(s)\, ds}$. By construction, Ψ is a non-zero function and the equation implies that $D(\Psi T) = 0$. By Exercise 25, for each scalar $\mu \in \mathbb{R}$, $\Psi T = T_\mu$ is a solution. Thus, $T = \Psi^{-1} T_\mu$, for each $\mu \in \mathbb{R}$ is the required solution of the equation. The solution makes sense because $\Psi^{-1} \in C^\infty(\mathbb{R})$. □

Exercise 32. For any given $\psi \in C^\infty(\mathbb{R})$ and $S \in \mathcal{D}'(\mathbb{R})$, find the solutions of the equation $DT + \psi T = S$ in $\mathcal{D}'(\mathbb{R})$.

Recall the topology on $C^\infty(\Omega)$ described in §1.3.4. The sequential description of the topology is the uniform convergence on compact sets. Thus, a sequence ψ_m is said to converge to 0 in $C^\infty(\Omega)$ if $D^\alpha \psi_m$, for all α, uniformly converges to 0 on all compact subsets of Ω.

Lemma 1.5.12. *If $\psi_m \to 0$ in $C^\infty(\Omega)$ then $\psi_m \phi \to 0$ in $\mathcal{D}(\Omega)$.*

Proof. Since $D^\alpha \psi_m \to 0$ uniformly on all compact subsets of Ω, in particular on $K = \operatorname{supp}(\phi)$. Thus, $D^\alpha \psi_m \phi \to 0$ uniformly on K, for all α, and $\operatorname{supp}(\psi_m \phi) \subseteq K$, for all m. □

Theorem 1.5.13. *If $T_m \to T$ in $\mathcal{D}'(\Omega)$ and $\psi_m \to \psi$ in $C^\infty(\Omega)$ then $\psi_m T_m \to \psi T$ in $\mathcal{D}'(\Omega)$.*

1.5.3 Support

The fact that we *cannot* have a product on $\mathcal{D}'(\Omega)$, generalising point-wise multiplication, that makes $\mathcal{D}'(\Omega)$ an algebra leads us to look for other ways to make $\mathcal{D}'(\Omega)$ an algebra. One such choice is the convolution operation and whether this can be extended to distributions. We shall see later than one can define a notion of convolution for distributions provided one of them has compact support! This motivates us to understand the support of a distribution which coincides for classical functions. Classically, support of a function is complement of the largest open set on which the function vanishes.

Definition 1.5.14 (Localisation)**.** *The* restriction *of $T \in \mathcal{D}'(\Omega)$ to an open subset $\omega \subset \Omega$, denoted as $T\mid_\omega$, is defined as*

$$T\mid_\omega(\phi) = T(\phi) \quad \forall \phi \in \mathcal{D}(\omega) \subset \mathcal{D}(\Omega).$$

We say a distribution T vanishes on an open set *$\omega \subset \Omega$ if $T\mid_\omega = 0$. More generally, two distributions S and T coincide on a open set ω if $S\mid_\omega = T\mid_\omega$.*

The inclusion $\mathcal{D}(\omega) \subset \mathcal{D}(\Omega)$ is continuous because any $\phi \in \mathcal{D}(\omega)$ can be extended by zero outside ω which belongs to $\mathcal{D}(\Omega)$. Since $T\mid_\omega \in \mathcal{D}'(\omega)$, T vanishing on ω is same as the zero distribution in $\mathcal{D}'(\omega)$.

Example 1.31. The restriction of the Dirac distribution δ_a to any open set $\omega \subset \Omega \setminus \{a\}$ is zero.

Proposition 1.5.15. *If $T \in \mathcal{D}'(\Omega)$ vanishes on the open subsets ω_i of Ω, then T vanishes on the union $\cup_{i\in I}\omega_i$.*

Proof. Let $\omega = \cup_{i\in I}\omega_i$. We choose the C^∞ locally finite partition of unity $\{\phi_i\}$ subordinate to $\{\omega_i\}$, i.e., $\mathrm{supp}(\phi_i) \subset \omega_i$ and $\sum_i \phi = 1$ (the summation is finite for each $x \in \omega$). Fix $\phi \in \mathcal{D}(\omega)$ and $K = \mathrm{supp}(\phi)$ then there exists finitely many $i_1, i_2, \ldots, i_k$ such that $K \cap \phi_{i_m} \neq \emptyset$. Thus, $\phi = \sum_{m=1}^k \phi\phi_{i_m}$ and, by linearity of T, $T(\phi) = \sum_{m=1}^k T(\phi\phi_{i_m})$. But $\mathrm{supp}(\phi\phi_{i_m}) \subset \omega_{i_m}$ and hence $T(\phi\phi_{i_m}) = 0$. Therefore, $T(\phi) = 0$ for all $\phi \in \mathcal{D}(\omega)$ and hence $T\mid_\omega = 0$. □

Definition 1.5.16. *The* support *of a distribution T is the complement of the largest open set ω such that $T\mid_\omega = 0$. Equivalently, the support of a distribution is the relative complement in Ω of the union of all open sets ω of Ω such that $T\mid_\omega = 0$.*

If E is the support of a distribution T, then $T(\phi) = 0$ for all $\phi \in \mathcal{D}(\Omega)$ such that $E \cap \mathrm{supp}(\phi) = \emptyset$ because all such ϕ is in $\mathcal{D}(E^c)$. Obviously, the support of the zero distribution is the empty set.

Exercise 33. Show that for any continuous function f, $\mathrm{supp}(T_f)$=$\mathrm{supp}(f)$. More generally, for any $f \in L^1_{\mathrm{loc}}(\Omega)$, there exists a measurable function g such that $g = f$ a.e. and $\mathrm{supp}(T_f) = \mathrm{supp}(g)$.

Example 1.32. The support of the Dirac distribution δ_a is the singleton set $\{a\}$. Similarly, the support of the derivatives of Dirac distribution δ_a^α is also the singleton set $\{a\}$. The open set $\Omega \setminus \{a\} = \cup_i B_i$, is the union of punctured open balls of rational radius r_i with centre $\{a\}$ removed. δ_a^α vanishes on B_i for each i and hence vanishes on $\Omega \setminus \{a\}$.

Exercise 34. Show that the support of $DT \in \mathcal{D}(\mathbb{R})$ is contained in the support of $T \in \mathcal{D}(\mathbb{R})$.

Proof. Let E and F denote the support of T and DT, respectively. We need to show that $F \subseteq E$, or equivalently, $E^c \subseteq F^c$. For any open set $\omega \subset E^c$, then $T \mid_\omega = 0$, i.e., for all $\phi \in \mathcal{D}(\omega)$ we have $T(\phi) = 0$. Consider $DT(\phi) = -T(\phi') = 0$. The second equality is due to the fact that if $\phi \in \mathcal{D}(\omega)$ then $\phi' \in \mathcal{D}(\omega)$ with $\text{supp}(\phi') \subseteq \text{supp}(\phi)$. □

A distribution is said to have *compact support*, if its support is a compact subset of Ω. A nice characterisation of distributions with compact support, which is a subclass of $\mathcal{D}'(\Omega)$, is that it can be identified as a dual of $C^\infty(\Omega)$ endowed with the topology of uniform convergence on compact sets. This observation is actually quite natural. Recall the motivation for the choice of $C_c^\infty(\Omega)$ as a test function. The compact support of test function was necessary to kill the boundary evaluation while applying integration by parts. If the function f itself is compact, to begin with, then we can expand the space of test functions to $C^\infty(\Omega)$. For convenience sake (for ease of notation), we set $C^\infty(\Omega) = \mathcal{E}(\Omega)$, the space of infinitely differentiable functions on Ω with the topology of uniform convergence on compact subsets. The inclusion $\mathcal{D}(\Omega) \subset \mathcal{E}(\Omega)$ is continuous w.r.t the respective topology. This is because if $\phi_m \to 0$ in $\mathcal{D}(\Omega)$ then $\phi_m \to 0$ in $\mathcal{E}(\Omega)$, as well. The following theorem asserts that any distribution $T : \mathcal{D}(\Omega) \to \mathbb{R}$ can be continuously extended as a functional to $\mathcal{E}(\Omega)$. Thus $T \in \mathcal{E}'(\Omega)$, for compact support T.

Theorem 1.5.17. *The dual space $\mathcal{E}'(\Omega)$ is the collection of all distributions with compact support.*

Proof. Let $T \in \mathcal{D}'(\Omega)$ such that support of T is compact. Set $K = \text{supp}(T)$. Thus, $T(\phi) = 0$ for all $\phi \in \mathcal{D}(\Omega)$ such that $\text{supp}(\phi) \subset K^c$. Choose $\chi \in \mathcal{D}(\Omega)$ such that $\chi \equiv 1$ on K. For all $\phi \in \mathcal{D}(\Omega)$, consider

$$T(\phi) - \chi T(\phi) = T(\phi - \chi\phi) = T((1 - \chi)\phi).$$

Since $1 - \chi = 0$ on K, $\text{supp}(1 - chi)\phi \subset K^c$ and hence $T = \chi T$ for any χ such that $\chi \equiv 1$ on K. This fact helps us to extend T as a functional on $\mathcal{E}(\Omega)$, because for all $\psi \in \mathcal{E}(\Omega)$, $\psi\chi \in \mathcal{D}(\Omega)$. We extend T to $\mathcal{E}(\Omega)$ by setting $T(\psi) = T(\chi\psi)$. The assignment is well-defined (it is independent of choice of χ). Suppose $\chi_1 \in \mathcal{D}(\Omega)$ is such that $\chi_1 \neq \chi$ and $\chi_1 \equiv 1$ on K, then $(\chi - \chi_1)\psi = 0$ on K. Therefore, $\text{supp}((\chi - \chi_1)\psi) \subset K^c$ and

$T((\chi - \chi_1)\psi) = 0$, By linearity of T, $T(\chi\psi) = T(\chi_1\psi)$. The conitnuity of T on $\mathcal{E}(\Omega)$ is a consequence of Lemma 1.5.12. Thus, $T \in \mathcal{E}'(\Omega)$.

Conversely, let $T \in \mathcal{E}'(\Omega)$. We need to show that $T \in \mathcal{D}'(\Omega)$ and has a compact support. The fact that $T \in \mathcal{D}'(\Omega)$ is obvious due to the continuous inclusion of $\mathcal{D}(\Omega)$ in $\mathcal{E}(\Omega)$. Let $\{K_i\}$ be the collection of compact subsets of Ω such that $K_i \subset K_{i+1}$ and $\Omega = \cup_{i=1}^{\infty} K_i$ (exhaustion subsets). Suppose that the support of T is not compact in Ω, then the support of T intersects $\Omega \setminus K_i$, for all i. Thus, for each i, there is a $\phi_i \in \mathcal{D}(\Omega)$ such that $\text{supp}(\phi_i) \subset \Omega \setminus K_i$ and $T(\phi_i) \neq 0$. Say $T(\phi_i) = \alpha_i \neq 0$. Then $T(\alpha_i^{-1}\phi_i) = 1$. Set $\psi_i = \alpha_i^{-1}\phi_i$. Thus, $\text{supp}(\psi_i) \subset \Omega \setminus K_i$ and $T(\psi_i) = 1$. Hence, the sequence $\psi_i \to 0$ in $\mathcal{E}(\Omega)$ because their support tend to empty set. Now, by continuity of T on $\mathcal{E}(\Omega)$, $T(\psi_i) \to 0$ which contradicts the fact that $T(\psi_i) \to 1$. Hence, T has a compact support. □

We have proved that $\mathcal{E}'(\Omega) \subset \mathcal{D}'(\Omega)$ is the class of all distributions whose support is compact. Distributions with compact support are always of finite order.

Proposition 1.5.18. *If $T \in \mathcal{E}'(\Omega)$ then order of T is finite.*

Proof. Since $T \in \mathcal{E}'(\Omega)$, T has compact support, say K. Let $\chi \in \mathcal{D}(\Omega)$ be such that $\chi \equiv 1$ on K, then we know that (cf. proof of last theorem) $T = \chi T$. Let K' be the support of χ and hence $K \subset K'$. Now, for any compact subset $K \subset \Omega$ and $\phi \in C^\infty(K)$, $\text{supp}(\phi\chi) \subset K'$. Consider,

$$|T(\phi)| = |\chi T(\phi)| = |T(\chi\phi)| \leq C_{K'}\|\chi\phi\|_{N_{K'}} \leq C_0\|\phi\|_{N_{K'}}.$$

The $N_{K'}$ is independent of all compact subsets K. Thus, T has finite order. □

1.5.4 Singular Support

The singular support of a distribution is the measure of the set on which the distribution fails to be smooth or C^∞.

Definition 1.5.19. *A distribution $T \in \mathcal{D}'(\Omega)$ is said to be C^∞ on $\omega \subset \Omega$ if there is a $f \in C^\infty(\omega)$ such that $T = T_f$ on $\mathcal{D}(\omega)$.*

Example 1.33. The Dirac distribution, δ_a, is C^∞ in $\Omega \setminus \{a\}$ because δ_a restricted to the open set $\Omega \setminus \{a\}$ is the zero function.

Local behaviour of "compatible" distributions can be patched up to get a global description of the distribution. This concept will facilitate the definition of singular support.

Theorem 1.5.20. *Let $T_i \in \mathcal{D}'(\omega_i)$, an arbitrary collection of distributions, such that $T_i \mid_{\omega_i \cap \omega_j} = T_j \mid_{\omega_i \cap \omega_j}$ then there is a unique distribution $T \in \mathcal{D}'(\Omega)$, where $\Omega = \cup_i \omega_i$, such that $T \mid_{\omega_i} = T_i$ for all i.*

Proof. Let $\{\phi_i\} \subset \mathcal{D}(\Omega)$ be a C^∞ locally finite partition of unity subordinate to $\{\omega_i\}$. Thus, $\text{supp}(\phi_i) \subset \omega_i$ and $1 = \sum_i \phi_i$. We define the functional $T : \mathcal{D}(\Omega) \to \mathbb{R}$ as $T(\phi) = \sum_i T_i(\phi\phi_i)$. We first check the continuity of T. One way is to show the sequential continuity. Alternately, For any $\phi \in C^\infty(K)$ (compact subset of Ω), there exist finitely many $i_1, i_2, \ldots, i_k$ such that $K \cap \text{supp}(\phi_{i_m}) \neq \emptyset$. Then

$$|T(\phi)| \leq \sum_{m=1}^{k} |T_{i_m}(\phi\phi_{i_m})| \leq \sum_m C_k \|\phi\phi_{i_m}\|_{N_k} \leq C_0 \|\phi\|_{N_0}.$$

It only remains to show that the restriction of T to ω_k is T_k. Let $\phi \in \mathcal{D}(\omega_k)$, then $\phi\phi_i \in \mathcal{D}(\omega_i \cap \omega_k)$ for all i. Thus,

$$T(\phi) = \sum_i T_i(\phi\phi_i) = \sum_i T_k(\phi\phi_i) = T_k(\sum_i \phi\phi_i) = T_k(\phi).$$

The uniqueness of T follows from Proposition 1.5.15. □

Corollary 1.5.21. *If $T \in \mathcal{D}'(\Omega)$ is C^∞ on an arbitrary collection of open subsets ω_i, then T is C^∞ on the union $\cup_{i \in I} \omega_i$.*

A consequence of the above corollary is the following definition.

Definition 1.5.22. *The* singular support *of $T \in \mathcal{D}'(\Omega)$ is the complement of the largest open set on which T is C^∞, denoted as sing.supp(T).*

A simple observation is that sing.supp$(T) \subset$ supp(T).

1.5.5 Shifting and Scaling

Given any function $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and a fixed $a \in \mathbb{R}^n$, we introduce the shift operator $\tau_a f(x) := f(x - a)$. In particular, $\tau_0 f = f$ for all f. Observe that $\tau_a \tau_b f(x) = \tau_a f(x - b) = f(x - b - a) = f(x - (b + a)) = \tau_{a+b} f(x)$.

Note that if $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ then $\tau_a f \in L^1_{\text{loc}}(\mathbb{R}^n)$. We have the following relation between T_f and $T_{\tau_a f}$. For any $\phi \in \mathcal{D}(\mathbb{R}^n)$, consider

$$\begin{aligned} T_{\tau_a f}(\phi) &= \int_{\mathbb{R}^n} \tau_a f(x)\phi(x)\,dx = \int_{\mathbb{R}^n} f(x-a)\phi(x)\,dx = \int_{\mathbb{R}^n} f(y)\phi(y+a)\,dy \\ &= \int_{\mathbb{R}^n} f(y)\tau_{-a}\phi(y)\,dy = T_f(\tau_{-a}\phi). \end{aligned}$$

The last equality is valid because $\tau_{-a}\phi \in \mathcal{D}(\mathbb{R}^n)$ whenever $\phi \in \mathcal{D}(\mathbb{R}^n)$. We wish to extend the notion of shift to distributions such that for $\tau_a T_f = T_{\tau_a f}$. This motivates the following definition of a shift of a distribution by $a \in \mathbb{R}^n$.

Definition 1.5.23. *For any $T \in \mathcal{D}'(\mathbb{R}^n)$, we define its* shift by $a \in \mathbb{R}^n$ *as* $\tau_a T(\phi) = T(\tau_{-a}\phi)$.

Given any function $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and a fixed scalar $\lambda \in (0,\infty)$, we introduce the scaling function $f_\lambda(x) := f(\lambda x)$. If $f = g + h$ then $f_\lambda(x) = f(\lambda x) = g(\lambda x) + h(\lambda x) = g_\lambda(x) + h_\lambda(x)$. Also, $\tau_a(f+g) = \tau_a f + \tau_a g$. Note that if $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ then $f_\lambda \in L^1_{\text{loc}}(\mathbb{R}^n)$. We have the following relation between T_f and T_{f_λ}. For any $\phi \in \mathcal{D}(\mathbb{R}^n)$, consider

$$\begin{aligned} T_{f_\lambda}(\phi) &= \int_{\mathbb{R}^n} f(\lambda x)\phi(x)\,dx = \lambda^{-n}\int_{\mathbb{R}^n} f(y)\phi\left(\frac{y}{\lambda}\right)\,dy \\ &= \lambda^{-n}\int_{\mathbb{R}^n} f(y)\phi_{1/\lambda}(y)\,dy = \lambda^{-n}T_f(\phi_{1/\lambda}). \end{aligned}$$

The last equality is valid because $\phi_{1/\lambda} \in \mathcal{D}(\mathbb{R}^n)$ whenever $\phi \in \mathcal{D}(\mathbb{R}^n)$. Similarly, one can argue for $\lambda \in (-\infty, 0)$ and deduce that

$$T_{f_\lambda}(\phi) = (-1)^n \lambda^{-n} T_f(\phi_{1/\lambda}).$$

Thus for any $\lambda \in \mathbb{R}\setminus\{0\}$, we have $T_{f_\lambda}(\phi) = |\lambda|^{-n}T_f(\phi_{1/\lambda})$. We wish to extend the notion of scaling to distributions such that for $(T_f)_\lambda = T_{f_\lambda}$. This motivates the following definition of a scaling of a distribution by $\lambda \in \mathbb{R}\setminus\{0\}$.

Definition 1.5.24. *For any $T \in \mathcal{D}'(\mathbb{R}^n)$, we define its* scaling by $\lambda \in \mathbb{R}\setminus\{0\}$ *as* $T_\lambda(\phi) = |\lambda|^{-n}T(\phi_{1/\lambda})$. *In particular, when $\lambda = -1$, we denote T_{-1} as $\check{T}$ and hence $\check{T}(\phi) = T(\check{\phi})$. We say $T \in \mathcal{D}'(\mathbb{R}^n)$ is even if $\check{T} = T$.*

Exercise 35. Show that $\check{\check{T}} = T$.

1.5.6 Convolution

Recall the definition of convolution of functions given in Definition 1.3.11. Thus, for any $x \in \mathbb{R}^n$,

$$(f * g)(x) := \int_{\mathbb{R}^n} f(x-y)g(y)\,dy.$$

In this section, we wish to extend the convolution operation to distributions such that for L^1 distributions the notion coincide. An obvious extension is visible when the convolution integral is rewritten using the shift and scaling operator. Note that, for a fixed $x \in \mathbb{R}^n$, $f(x-y) = f(-(y-x)) = \check{f}(y-x) = \tau_x\check{f}(y)$. With this notation the convolution is rewritten as

$$(f * g)(x) = \int_{\mathbb{R}^n} \tau_x\check{f}(y)g(y)\,dy = \int_{\mathbb{R}^n} f(y)\tau_x\check{g}(y)\,dy.$$

The second equality is due to the commutativity of convolution operation. The reformulation above motivates the following definition of convolution between a distribution and a test function.

Definition 1.5.25. *Let $T \in \mathcal{D}'(\mathbb{R}^n)$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$. The* convolution $T*\phi : \mathbb{R}^n \to \mathbb{R}$ *is defined as*

$$(T * \phi)(x) = T(\tau_x\check{\phi}).$$

The above definition is meaningful because $\tau_x\check{\phi} \in \mathcal{D}(\mathbb{R}^n)$, for all $x \in \mathbb{R}^n$, whenever $\phi \in \mathcal{D}(\mathbb{R}^n)$. In particular, $(T * \check{\phi})(x) = T(\tau_x\phi)$ since $\check{\check{\phi}} = \phi$. Thus, $T(\phi) = (T * \check{\phi})(0)$. As a consequence, if $T * \phi = 0$ for all $\phi \in \mathcal{D}(\mathbb{R}^n)$ then $T = 0$ in $\mathcal{D}'(\mathbb{R}^n)$.

Exercise 36. (i) Show that $T * (\phi_1 + \phi_2) = T * \phi_1 + T * \phi_2$.

(ii) Show that $(S + T) * \phi = S * \phi + T * \phi$.

Proof. (i) Consider

$$T*(\phi_1+\phi_2)(x) = T(\tau_x(\phi_1+\phi_2)\check{}) = T(\tau_x\check{\phi}_1+\tau_x\check{\phi}_2) = (T*\phi_1+T*\phi_2)(x).$$

(ii) Consider

$$(S + T) * \phi = (S + T)(\tau_x\check{\phi}) = S * \phi + T * \phi.$$

□

Exercise 37. Let $T \in \mathcal{D}'(\mathbb{R}^n)$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$, then

(i) for any $a \in \mathbb{R}^n$, show that $\tau_a(T * \phi) = \tau_a T * \phi = T * \tau_a \phi$.

(ii) show that, for any multi-index α, $D^\alpha(T * \phi) = D^\alpha T * \phi = T * D^\alpha \phi$. In particular, $T * \phi \in \mathcal{E}(\mathbb{R}^n)$.

(iii) for $\chi \in \mathcal{D}(\mathbb{R}^n)$, $T * (\phi * \chi) = (T * \phi) * \chi$.

For any $\psi \in \mathcal{E}(\mathbb{R}^n)$, $\tau_x \check{\psi} \in \mathcal{E}(\mathbb{R}^n)$. Thus, the convolution notion can be extended to any $\psi \in \mathcal{E}(\mathbb{R}^n)$ and $T \in \mathcal{E}'(\mathbb{R}^n)$.

Definition 1.5.26. *Let $T \in \mathcal{E}'(\mathbb{R}^n)$ and $\psi \in \mathcal{E}(\mathbb{R}^n)$. The* convolution $T * \psi : \mathbb{R}^n \to \mathbb{R}$ *is defined as*

$$(T * \psi)(x) = T(\tau_x \check{\psi}).$$

Exercise 38. Let $T \in \mathcal{E}'(\mathbb{R}^n)$ and $\psi \in \mathcal{E}(\mathbb{R}^n)$, then

(i) for any $a \in \mathbb{R}^n$, show that $\tau_a(T * \psi) = \tau_a T * \psi = T * \tau_a \psi$.

(ii) show that, for any multi-index α, $D^\alpha(T * \psi) = D^\alpha T * \psi = T * D^\alpha \psi$. In particular, $T * \phi \in \mathcal{E}(\mathbb{R}^n)$. Further, if $\phi \in \mathcal{D}(\mathbb{R}^n)$ then $T * \phi \in \mathcal{D}(\mathbb{R}^n)$ and

$$T * (\psi * \phi) = (T * \psi) * \phi = (T * \phi) * \psi.$$

Definition 1.5.27. *Let $S, T \in \mathcal{D}'(\mathbb{R}^n)$ such that one of them has compact support, i.e. either T or S is in $\mathcal{E}'(\mathbb{R}^n)$. We define the* convolution $S * T$ *as,*

$$(S * T)(\phi) = (S * (T * \check{\phi}))(0), \quad \forall \phi \in \mathcal{D}(\mathbb{R}^n).$$

Equivalently, one can also define the convolution $S * T$ to be

$$(S * T) * \phi = S * (T * \phi) \quad \forall \phi \in \mathcal{D}(\mathbb{R}^n).$$

Exercise 39. Let $S, T \in \mathcal{D}'(\mathbb{R}^n)$ such that one of them is in $\mathcal{E}'(\mathbb{R}^n)$.

(i) Show that $T * S = S * T$.

(ii) $\operatorname{supp}(S * T) \subset \operatorname{supp}(S) + \operatorname{supp}(T)$.

(iii) For any multi-index α, $D^\alpha(T * S) = D^\alpha T * S = T * D^\alpha S$.

Exercise 40. For any three distributions $T_1, T_2, T_3 \in \mathcal{D}'(\mathbb{R}^n)$ such that at least two of them have compact support then

$$T_1 * (T_2 * T_3) = (T_1 * T_2) * T_3.$$

Theorem 1.5.28. *Let $T \in \mathcal{D}'(\mathbb{R}^n)$, then $T = T * \delta_0 = \delta_0 * T$. Also, for any multi-index α, $D^\alpha T = D^\alpha \delta_0 * T$.*

Proof. Since δ_0 has compact support $\{0\}$, the convolution makes sense. By commutativity, $T * \delta_0 = \delta_0 * T$. For $\phi \in \mathcal{D}(\mathbb{R}^n)$, consider $(T * \delta_0)(\phi) = (T * (\delta_0 * \check{\phi}))(0)$. Set $\psi := \delta_0 * \check{\phi}$. Then $(T * \delta_0)(\phi) = T(\check{\psi})$. Note that

$$\check{\psi}(x) = \psi(-x) = (\delta_0 * \check{\phi})(-x) = \delta_0(\tau_{-x}\phi) = \tau_{-x}\phi(0) = \phi(x).$$

Hence, $(T * \delta_0)(\phi) = T(\phi)$. □

Theorem 1.5.29. *$\mathcal{D}(\Omega)$ is dense in $\mathcal{D}'(\Omega)$.*

The following exercise is same as Exercise 25. However, using convolution one can give a simpler proof to Exercise 25.

Exercise 41. Let $T \in \mathcal{D}'(\mathbb{R})$ be such that $DT = 0$, then show that T is a regular distribution generated by a constant function.

Proof. For a given sequence of mollifiers, let $T_\varepsilon(x) = (T * \rho_\varepsilon)(x)$. Thus, $T_\varepsilon \in \mathcal{E}(\mathbb{R})$ and $DT_\varepsilon = DT * \rho_\varepsilon = 0$. Hence, $T_\varepsilon = \lambda_\varepsilon$, where λ_ε is a constant function for each ε. Also, $\lambda_\varepsilon \rightharpoonup T$ in $\mathcal{D}'(\mathbb{R})$. In particular, choose $\phi \in \mathcal{D}(\mathbb{R})$ such that $\int \phi = 1$, then the sequence of real numbers λ_ε converges to some λ. Thus, $T = T_\lambda$. □

Exercise 42. Let T denote the distribution corresponding to the constant function 1, $\delta_0^{(1)}$ be the dipole distribution at 0 and H be the Heaviside function

$$H(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0. \end{cases}$$

Show that $(T * \delta_0^{(1)}) * H \neq T * (\delta_0^{(1)} * H)$.

Proof. Consider

$$(1 * \delta_0^{(1)})(\phi) = 1 * (\delta_0^{(1)} * \check{\phi})(0) = (1 * \psi)(0) = T_1(\check{\psi}) = \int_{\mathbb{R}} \check{\psi}(x)\, dx.$$

But $\check{\psi}(x) = \delta_0^{(1)} * \check{\phi}(-x) = \delta_0^{(1)}(\tau_{-x}\phi) = -\tau_{-x}\phi'(0) = -\phi'(x)$. Hence $(1 * \delta_0^{(1)})(\phi) = -\int_{\mathbb{R}} \phi'(x)\,dx = 0$. Further, $0 * H = 0$. Now, consider

$$(\delta_0^{(1)} * H)(\phi) = \delta_0^{(1)} * (H * \check{\phi})(0) = (\delta_0^{(1)} * \psi)(0) = \delta_0^{(1)}(\check{\psi}) = (\check{\psi})'(0) = (\check{\psi})'(0).$$

But $(\check{\psi})'(0) = (H' * \check{\phi})(0) = (\delta_0 * \check{\phi})(0) = \delta_0(\phi) = \phi(0)$. Since δ_0 is the identity distribution under convolution operation, $1 * \delta_0 = 1$.

Alternately, notice that by derivative of convolution $(1 * \delta_0^{(1)}) * H = (D1 * \delta_0) * H = D1 * H = 0 * H = 0$. On the other hand, $1 * (\delta_0^{(1)} * H) = 1 * (\delta_0 * DH) = 1 * (\delta_0 * \delta_0) = 1 * \delta_0 = 1$. □

1.6 Tempered Distributions

Recall the definition of Schwartz space from Definition **??**. By definition, $\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$. The inclusion is also dense because, for any $f \in \mathcal{S}(\mathbb{R}^n)$, $\phi_m f \to f$ in $\mathcal{S}$ where $\phi_m(x) = \phi(x/m)$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$ is such that $\phi \equiv 1$ on the unit ball in $\mathbb{R}^n$. Hence the dual of $\mathcal{S}$, denoted $\mathcal{S}'$, can be identified with a subspace of $\mathcal{D}'(\mathbb{R}^n)$.

Definition 1.6.1. *The space of distributions $\mathcal{S}'$ is called the space of* tempered distributions.

Example 1.34. Any distribution with compact support is tempered, i.e., $\mathcal{E}' \subset \mathcal{S}'$ because $\mathcal{D} \subset \mathcal{S} \subset \mathcal{E}$ and all inclusions are dense.

Example 1.35. Any slowly increasing measure induces a tempered distribution. A measure μ on $\mathbb{R}^n$ is said to be *slowly increasing* if, for some integer $k \geq 0$,

$$\int_{\mathbb{R}^n} \frac{d\mu(x)}{(1 + |x|^2)^k} < +\infty.$$

For instance, any bounded measure is slowly increasing. Set

$$T_\mu(f) = \int_{\mathbb{R}^n} f(x)\,d\mu(x), \quad f \in \mathcal{S}(\mathbb{R}^n).$$

Then T_μ is a linear functional on $\mathcal{S}(\mathbb{R}^n)$ and

$$|T_\mu(f)| \leq \left(\sup_{x \in \mathbb{R}^n} |f(x)(1 + |x|^2)^k| \right) \int_{\mathbb{R}^n} \frac{d\mu(x)}{(1 + |x|^2)^k}.$$

Further, it follows that if $f_m \to 0$ in $\mathcal{S}(\mathbb{R}^n)$ then $T_\mu(f_m) \to 0$. Thus, $T_\mu \in \mathcal{S}'(\mathbb{R}^n)$.

Example 1.36. If $1 \leq p \leq \infty$ then $L^p(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$. If $f \in L^p(\mathbb{R}^n)$, define

$$T_f(\phi) = \int_{\mathbb{R}^n} f\phi \, dx \quad \forall \phi \in \mathcal{S}(\mathbb{R}^n).$$

If $1 < p < \infty$, let q be the conjugate exponent of p given as $(1/p)+(1/q) = 1$. Then for $k > \frac{n}{2q}$, $g(x) = (1+|x|^2)^{-k} \in L^q(\mathbb{R}^n)$. Hence, by Hölder's inequality,

$$|T_f(\phi)| \leq \left(\sup_{x \in \mathbb{R}^n} |(1+|x|^2)^k \phi(x)| \right) \|f\|_{L^p(\mathbb{R}^n)} \|g\|_{L^q(\mathbb{R}^n)}.$$

Thus, T_f is tempered, and $f \mapsto T_f$ is continuous on $L^p(\mathbb{R}^n)$. If $p = 1$,

$$|T_f(\phi)| \leq \|\phi\|_{\infty,\mathbb{R}^n} \|f\|_{1,\mathbb{R}^n}$$

and if $p = \infty$,

$$|T_f(\phi)| < \sup_{x \in \mathbb{R}^n} \left|(1+|x|^2)^k \phi(x)\right| \|f\|_{\infty\,\mathbb{R}^n} \|g\|_{1,(\mathbb{R}^n)}$$

where g is as above with $k > \frac{n}{2}$.

Definition 1.6.2. *Let $T \in \mathcal{S}'(\mathbb{R}^n)$. The* Fourier transform *of T, denoted as $\hat{T}$, is defined as $\hat{T}(f) = T(\hat{f})$ for all $f \in \mathcal{S}(\mathbb{R}^n)$.*

Since $f \mapsto \hat{f}$ is continuous on $\mathcal{S}$, it follows that $\hat{T} \in \mathcal{S}'$.

Remark 1.6.3. Since $\mathcal{S}(\mathbb{R}^n) \subset L^1(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$. It seems that, a priori, there are two definitions of the Fourier transform on $\mathcal{S}(\mathbb{R}^n)$, each inherited from $L^1(\mathcal{R}^n)$ and $\mathcal{S}'(\mathbb{R}^n)$, respectively. But they are one and the same because if $f \in \mathcal{S}(\mathbb{R}^n)$ then, for any $g \in \mathcal{S}(\mathbb{R}^n)$, by Parseval relation,

$$\hat{T_f}(g) = \int_{\mathbb{R}^n} f\hat{g} = \int_{\mathbb{R}^n} \hat{f}g = T_{\hat{f}}(g).$$

Hence $\hat{T_f} = T_{\hat{f}}$ and, hence, both definitions of the Fourier transform coincide on $\mathcal{S}(\mathbb{R}^n)$.

Recall that $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$, defined as $\mathcal{F}(f) = \hat{f}$, is an isomorphism on $\mathcal{S}(\mathbb{R}^n)$, it extends as a bijection to $\mathcal{S}'(\mathbb{R}^n)$, as well.

Remark 1.6.4. Since $\mathcal{S}(\mathbb{R}^n) \subset L^2(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$, then there are two definitions of Fourier transform on $L^2(\mathbb{R}^n)$ inherited from $\mathcal{S}'(\mathbb{R}^n)$ and the other by extending from $\mathcal{S}(\mathbb{R}^n)$. It turns out that both are same. Consider the weak-* convergence of tempered distributions, i.e., $T_m \to T$ weak-* sense if, for every $\phi \in \mathcal{S}(\mathbb{R}^n)$, $T_m(\phi) \to T(\phi)$. If $T_m \to T$ in $\mathcal{S}'(\mathbb{R}^n)$ then $\hat{T_m} \to \hat{T}$ in $\mathcal{S}'$ because

$$\hat{T}_m(\phi) = T_m(\hat{\phi}) \to T(\hat{\phi}) = \hat{T}(\phi).$$

Let $f \in L^2(\mathbb{R}^n)$ and let $f_k \in \mathcal{S}(\mathbb{R}^n)$ such that $f_k \to f$ in $L^2(\mathbb{R}^n)$. Since the inclusion of $L^2(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$ is continuous, $f_k \to f$ in $\mathcal{S}'(\mathbb{R}^n)$, as well. Hence $\hat{f}_k \to \hat{f}$ in $\mathcal{S}'(\mathbb{R}^n)$. On the other hand, $\mathcal{F}(f_k) \to \mathcal{F}(f)$ in $L^2(\mathbb{R}^n)$ and hence in $\mathcal{S}'(\mathbb{R}^n)$. But as $f_k \in \mathcal{S}(\mathbb{R}^n)$, we know that $\mathcal{F}(f_k) = \hat{f}_k$. Now, by uniqueness of the weak-* limit $\hat{f} = \mathcal{F}(f)$.

By Proposition **??**, for any polynomial P and mulit-index α, if $T \in \mathcal{S}'(\mathbb{R}^n)$ then $PT \in \mathcal{S}'(\mathbb{R}^n)$ and $D^\alpha T \in \mathcal{S}'(\mathbb{R}^n)$.

We now extend the result of Theorem **??** to tempered distribution.

Theorem 1.6.5. *Let $T \in \mathcal{S}'(\mathbb{R}^n)$ and let α be a multi-index. Then*

$$D^\alpha \hat{T} = (-\imath)^{|\alpha|}\widehat{(x^\alpha T)}$$

and

$$\widehat{(D^\alpha T)} = (\imath)^{|\alpha|}\xi^\alpha \hat{T}.$$

Proof. Let $f \in \mathcal{S}$. Then

$$\begin{aligned}
\widehat{(x^\alpha T)}(f) &= (x^\alpha T)(\hat{f}) = T(x^\alpha \hat{f}(x)) \\
&= \frac{1}{(\imath)^{|\alpha|}} T(\widehat{D^\alpha f}) = \frac{1}{(\imath)^{|\alpha|}} \hat{T}(D^\alpha f) \\
&= \frac{1}{(-\imath)^{|\alpha|}} D^\alpha \hat{T}(f).
\end{aligned}$$

Thus, the first relation is proved. The second relation can be proved similarly. □

Example 1.37. $\hat{\delta}_0 = 1$, $\widehat{\left(\frac{\partial \delta_0}{\partial x_k}\right)} = \imath\xi_k$ and $\hat{1} = \delta_0$. Let $\phi \in \mathcal{S}(\mathbb{R}^n)$. Then

$$\hat{\delta}_0(\phi) = \delta_0(\hat{\phi}) = \hat{\phi}(0) = \int_{\mathbb{R}^n} \phi(x)\, dx.$$

Thus, $\hat{\delta_0} = 1$, i.e., the distribution induced by the constant function 1. Further,

$$\widehat{\left(\frac{\partial \delta_0}{\partial x_k}\right)} = \imath \xi_k \hat{\delta_0} = \imath \xi_k.$$

The $\hat{1} = \delta_0$ follows by the Fourier inversion formula.

1.7 Fundamental Solution

Let $L = \sum_{|\alpha| \leq n} c_\alpha D^\alpha$ be a n-th order linear differential operator with constant coefficients. Thus, any differential equation is of the form $LT = S$, in the distribution sense for any given distribution S and T is the distribution solution of the differential equation.

Definition 1.7.1. *A distribution K is said to be a* fundamental solution *of the operator L if $L(K) = \delta_0$, where δ_0 is the Dirac distribution at $\{0\}$.*

The existence of fundamental solution of a linear differential operator for constant coefficients is the famous result of Malgrange and Ehrenpreis (cf. [Rud91]).

Suppose we wish to solve $LT = S$ for a given distribution S, we first find the fundamental solution $LK = \delta_0$. Then $K * S$ (as long as the convolution makes sense) is a solution of the equation $L(\cdot) = S$, because $L(K * S) = LK * S = \delta_0 * S = S$. The fundamental solution is not unique because any solution U of the homogeneous equation $L(U) = 0$ can be added to the fundamental solution K to obtain other fundamental solutions, since $L(U + K) = L(U) + L(K) = \delta_0$.

Definition 1.7.2. *A differential operator $L = \sum_{|\alpha| \leq n} c_\alpha D^\alpha$ with $c_\alpha \in C^\infty$ is said to be* hypoelliptic *whenever $Lu \in C^\infty$ implies that u is C^∞.*

Theorem 1.7.3. *For a differential operator with constant coefficients, the following are equivalent:*

(i) L is hypoelliptic.

(ii) Every fundamental solution of L is C^∞ on $\mathbb{R}^n \setminus \{0\}$.

(iii) Some fundamental solution of L is C^∞ on $\mathbb{R}^n \setminus \{0\}$.

1.8 Further Reading

Most of the concepts discussed in this chapter can be found in detail in [Kes89, Rud91, Hor70, Tre70].

Chapter 2

Sobolev Spaces

In this chapter we develop a class of functional spaces which form the right setting to study partial differential equations. Recall that $\Omega \subset \mathbb{R}^n$ is open and not necessarily bounded. The situations where Ω needs to be bounded will be specified clearly. Before we describe Sobolev spaces in detail, we introduce a classical functional space very useful in the theory of Sobolev spaces.

2.1 Space of Continuous Functions

Let $C(X)$ denote the class of all real valued continuous functions on the topological space X.

Definition 2.1.1. *A sequence $\{f_n\} \subset C(X)$ is said to converge* uniformly *to f if $\lim_{n\to\infty} \|f_n - f\|_\infty = 0$, where $\|f\|_\infty = \sup_{x\in X} |f(x)|$. The corresponding topology is called* uniform topology.

Exercise 43. For a compact topological space X, $C(X)$ is a Banach space.

Exercise 44. Let $I \subset \mathbb{R}$ be a closed bounded interval of $\mathbb{R}$. If $\{f_n\}$ is a monotone sequence of continuous real valued functions on I which converge point-wise to a continuous function f, then the convergence is uniform on I.

Lemma 2.1.2. *There is a sequence of polynomials $\{p_n\}$ which converge uniformly to $|x|$ on $[-1, 1]$.*

Proof. Let $p_0 = 1$ and

$$p_{n+1}(x) = \frac{1}{2}(x^2 + 2p_n(x) - p_n^2(x)) \quad \forall n = 0, 1, 2, \ldots.$$

Note that each p_n is a polynomial. Further, the following recursive relations hold

$$p_n(x) - p_{n+1}(x) = \frac{1}{2}(p_n^2(x) - x^2)$$

and

$$p_{n+1} - |x| = \frac{1}{2}(x^2 - 2|x| + 2p_n - p_n^2) = \frac{1}{2}\left[(1-|x|)^2 - (1-p_n)^2\right]$$

Since $|x| \leq p_0 = 1$, we have $|x| \leq p_1 \leq p_0 = 1$. By induction, we have $|x| \leq p_{n+1} \leq p_n$ for all n. Hence $p_n(x)$ converges for every $x \in [-1,1]$ (decreasing and bounded below). Set $p(x) := \lim_n p_n(x)$, then using the recursive formula $p(x) = \frac{1}{2}(x^2 + 2p(x) - p^2(x))$ we get $p^2(x) = x^2$. Since p is limit of positive sequences $p \geq 0$ and hence $p(x) = |x|$. The convergence is uniform because the sequence is monotone. □

Lemma 2.1.3. *For any $c \in \mathbb{R}$, there exists a sequence $\{p_n\}$ of polynomials which converge to $|x-c|$ uniformly on every compact subset of $\mathbb{R}$.*

Proof. In $[-1,1]$ there is a sequence P_n such that $|P_n(x) - |x|| < 1/n^2$ for all $x \in [-1,1]$. Define $p_n(x) = nP_n[(x-c)/n]$, then

$$|p_n(x) - |x-c|| = |nP_n[(x-c)/n] - |x-c|| = n|P_n[(x-c)/n] - |x-c|/n| < 1/n$$

for all $|x-c|/n < 1$ or, equivalently, $x \in n[c-1, c+1]$. □

Definition 2.1.4. *A subset $A \subset C(X)$ is said to* separate points *of X if, for any $x, y \in X$, such that $x \neq y$ there exists $f \in A$ such that $f(x) \neq f(y)$.*

Lemma 2.1.5. *Let $A \subset C(X)$ satisfy the following properties:*

(i) A is a vector (linear) subspace of $C(X)$;

(ii) every constant function is in A; and

(iii) A separates points.

Then, for any $x, y \in X$ with $x \neq y$ and $a, b \in \mathbb{R}$, there exists a $f \in A$ such that $f(x) = a$ and $f(y) = b$.

Proof. Since A separates points, there is a $g \in C(X)$ such that $g(x) = \alpha$ and $g(y) = \beta$ and $\alpha \neq \beta$. Choose scalars $s := \frac{b-a}{\beta-\alpha}, t := \frac{\beta a - \alpha b}{\beta - \alpha} \in \mathbb{R}$ (motivation for the choice will be clear soon), then $sg + t \in A$ because A is a linear space and the function $1 \in A$. Set $f := sg + t$. Then, note that $f(x) = s\alpha + t = a$ and $f(y) = s\beta + t = b$. □

We remark that in the above result a and b could be same and not necessarily $a \neq b$.

Theorem 2.1.6. *Let X be a compact topological space and $A \subset C(X)$ satisfies the properties as in Lemma 2.1.5 and also is a lattice, i.e., $f \vee g \in A$ and $f \wedge g \in A$ whenever $f, g \in A$. Then A is dense in $C(X)$ under the uniform topology.*

Proof. Let $f \in C(X)$. Given $\varepsilon > 0$, we must get an $g \in A$ such that $\|f - g\| < \varepsilon$. Let $x, y \in X$ be such that $x \neq y$ and apply Lemma 2.1.5 for $a := f(x)$ and $b := f(y)$. Thus, we have a $g_{xy} \in A$ such that $g_{xy}(x) = f(x)$ and $g_{xy}(y) = f(y)$. Fix $x \in X$ and for each $y \in X$ with $y \neq x$, by the continuity of $g_{xy} - f$ at y, for the given $\varepsilon > 0$, there is an open set $U_{xy} \in X$ such that $|g_{xy}(z) - f(z)| < \varepsilon$ for all $z \in U_{xy}$. In particular, $g_{xy}(z) < f(z) + \varepsilon$ for all $z \in U_{xy}$. For the fixed $x \in X$, the open sets U_{xy} form an open cover of X and, since X is compact, we have finite collection of $\{y_i\}_{i=1}^n \subset X$ such that $X = \cup_{i=1}^n U_{xy_i}$. Set $g_x := g_{xy_1} \wedge \cdots \wedge g_{xy_n}$, then $g_x \in A$. Since $g_{xy_i}(z) < f(z) + \varepsilon$, we have $g_x(z) < f(z) + \varepsilon$ for all $z \in X$. Moreover, $g_x(x) = f(x)$.

Now, for each fixed $x \in X$, by the continuity of $g_x - f$ at x, for the given $\varepsilon > 0$, there is an open set $V_x \in X$ such that $|g_x(z) - f(z)| < \varepsilon$ for all $z \in V_x$. In particular, $g_x(z) > f(z) - \varepsilon$ for all $z \in V_x$. The open sets V_x form an open cover of X and, since X is compact, we have finite collection of $\{x_i\}_{i=1}^m \subset X$ such that $X = \cup_{i=1}^m V_{x_i}$. Set $g := g_{x_1} \vee \cdots \vee g_{x_n}$, then $g \in A$. Since $g_{x_i}(z) > f(z) - \varepsilon$, we have $g(z) > f(z) - \varepsilon$ for all $z \in X$. Therefore $|f(z) - g(z)| < \varepsilon$ for all $z \in X$ and hence $\|f - g\| < \varepsilon$. □

Exercise 45. A linear subspace $A \subset C(X)$ is a lattice iff $f \in A$ implies $|f| \in A$.

Proof. If A is a lattice and $f \in A$, then $|f| = f \vee (-f)$. Conversely, if $|f| \in A$ whenever $f \in A$, then

$$f \vee g = \frac{f+g}{2} + \frac{|f-g|}{2} \text{ and } f \wedge g = \frac{f+g}{2} - \frac{|f-g|}{2}.$$

□

Theorem 2.1.7 (Stone-Weierstrass)**.** *Let X be a compact topological space and $A \subset C(X)$ satisfies the properties as in Lemma 2.1.5 and, in addition, satisfies the property that $fg \in A$ whenever $f, g \in A$. Then A is dense in $C(X)$ under the uniform topology.*

Proof. We first consider the closure of A and denote it as $\overline{A}$. Note that $\overline{A}$ satisfies all the hypotheses of A. It is enough to show that $\overline{A}$ is a lattice or, equivalently, $|f| \in \overline{A}$ whenever $f \in \overline{A}$. We introduce the notation

$$p(f) := a_n f^n + a_{n-1} f^{n-1} + \cdots + a_1 f + a_0$$

for any real polynomial $p(x) = \sum_{i=0}^{n} a_i x^i$. Let us assume $f \in \overline{A}$. We need to show that $|f| \in \overline{A}$. Since $f \in \overline{A}$, we have $p(f) \in \overline{A}$. By Lemma 2.1.3, we have a sequence of polynomials p_n converging uniformly on compact subsets of $\mathbb{R}$ to $|x|$. Thus, we have $p_n(f)$ converge uniformly to $|f|$ on X. Hence $|f|$ is in $\overline{A}$, since $\overline{A}$ is closed. Since $\overline{A}$ satisfies all the hypotheses of A, by Theorem 2.1.6, $\overline{A}$ is dense in $C(X)$. Thus, $\overline{A} = C(X)$ and hence A is dense in $C(X)$. □

Corollary 2.1.8 (Weierstrass Approximation)**.** *The polynomial functions are dense in $C[a, b]$ with the uniform topology.*

Corollary 2.1.9. *Every continuous function on $[-\pi, \pi]$ can be approximated uniformly by trigonometric polynomials.*

Definition 2.1.10. *Let X be a topological space. A set $E \subset X$ is said to be totally bounded if, for every given $\varepsilon > 0$, there exists a finite collection of points $\{x_1, x_2, \cdots, x_n\} \subset X$ such that $E \subset \cup_{i=1}^{n} B_\varepsilon(x_i)$.*

Exercise 46. If $E \subset X$ is totally bounded then $E^n \subset X^n$ is also totally bounded.

Definition 2.1.11. *A subset $A \subset C(X)$ is said to be* bounded *if there exists a $M \in \mathbb{N}$ such that $\|f\|_\infty \leq M$ for all $f \in A$.*

Definition 2.1.12. *A subset $A \subset C(X)$ is said to be* equicontinuous *at $x_0 \in X$ if, for every given $\varepsilon > 0$, there is an open set U of x_0 such that*

$$|f(x) - f(x_0)| < \varepsilon \quad \forall x \in U; f \in A.$$

A is said to be equicontinuous *if it is equicontinuous at every point of X.*

Theorem 2.1.13. *Let X be a compact topological space and $A \subset C(X)$. If A is totally bounded then A is equicontinuous.*

Proof. Let A be totally bounded. Then, for given $\varepsilon > 0$, there is a collection of $\{f_1, f_2, \cdots, f_m\} \subset C(X)$ such that $A \subset \cup_{j=1}^m B_{\varepsilon/3}(f_j)$. By the continuity of f_j, for each $x \in X$, there is an open set U_j^x containing x such that $|f_j(y) - f_j(x)| < \varepsilon/3$ for all $y \in U_j^x$. Let $U_x := \cap_{j=1}^m U_j^x$ which is an open set containing x. Now, for any $f \in A$, choose j such that $f \in B_{\varepsilon/3}(f_j)$. Then, for all $y \in U_x$, we have

$$|f(x) - f(y)| \leq |f(x) - f_j(x)| + |f_j(x) - f_j(y)| + |f_j(y) - f(y)| < \varepsilon.$$

The first and third term is smaller that $\varepsilon/3$, by the total boundedness of A and the second term is smaller than $\varepsilon/3$ by the continuity of f_j. Hence A is equicontinuous. □

Corollary 2.1.14 (part of Ascoli-Arzela Theorem)**.** *Let X be a compact topological space. If a subset $A \subset C(X)$ is compact then A is closed and equicontinuous.*

Proof. Since $C(X)$ is a metric space and A is compact we have that A is closed and totally bounded. By above theorem, A is equicontinuous. □

The converse of the Theorem proved above is true with some restriction on the range.

Theorem 2.1.15. *Let X be a compact topological space and Y be a totally bounded metric space. If a subset $A \subset C(X, Y)$ is equicontinuous then A is totally bounded.*

Proof. Let A be equicontinuous and $\varepsilon > 0$. Then, for each $x \in X$, there is a open set U_x containing x such that

$$|f(y) - f(x)| < \frac{\varepsilon}{3} \quad \forall y \in U_x; f \in A.$$

Since X is compact, there is a finite set of points $\{x_i\}_1^n \subset X$ such that $X = \cup_{i=1}^n U_{x_i}$. Define the subset E_A of Y^n as,

$$E_A := \{(f(x_1), f(x_2), \cdots, f(x_n)) \mid f \in A\}$$

which is endowed with the product metric, i.e.,

$$d(y, z) = \max_{1 \leq i \leq n} \{|y_i - z_i|\}$$

where $y, z \in Y^n$ are n-tuples. Since Y is totally bounded, Y^n is also totally bounded (cf. Exercise 46). Thus, E_A is totally bounded and there are m number of n-tuples, $y_j := (f_j(x_1), f_j(x_2), \cdots, f_j(x_n)) \in Y^n$, for each $1 \leq j \leq m$, such that $E_A \subset \cup_{j=1}^m B_{\varepsilon/3}(k_j)$. For any $f \in A$, there is a j such that $d(y_j, z_f) < \frac{\varepsilon}{3}$ where $z_f = (f(x_1), f(x_2), \cdots, f(x_n))$. In particular, given any $f \in A$, there is a j such that, for all $1 \leq i \leq n$,

$$|f_j(x_i) - f(x_i)| < \frac{\varepsilon}{3}.$$

Given $f \in A$, fix the j as chosen above.Now, for any given $x \in X$, there is a i such that $x \in U_{x_i}$. For this choice of i, j, we have

$$|f(x) - f_j(x)| \leq |f(x) - f(x_i)| + |f(x_i) - f_j(x_i)| + |f_j(x_i) - f_j(x)|$$

The first and third term is smaller that $\varepsilon/3$ by the continuity of f and f_j, respectively, and the second term is smaller than $\varepsilon/3$ by choice of f_j. Hence A is totally bounded, i.e., $A \subset \cup_{j=1}^m B_\varepsilon(f_j)$, equivalently, for any $f \in A$ there is a j such that $\|f - f_j\|_\infty < \varepsilon$. □

Lemma 2.1.16. *Let X be compact topological space. If $A \subset C(X)$ is bounded then there is a compact subset $K \subset \mathbb{R}$ such that $f(x) \in K$ for all $f \in A$ and $x \in X$.*

Proof. Choose an element $g \in A$. Since A is bounded in the uniform topology, there is a M such that $\|f - g\|_\infty < M$ for all $f \in A$. Since X is compact, $g(X)$ is compact. Hence there is a $N > 0$ such that $g(X) \subset [-N, N]$. Then $f(X) \subset [-M - N, M + N]$ for all $f \in A$. Set $K := [-M - N, M + N]$ and we are done. □

Corollary 2.1.17 (other part of Ascoli-Arzela Theorem)**.** *Let X be a compact topological space. If a subset $A \subset C(X)$ is closed, bounded and equicontinuous then A is compact.*

Proof. Since A is bounded, by Lemma above, we have $A \subset C(X, K) \subset C(X)$ for some compact subset $K \subset \mathbb{R}$. Then, by the Theorem above, A is totally bounded. Since A is a closed and totally bounded subset of the metric space $C(X)$, A is compact. □

2.2 Hölder Spaces

In this section, we introduce some class of function spaces in $C^k(\Omega)$, for all integers $k \geq 0$, which can viewed in some sense as spaces of "fractional" derivatives. For any $\gamma \in (0, \infty)$ and $x_0 \in \Omega$, a function $u : \Omega \to \mathbb{R}$ satisfies at x_0, for all $x \in \Omega$, the estimate

$$|u(x) - u(x_0)| = \frac{|u(x) - u(x_0)|}{|x - x_0|^\gamma}|x - x_0|^\gamma \leq \sup_{\substack{x \in \Omega \\ x \neq x_0}} \left\{ \frac{|u(x) - u(x_0)|}{|x - x_0|^\gamma} \right\} |x - x_0|^\gamma.$$

However, it is not always necessary that the supremum is finite. Note that the modulus in the numerator and denominator are in $\mathbb{R}$ and $\mathbb{R}^n$, respectively.

Definition 2.2.1. *Let $\gamma \in (0, 1]$. We say a function $u : \Omega \to \mathbb{R}$ is* Hölder continuous *of exponent γ at $x_0 \in \Omega$, if*

$$p_\gamma(u)(x_0) := \sup_{\substack{x \in \Omega \\ x \neq x_0}} \left\{ \frac{|u(x) - u(x_0)|}{|x - x_0|^\gamma} \right\} < +\infty.$$

Note that any Hölder continuous function at $x_0 \in \Omega$ satisfies the estimate

$$|u(x) - u(x_0)| \leq p_\gamma(u)(x_0)|x - x_0|^\gamma, \quad \text{for } x \in \Omega. \tag{2.2.1}$$

The constant $p_\gamma(u)(x_0)$ may depend on u, Ω, γ and x_0. It follows from (2.2.1) that any Hölder continuous function at x_0, for any exponent γ, is also continuous at x_0. Because, for every $\varepsilon > 0$, we can choose $\delta = \left[\frac{\varepsilon}{p_\gamma(u)(x_0)} \right]^{1/\gamma}$. The case when $\gamma = 1$ corresponds to the Lipshitz continuity of u at x_0.

Definition 2.2.2. *Let $\gamma \in (0, 1]$. We say a function $u : \Omega \to \mathbb{R}$ is* uniformly Hölder continuous *of exponent γ, if*

$$p_\gamma(u) := \sup_{\substack{x, y \in \Omega, \\ x \neq y}} \left\{ \frac{|u(x) - u(y)|}{|x - y|^\gamma} \right\} < +\infty \tag{2.2.2}$$

and is denoted by $C^{0,\gamma}(\Omega)$. The quantity $p_\gamma(u)$ is called the γ-th Hölder coefficient *of u. If the Hölder coefficient is finite on every compact subsets of Ω, then u is said to be* locally *Hölder continuous with exponent γ, denoted as $C^{0,\gamma}_{loc}(\Omega)$.*

Note that $p_\gamma(u) = \sup_{x_0\in\Omega} p_\gamma(u)(x_0)$. In other words, uniformly Hölder continuous function do not include all those that are Hölder continuous at all points of Ω. However, bounded locally Hölder continuous functions are Hölder continuous at all points of Ω.

Example 2.1. The functions $u(x) = |x|^\delta$ on the open ball $B_1(0) \subset \mathbb{R}^n$, for $\delta \in (0, 1]$, is Hölder continuous, with exponent γ, for all $0 < \gamma \le \delta$, but is not Hölder continuous for $\gamma > \delta$. In fact, $|x|^\delta \in C^{0,\delta}(B_1(0))$ for each $\delta \in [0, 1]$. Thus, we have one example from each of the space $C^{0,\delta}(B_1(0))$. We first check the Hölder continuity with $y = 0$. The γ-th Hölder coefficient is

$$\sup_{x\in B_1(0)} \frac{|x|^\delta}{|x|^\gamma} = |x|^{\delta-\gamma} \le 1 \quad \text{for } \gamma \le \delta.$$

For $\gamma > \delta$ the γ-th Hölder coefficient blows up. More generally, for $x \neq 0$ and $|x| > |y|$ (wlog),

$$\frac{\big||x|^\delta - |y|^\delta\big|}{|x-y|^\gamma} = \frac{|x|^{\delta-\gamma}\left|1 - \left(\frac{|y|}{|x|}\right)^\delta\right|}{\left|\frac{x}{|x|} - \frac{y}{|x|}\right|^\gamma}.$$

Since $0 < 1 - \frac{|y|}{|x|} < 1$ and $0 < \delta \le 1$, $\left|1 - \left(\frac{|y|}{|x|}\right)^\delta\right| \le \left|1 - \frac{|y|}{|x|}\right|$. Thus,

$$\frac{|x|^{\delta-\gamma}\left|1 - \left(\frac{|y|}{|x|}\right)^\delta\right|}{\left|\frac{x}{|x|} - \frac{y}{|x|}\right|^\gamma} \le \frac{|x|^{\delta-\gamma}\left|1 - \frac{|y|}{|x|}\right|}{\left|\frac{x}{|x|} - \frac{y}{|x|}\right|^\gamma} \le \frac{||x| - |y||}{|x|^{1-\gamma}|x-y|^\gamma}.$$

The last inequality is true for $\gamma \le \delta$ and we have used $|x|^{\delta-\gamma} \le 1$. For one dimension, the last quantity is equal to 1 because

$$\frac{||x| - |y||}{|x|^{1-\gamma}|x-y|^\gamma} = \frac{|x|\left|1 - \frac{|y|}{|x|}\right|}{|x||1 - \frac{y}{x}|^\gamma} \le \frac{\left|1 - \frac{y}{x}\right|}{|1 - \frac{y}{x}|} = 1.$$

For higher dimensions, INCOMPLETE!!!

Example 2.2. The Cantor function $f_C \in C^{0,\gamma}([0, 1])$, i.e., is Hölder continuous with exponent $0 < \gamma \le \log_3 2$.[1] Geometrically, the graph of Hölder continuous function have fractal appearance which increases with smaller γ.

[1] $\log_3 2 \approx 0.6309\ldots$

The case $\gamma = 0$ corresponds to bounded functions on Ω and hence is ignored as an possible exponent. In fact, *continuous* functions in $C^{0,0}(\Omega)$ can be identified with the space $C_b(\Omega)$ of bounded continuous functions on Ω. The space $C(\Omega)$ can be identified with continuous functions in $C^{0,0}_{\text{loc}}(\Omega)$, the space of all locally Hölder continuous with exponent $\gamma = 0$. The case $\gamma > 1$ is also ignored because only constant functions, on each connected component of Ω, can satisfy Hölder estimate with exponent greater than one.

Exercise 47. If $u \in C^{0,\gamma}(\Omega)$ with exponent $\gamma > 1$, then u is constant in each of the connected component of Ω.

Proof. Let $u \in C^{0,\gamma}(\Omega)$. For any $x \in \Omega$,

$$|u_{x_i}(x)| = \lim_{t\to 0} \frac{1}{|t|}|u(x+te_i)-u(x)| \leq \lim_{t\to 0} \frac{p_\gamma(u)(x)}{|t|}|t|^\gamma = \lim_{t\to 0} p_\gamma(u)(x)|t|^{\gamma-1} = 0.$$

Therefore, the gradient $Du(x) = 0$ for all $x \in \Omega$ and hence u is constant in each connected components of Ω. □

The case $\gamma = 1$ corresponds to u being *Lipschitz* continuous. The space of all Lipschitz functions is denoted as $\text{Lip}(\Omega) = C^{0,1}(\Omega)$.

Exercise 48. If Ω has finite diameter, then for any $0 < \gamma < \delta \leq 1$ we have $C^{0,1}(\Omega) \subseteq C^{0,\delta}(\Omega) \subsetneq C^{0,\gamma}(\Omega) \subsetneq C(\Omega)$.

Proof. The last inclusion is trivial because every $u \in C^{0,\gamma}(\Omega)$ is uniformly continuous by choosing $\delta = \left[\frac{\varepsilon}{p_\gamma(u)}\right]^{1/\gamma}$, for any given $\varepsilon > 0$. Thus, $C^{0,\gamma}(\Omega) \subsetneq C(\Omega)$ for all $0 < \gamma \leq 1$. On the other hand, let $u \in C^{0,\delta}(\Omega)$. Consider

$$\begin{aligned} p_\gamma(u) &= \sup_{\substack{x,y\in\Omega \\ x\neq y}} \left\{ \frac{|u(x)-u(y)|}{|x-y|^\gamma} \right\} \\ &= \sup_{\substack{x,y\in\Omega \\ x\neq y}} \left\{ \frac{|u(x)-u(y)|}{|x-y|^\delta} |x-y|^{\delta-\gamma} \right\} \\ &\leq \text{diam}(\Omega)^{\delta-\gamma} p_\delta(u). \end{aligned}$$

Thus, $p_\gamma(u) < +\infty$ and $u \in C^{0,\gamma}(\Omega)$. The inclusions are strict by Example 2.1. If $\gamma < \delta$, then $u(x) = |x|^\gamma$ is in $C^{0,\gamma}(-1,1)$ but not in $C^{0,\delta}(-1,1)$. In particular, $\sqrt{x} \in C^{0,1/2}(-1.1)$ but is not Lipschitz, i.e. do not belong to $C^{0,1}(-1,1)$. □

A natural question at this juncture is: what is the relation between the spaces $C^1(\Omega)$ and $C^{0,1}(\Omega)$. Of course $|x| \in C^{0,1}(-1,1)$ and not in $C(-1,1)$. Given the inclusion relation in above exercise, we would tend to believe that $C^1(\Omega)$ is contained in $C^{0,1}(\Omega)$, but this is not true. In fact, only a subclass of $C^1(\Omega)$ belong to Lipschitz class.

Theorem 2.2.3. *Let Ω be a convex domain. If $u \in C^1(\Omega)$ and $D^\alpha u$ is bounded on Ω, for each $|\alpha| = 1$, then $u \in C^{0,1}(\Omega)$.*

Proof. Let $u \in C^1(\Omega)$ such that there is a $C_0 > 0$ such that $|\nabla u(x)| \leq C_0$ for all $x \in \Omega$. For any two given points $x, y \in \Omega$, we define the function $F : [0,1] \to \mathbb{R}$ as $F(t) = u((1-t)x + ty)$. F is well defined due to the convexity of Ω. Since u is differentiable, F is differentiable in $(0,1)$. Thus, for all $t \in (0,1)$, $F'(t) = \nabla u[(1-t)x+ty]\cdot(y-x)$. Therefore, $|F'(t)| \leq C_0|y-x|$. By mean value theorem, there is a $\xi \in (0,1)$ such that $F'(\xi) = F(1) - F(0)$. Thus,

$$|u(y) - u(x)| = |F(1) - F(0)| = |F'(\xi)| \leq C_0|y - x|.$$

Hence, u is Lipschitz continuous. □

What fails in the converse of the above theorem is the fact that a Lipschitz function can fail to be differentiable. If we assume, in addition to Lipschitz continuity, that u is differentiable then $u \in C^1$ and derivative is bounded.

Exercise 49. Prove the above theorem when Ω is path connected.

Theorem 2.2.4. *For any $\gamma \in (0,1]$, the space $C^{0,\gamma}(\Omega)$ can be identified with the space $C^{0,\gamma}(\overline{\Omega})$.*

Proof. The space $C^{0,\gamma}(\overline{\Omega})$ can be identified with a subset of $C^{0,\gamma}(\Omega)$, by identifying any $u \in C^{0,\gamma}(\overline{\Omega})$ with $u\,|_\Omega \in C^{0,\gamma}(\Omega)$. On the other hand, for any $u \in C^{0,\gamma}(\Omega)$ we wish to extend it uniquely to a function $\tilde{u} \in C^{0,\gamma}(\overline{\Omega})$ such that $\tilde{u}\,|_\Omega = u$. Let $u \in C^{0,\gamma}(\Omega)$. Set $\tilde{u}(x) = u(x)$ for all $x \in \Omega$. Now, for any $x_0 \in \partial\Omega$, choose a sequence $\{x_m\} \subset \Omega$ such that $\lim_m x_m = x_0$. Then, we have

$$|u(x_k) - u(x_l)| \leq C|x_k - x_l|^\gamma \to 0 \text{ as } k, l \to \infty.$$

Thus, $\{u(x_m)\}$ is a Cauchy sequence and converges in $\mathbb{R}$. Set $\tilde{u}(x_0) = \lim_m u(x_m)$. We now show that the definition of $\tilde{u}(x_0)$ is independent of

the choice of the sequence. If $\{y_m\} \subset \Omega$ is any other sequence converging to x_0, then

$$|u(x_m) - u(y_m)| \leq C|x_m - y_m|^\gamma \to 0 \text{ as } m \to \infty.$$

Thus, $\tilde{u}(x_0)$ is well-defined for all $x_0 \in \partial\Omega$. Moreover, for all $x, y \in \overline{\Omega}$, we have

$$|\tilde{u}(x) - \tilde{u}(y)| = \lim_m |u(x_m - u(y_m)| \leq C \lim_m |x_m - y_m|^\gamma = C|x - y|^\gamma.$$

Hence, $\tilde{u}$ satisfies the Hölder estimate with exponent γ and is in $C^{0,\gamma}(\overline{\Omega})$. □

The above result is very special to uniformly continuous spaces and, hence, Hölder spaces. For instance, a similar result is not true between $C(\Omega)$ and $C(\overline{\Omega})$. If $\overline{\Omega}$ is compact then every function in $C(\overline{\Omega})$ is bounded, where as $C(\Omega)$ might have unbounded functions. An interesting point in the proof of above result is that the Hölder coefficient are same for the extension, i.e., $p_\gamma(u) = p_\gamma(\tilde{u})$. An advantage of above theorem is that, without loss of generality, we can use the space $C^{0,\gamma}(\Omega)$ and $C^{0,\gamma}(\overline{\Omega})$ interchangeably.

Exercise 50. Show that the Hölder coefficient, p_γ, defines a semi-norm on $C^{0,\gamma}(\Omega)$.

Proof. For any constant function u, $p_\gamma(u) = 0$. Also,

$$\begin{aligned}
p_\gamma(u+v) &= \sup_{\substack{x,y\in\overline{\Omega}\\ x\neq y}} \left\{ \frac{|u(x)+v(x)-u(y)-v(y)|}{|x-y|^\gamma} \right\} \\
&= \sup_{\substack{x,y\in\overline{\Omega}\\ x\neq y}} \left\{ \frac{|u(x)-u(y)+v(x)-v(y)|}{|x-y|^\gamma} \right\} \\
&\leq \sup_{\substack{x,y\in\overline{\Omega}\\ x\neq y}} \left\{ \frac{|u(x)-u(y)|+|v(x)-v(y)|}{|x-y|^\gamma} \right\} \\
&\leq p_\gamma(u) + p_\gamma(v).
\end{aligned}$$

Further $p_\gamma(\lambda u) = |\lambda| p_\gamma(u)$. □

Note that for bounded open subsets Ω, the space $C^{0,\gamma}(\Omega)$ inherits the uniform topology from $C(\Omega)$.

Exercise 51. For any bounded open set $\Omega \subset \mathbb{R}^n$, show that $C^{0,\gamma}(\Omega)$ is dense in $C(\Omega)$.

To make the space $C^{0,\gamma}(\Omega)$ complete on bounded Ω, we define the γ-Hölder norm on $C^{0,\gamma}(\overline{\Omega})$ as

$$\|u\|_{C^{0,\gamma}(\Omega)} := \|u\|_\infty + p_\gamma(u),$$

where $\|u\|_\infty := \sup_{x\in\overline{\Omega}} |u(x)|$.

Exercise 52. For any bounded open set $\Omega \subset \mathbb{R}^n$, show that $\|\cdot\|_{C^{0,\gamma}(\Omega)}$ is a norm on $C^{0,\gamma}(\Omega)$.

Theorem 2.2.5. *For any bounded open set $\Omega \subset \mathbb{R}^n$, the space $C^{0,\gamma}(\Omega)$ is a Banach space with norm $\|\cdot\|_{C^{0,\gamma}(\Omega)}$.*

Proof. We need to prove the completeness of the space $C^{0,\gamma}(\Omega)$ w.r.t the norm $\|\cdot\|_{C^{0,\gamma}(\Omega)}$. Let $\{u_m\}$ be a Cauchy sequence in $C^{0,\gamma}(\Omega)$, then $\{u_m\} \subset C(\Omega)$ is Cauchy w.r.t the supremum norm. Thus, there is a $u \in C(\Omega)$ such that $\|u_m - u\|_\infty \to 0$, as $m \to \infty$. We first show that $u \in C^{0,\gamma}(\Omega)$. For $x, y \in \Omega$ with $x \neq y$, consider

$$\frac{|u(x)-u(y)|}{|x-y|^\gamma} = \lim_m \frac{|u_m(x)-u_m(y)|}{|x-y|^\gamma} \leq \limsup_m p_\gamma(u_m) \leq \lim_m \|u_m\|_{C^{0,\gamma}(\Omega)}.$$

Since $\{u_m\}$ is Cauchy, $\lim_m \|u_m\|_{C^{0,\gamma}(\Omega)} < \infty$. Hence $u \in C^{0,\gamma}(\Omega)$. Finally, we show that the sequence $\{u_m\}$ converges to u in $C^{0,\gamma}(\Omega)$ w.r.t the γ-Hölder norm. Consider,

$$\begin{aligned}\frac{|u_m(x)-u(x)-u_m(y)+u(y)|}{|x-y|^\gamma} &= \lim_k \frac{|u_m(x)-u_k(x)-u_m(y)+u_k(y)|}{|x-y|^\gamma} \\ &\leq \limsup_k p_\gamma(u_m-u_k).\end{aligned}$$

Therefore, $\lim_m p_\gamma(u_m - u) \leq \lim_m \limsup_k p_\gamma(u_m - u_k)$ and the RHS converges to 0 since the sequence is Cauchy. Hence, $\|u_m - u\|_{C^{0,\gamma}(\Omega)} \to 0$. □

Exercise 53. The space $C^{0,\gamma}(\Omega)$ norm is not separable.[2]

For an unbounded open set Ω, we consider a sequence of exhaustion compact sets, as described in §1.3.4, and make the space $C^{0,\gamma}(\Omega)$ a Fréchet space.

[2]The uniform norm in $C(\Omega)$ is separable

Definition 2.2.6. *Let X and Y be Banach spaces. A continuous (bounded) linear operator $L : X \to Y$ is said to be* compact *if L maps every bounded subset of X to precompact (closure compact) subsets of Y. Equivalently, L maps bounded sequences of X to sequences in Y that admit convergent subsequences.*

Definition 2.2.7. *Let X and Y be Banach spaces such that $X \subset Y$. We say X is* continuously imbedded *in Y (denoted as $X \hookrightarrow Y$), if there is a constant such that*

$$\|x\|_Y \leq C\|x\|_X \quad \forall x \in X.$$

Further, we say X is compactly imbedded *in Y (denoted as $X \subset\subset Y$) if in addition to being continuously imbedded in Y, every bounded set in X (w.r.t the norm in X) is compact in Y (w.r.t the norm in Y).*

Theorem 2.2.8. *Let Ω be a bounded open subset of $\mathbb{R}^n$. For any $0 < \gamma < \delta \leq 1$, the inclusion map $I : C^{0,\delta}(\Omega) \to C^{0,\gamma}(\Omega)$ is continuous and compact. Further, the inclusion map $I : C^{0,\gamma}(\Omega) \to C(\Omega)$ is compact, for all $0 < \gamma \leq 1$.*

Proof. The continuity of I follows from the inequality proved in Exercise 48 and hence I is a bounded linear operator. We need to show I is compact. Let $\{u_m\}$ be a sequence bounded in $C^{0,\delta}(\Omega)$. Without loss of generality, we can assume that $\|u_m\|_{C^{0,\delta}(\Omega)} \leq 1$. Therefore, $p_\delta(u_m) \leq 1$ for all m which implies that $\{u_m\}$ is an equicontinuous sequence in $C(\Omega)$. By Arzelà-Ascoli theorem, there exists a subsequence $\{u_{m_k}\}$ of $\{u_m\}$ and a $u \in C(\Omega)$ such that $\|u_{m_k} - u\|_\infty \to 0$, as $k \to \infty$. The $u \in C(\Omega)$ is, in fact, in $C^{0,\delta}(\Omega) \subset C^{0,\gamma}(\Omega)$ because

$$\frac{|u(x) - u(y)|}{|x - y|^\delta} = \lim_k \frac{|u_{m_k}(x) - u_{m_k}(y)|}{|x - y|^\delta} \leq 1$$

and, hence, $p_\delta(u) \leq 1$. We now show that the u_{m_k} converges to u in $C^{0,\gamma}(\Omega)$. For simplicity, set $v_k := u_{m_k} - u$ and we will show that $\{v_k\}$ converges to 0 in $C^{0,\gamma}(\Omega)$. Obviously, $\|v_k\|_\infty \to 0$ thus it is enough to show that $p_\gamma(v_k) \to 0$. Note that, for every given $\varepsilon > 0$, we have $p_\gamma(v_k) \leq S_k^\varepsilon + T_k^\varepsilon$, where

$$S_k^\varepsilon = \sup_{\substack{x,y\in\Omega \\ x\neq y;|x-y|\leq\varepsilon}} \left\{ \frac{|v_k(x) - v_k(y)|}{|x - y|^\gamma} \right\}$$

and

$$T_k^\varepsilon = \sup_{\substack{x,y\in\Omega \\ |x-y|>\varepsilon}} \left\{ \frac{|v_k(x) - v_k(y)|}{|x - y|^\gamma} \right\}.$$

Consider

$$S_k^\varepsilon = \sup_{\substack{x,y\in\Omega \\ x\neq y;|x-y|\leq\varepsilon}} \left\{ \frac{|v_k(x)-v_k(y)|}{|x-y|^\delta}|x-y|^{\delta-\gamma} \right\} \leq \varepsilon^{\delta-\gamma} p_\delta(v_k) \leq 2\varepsilon^{\delta-\gamma}.$$

Similarly,

$$T_k^\varepsilon \leq 2\varepsilon^{-\gamma}\|v_k\|_\infty.$$

Hence, $\limsup_k p_\gamma(v_k) \leq 2\varepsilon^{\delta-\gamma} + 2\varepsilon^{-\gamma}\limsup_k \|v_k\|_\infty = 2\varepsilon^{\delta-\gamma} + 0$. Since ε can be made as small as possible, we have $p_\gamma(v_k) \to 0$. □

We extend the Hölder continuous functions in to all differentiable class of functions. We denote by $C^{k,\gamma}(\Omega)$ the space of all $C^k(\Omega)$, k-times continuously differentiable, functions such that $D^\alpha u$ is Hölder continuous with exponent γ, i.e., $D^\alpha u \in C^{0,\gamma}(\Omega)$ for all $|\alpha| = k$. For a bounded open set Ω, we give the γ-th Hölder norm on $C^{k,\gamma}(\Omega)$ as

$$\|u\|_{C^{k,\gamma}(\Omega)} := \sum_{|\alpha|=0}^{k} \|D^\alpha u\|_\infty + \sum_{|\alpha|=k} p_\gamma(D^\alpha u). \tag{2.2.3}$$

It is enough to consider the Hölder coefficient of only the k-th derivative because this is enough to complete the space.

Exercise 54. For any bounded open set $\Omega \subset \mathbb{R}^n$, show that $\|\cdot\|_{C^{k,\gamma}(\Omega)}$ is a norm on $C^{k,\gamma}(\Omega)$. and the space $C^{k,\gamma}(\Omega)$ is a Banach space with norm $\|\cdot\|_{C^{k,\gamma}(\Omega)}$.

Proof. Let $\{u_m\}$ be a Cauchy sequence in $C^{k,\gamma}(\Omega)$. Then, $\{u_m\} \subset C^k(\Omega)$ is Cauchy in the supremum norm. Thus, there is a $u \in C^k(\Omega)$ such that $\|u_m - u\|_{C^k(\Omega)} \to 0$, as $m \to \infty$. The fact that $u \in C^{k,\gamma}(\Omega)$ and that u is a limit, in the γ-Hölder norm, of the Cauchy sequence is similar to case $k = 0$ proved in Theorem 2.2.5. □

As seen before, for an unbounded open set Ω, we consider a sequence of exhaustion compact sets, as described in §1.3.4, and make the space $C^{k,\gamma}(\Omega)$ a Fréchet space. We end this section with a final result on the inclusion between different order Hölder spaces.

Theorem 2.2.9. *Let Ω be a bounded, convex open subset of $\mathbb{R}^n$. For any $0 < \gamma, \delta \leq 1$ and $k, \ell \in \mathbb{N} \cup \{0\}$ such that $k + \gamma < \ell + \delta$, then the inclusion map $I : C^{\ell,\delta}(\Omega) \to C^{k,\gamma}(\Omega)$ is continuous and compact.*

Proof. If $\gamma = \delta$ and $k = \ell$ then there is nothing to prove. Alternately, without loss of generality we assume $0 < \gamma < \delta \leq 1$ and $k < \ell$. Then, we know from Theorem 2.2.8 that $C^{\ell,\delta}(\Omega) \subset C^{\ell,\gamma}(\Omega)$ and the inclusion is continuous and compact. Let $u \in C^{\ell,\gamma}$. We need to show that $u \in C^{k,\gamma}(\Omega)$. i.e., $p_\gamma(D^k u) < \infty$. Note that it is enough to show that $p_1(D^k u) < \infty$. The fact that $u \in C^{\ell,\gamma}$ implies that $u \in C^\ell(\Omega)$ and $D^{\ell-1}u$ is bounded. Thus, by Theorem 2.2.3, we have $u \in C^{\ell-1,1}(\Omega)$ and hence is in $C^{\ell-1,\gamma}(\Omega)$. By repeating the argument for each derivative of $D^{\ell-i}$ for each $i = 1, 2, \ldots, k$, we have the required result. The inclusion $C^{\ell,\gamma}(\Omega) \subset C^{k,\gamma}(\Omega)$ is continuous. The composition of a continuous and compact operator is compact. Thus, the inclusion I is compact. □

2.3 $W^{k,p}(\Omega)$ Spaces

For each $1 \leq p \leq \infty$, we define its *conjugate* exponent q to be,

$$q = \begin{cases} \frac{p}{p-1} & \text{if } 1 < p < +\infty \\ +\infty & \text{if } p = 1 \\ 1 & \text{if } p = +\infty. \end{cases}$$

In short, q is precisely that number corresponding to p for which $1/p + 1/q = 1$.

In this section we shall define spaces analogous to $C^k(\Omega)$ in the spaces $L^p(\Omega)$ using the generalised notion of derivative, viz. weak derivative (cf. Definition 1.5.2). We know that every $u \in L^p(\Omega)$ being locally integrable induces a distribution T_u. Further, the distribution T_u is differentiable for all multi-indices α. However, we have already seen that T_u need not be weakly differentiable. For a fixed multi-index α, if there exists a $v_\alpha \in L^p(\Omega)$ such that $T_{v_\alpha} = D^\alpha T_u$, then we denote the v_α as $D^\alpha u$. We know that such a v_α is unique up to a set of measure zero. As usual, Ω is an open subset of $\mathbb{R}^n$.

Definition 2.3.1. *Let $k \geq 0$ be an integer and $1 \leq p \leq \infty$. The* Sobolev *space $W^{k,p}(\Omega)$ is the subclass of all $u \in L^p(\Omega)$ such that there exists a $v_\alpha \in L^p(\Omega)$, for all $0 \leq |\alpha| \leq k$, such that*

$$\int_\Omega v_\alpha \phi \, dx = (-1)^{|\alpha|} \int_\Omega u D^\alpha \phi \, dx \quad \forall \phi \in \mathcal{D}(\Omega).$$

Equivalently,

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) \mid D^\alpha u \in L^p(\Omega) \quad \forall 0 \leq |\alpha| \leq k\}.$$

Obviously, $W^{0,p}(\Omega) = L^p(\Omega)$. Also, we have the inclusion $W^{k,p}(\Omega) \subsetneq W^{\ell,p}(\Omega)$ for all $\ell < k$. The inclusion is strict as seen from examples below.

Example 2.3. On a bounded subset $\Omega = (-a, a) \subset \mathbb{R}$, the function f defined as

$$f(x) = \begin{cases} 1 & x \in [0, a) \\ 0 & x \in (-a, 0). \end{cases}$$

is in $L^p(-a, a)$, for all p. However, the distributional derivative of f is the Dirac measure at 0, δ_0, which is a singular distribution. Hence $f \notin W^{k,p}(\Omega)$, for all $k > 0$.

Example 2.4. Consider the function g defined as

$$g(x) = \begin{cases} x & x \in (0, a) \\ 0 & x \in (-a, 0]. \end{cases}$$

is in $W^{1,p}(-a, a)$, for all p, since both g and its distributional derivative $Dg = f$ are in $L^p(-a, a)$, for all p (previous example).Hence $g \notin W^{k,p}(\Omega)$, for all $k \geq 2$.

Exercise 55. If $u \in L^p(\Omega)$ is such that $D^\alpha u \in L^p(\Omega)$ for $|\alpha| = k$, then is it true that $D^\alpha u \in L^p(\Omega)$ for all $|\alpha| = 1, 2, \ldots, k-1$?

Exercise 56. Show that the spaces $W^{k,p}(\Omega)$ are all (real) vector spaces.

Exercise 57. Show that $u \mapsto \sum_{|\alpha|=k} \|D^\alpha u\|_p$ defines a semi-norm on $W^{k,p}(\Omega)$ for $k > 0$. We denote the semi-norm by $|u|_{k,p,\Omega}$.

For $1 \leq p < \infty$, we endow the space $W^{k,p}(\Omega)$ with the natural norm,

$$\|u\|_{k,p,\Omega} := \sum_{|\alpha|=0}^{k} \|D^\alpha u\|_p = \sum_{|\alpha|=0}^{k} \left(\int_\Omega |D^\alpha u|^p \right)^{1/p}.$$

For $p = \infty$, we define the norm on $W^{k,\infty}(\Omega)$ to be,

$$\|u\|_{k,\infty,\Omega} = \sum_{|\alpha|=0}^{k} \text{ess sup}_\Omega |D^\alpha u|.$$

Observe that $\|u\|_{0,p,\Omega} = |u|_{0,p,\Omega} = \|u\|_p$, the usual L^p-norm.

Exercise 58. Show that the norm $\|u\|_{k,p,\Omega}$ defined above is equivalent to the norm

$$\left(\sum_{|\alpha|=0}^{k} \|D^\alpha u\|_p^p\right)^{1/p} = \left(\sum_{|\alpha|=0}^{k} \int_\Omega |D^\alpha u|^p\right)^{1/p}, \quad \text{for } 1 < p < \infty.$$

For $p = 1$, the norms are same.

Frequently, we may skip the domain subscript in the norm where there is no confusion on the domain of function. Also, we shall tend to use the symbol $\|\cdot\|_p$ for $\|\cdot\|_{0,p}$ in L^p-spaces. We set $H^k(\Omega) = W^{k,2}(\Omega)$. For $u \in H^k(\Omega)$, we shall denote the norm as $\|\cdot\|_{k,\Omega}$.

Exercise 59. Show that

$$\langle u, v\rangle_{k,\Omega} = \sum_{|\alpha|=0}^{k} \int_\Omega D^\alpha u D^\alpha v, \quad \forall u, v \in H^k(\Omega)$$

defines an inner-product in $H^k(\Omega)$.

Theorem 2.3.2. *For every $1 \le p \le \infty$, the space $W^{k,p}(\Omega)$ is a Banach space. If $1 < p < \infty$, it is reflexive and if $1 \le p < \infty$, it is separable. In particular, $H^k(\Omega)$ is a separable Hilbert space.*

Proof. Let $\{u_m\}$ be a Cauchy sequence in $W^{k,p}(\Omega)$. Thus, by the definition of the norm on $W^{k,p}(\Omega)$, $\{u_m\}$ and $\{D^\alpha u_m\}$, for $1 < |\alpha| \le k$ are Cauchy in $L^p(\Omega)$. Since $L^p(\Omega)$ is complete, there exist function $u, v_\alpha \in L^p(\Omega)$ such that

$$u_m \to u \quad \text{and } D^\alpha u_m \to v_\alpha \quad \forall 1 < |\alpha| \le k.$$

To show that $W^{k,p}(\Omega)$ is complete, it is enough to show that $D^\alpha u = v_\alpha$, for $1 < |\alpha| \le k$. Let $\phi \in \mathcal{D}(\Omega)$, then for each α such that $1 < |\alpha| \le k$, we have

$$\begin{aligned}
\int_\Omega u D^\alpha \phi \, dx &= \lim_{m\to\infty} \int_\Omega u_m D^\alpha \phi \, dx \\
&= \lim_{m\to\infty} (-1)^{|\alpha|} \int_\Omega D^\alpha u_m \phi \, dx \\
&= (-1)^{|\alpha|} \int_\Omega v_\alpha \phi \, dx
\end{aligned}$$

Thus, $D^\alpha u = v_\alpha$, i.e., v_α is the α weak derivative of u.

Note that we have an isometry from $W^{k,p}(\Omega)$ to $(L^p(\Omega))^\beta$, where $\beta := \sum_{i=0}^{k} n^i$ and the norm endowed on $(L^p(\Omega))^\beta$ is $\|u\| := \left(\sum_{i=1}^{\beta} \|u_i\|_p^p\right)^{1/p}$ with $u = (u_1, u_2, \ldots, u_\beta)$. The image of $W^{k,p}(\Omega)$ under this isometry is a closed subspace of $(L^p(\Omega))^\beta$. The reflexivity and separability of $W^{k,p}(\Omega)$ is inherited from reflexivity and separability of $(L^p(\Omega))^\beta$, for $1 < p < \infty$ and $1 \leq p < \infty$, respectively. □

2.4 Smooth Approximations

Recall that in Theorem 1.3.20, we proved that $C_c^\infty(\Omega)$ is dense in $L^p(\Omega)$ for any open subset $\Omega \subseteq \mathbb{R}^n$. We are now interested in knowing if the density of $C_c^\infty(\Omega)$ holds true in the Sobolev space $W^{k,p}(\Omega)$, for any $k \geq 1$[3].

Theorem 2.4.1. *For $1 \leq p < \infty$, $C_c^\infty(\mathbb{R}^n)$ is dense in $W^{k,p}(\mathbb{R}^n)$, for all $k \geq 1$.*

Proof. Let $k \geq 1$ and $u \in W^{k,p}(\mathbb{R}^n)$. The $D^\alpha u \in L^p(\mathbb{R}^n)$ for all $0 \leq |\alpha| \leq k$. By Theorem 1.3.19, we know that $\{\rho_m * u\} \subset C^\infty(\mathbb{R}^n)$ converges to u in the $W^{k,p}(\Omega)$-norm, for all $k \geq 0$. Choose a function $\phi \in C_c^\infty(\mathbb{R}^n)$ such that $\phi \equiv 1$ on $B(0;1)$, $\phi \equiv 0$ on $\mathbb{R}^n \setminus B(0;2)$ and $0 \leq \phi \leq 1$ on $B(0;2) \setminus B(0;1)$. Note that $|D^\alpha \phi|$ is bounded. In fact, $|D^\alpha \phi(x)| \leq 1$ for all $0 \leq |\alpha| \leq k$. Now set $\phi_m = \phi(x/m)$. This is one choice of cut-off functions as introduced in

[3]The case $k = 0$ is precisely the result of Theorem 1.3.20

Theorem 1.3.20. Set $u_m := \phi_m(\rho_m * u)$ in $\mathbb{R}^n$, then

$$\begin{aligned}
\|D^\alpha u_m - D^\alpha u\|_p &\leq \sum_{\beta<\alpha} \left\|C_\alpha (D^{\alpha-\beta}\phi_m)(\rho_m * D^\beta u)\right\|_p \\
&\quad + \sum_{\beta=\alpha} \|\phi_m(\rho_m * D^\alpha u) - D^\alpha u\|_p \\
&\leq \sum_{\beta<\alpha} |C_\alpha| \|D^{\alpha-\beta}\phi_m\|_\infty \|\rho_m * D^\beta u\|_p \\
&\quad + \sum_{\beta=\alpha} \|\phi_m(\rho_m * D^\alpha u - D^\alpha u)\|_p \\
&\quad + \sum_{\beta=\alpha} \|\phi_m D^\alpha u - D^\alpha u\|_p \\
&\leq \frac{M}{m^{|\alpha-\beta|}} \sum_{\beta<\alpha} |C_\alpha| \|D^\beta u\|_p \\
&\quad + \sum_{\beta=\alpha} \|\rho_m * D^\alpha u - D^\alpha u\|_p \\
&\quad + \sum_{\beta=\alpha} \|\phi_m D^\alpha u - D^\alpha u\|_p .
\end{aligned}$$

The first term with $1/m$ goes to zero, the second term tends to zero by Theorem 1.3.19 and the last term goes to zero by Dominated convergence theorem. Since

$$\begin{aligned}
\|u_m - u\|_{k,p} &= \sum_{|\alpha|=0}^{k} \|D^\alpha u_m - D^\alpha u\|_p \\
&= \sum_{|\alpha|=0}^{k} \left\| \sum_{\beta\leq\alpha} C_\alpha (D^{\alpha-\beta}\phi_m)(\rho_m * D^\beta u) - D^\alpha u \right\|_p
\end{aligned}$$

and each term in the sum goes to zero as m increases, we have u_m converges to u in $W^{k,p}(\mathbb{R}^n)$ norm. □

The real trouble in approximating $W^{k,p}(\Omega)$ by $C_c^\infty(\Omega)$ functions is that the approximation approach boundary continuously, if Ω has a boundary.

Theorem 2.4.2. *Let $1 \leq p < \infty$ and $\Omega \subset \mathbb{R}^n$. Then $C_c^\infty(\mathbb{R}^n)$ is dense in $W_{loc}^{k,p}(\Omega)$, i.e., for each $u \in W^{k,p}(\Omega)$ there is a sequence $v_m \in C_c^\infty(\mathbb{R}^n)$ such*

that $\|v_m - u\|_{k,p,\omega}$*, for all* $k \geq 1$ *and for all* $\omega \subset\subset \Omega$ *(relatively compact in* Ω*).*

Proof. Let $u \in W^{k,p}(\Omega)$. Fix ω relatively compact subset of Ω and a α such that $1 \leq |\alpha| \leq k$. Let $\tilde{u}$ denote the extension of u by zero in Ω^c. Choose a $\phi \in C_c^\infty(\Omega)$ such that $\phi \equiv 1$ on ω and $0 \leq \phi \leq 1$ in Ω. Then $\text{supp}(\phi u) \subset \Omega$ and we extend ϕu to all of $\mathbb{R}^n$ by zero in Ω^c and denote it as $\widetilde{\phi u}$. Set $v = \widetilde{\phi u}$ on $\mathbb{R}^n$. Note that $v \in W^{k,p}(\mathbb{R}^n)$. By Theorem 2.4.1, we have the sequence $v_m := \phi_m(\rho_m * v) \in C_c^\infty(\mathbb{R}^n)$ converging to v in α Sobolev norm. Since on ω, $v = u$, we have

$$\|v_m - u\|_{k,p,\omega} \leq \|v_m - v\|_{k,p,\mathbb{R}^n} \to 0.$$

This is true for all relatively compact subset ω of Ω. □

Note that the restriction 'relatively compact set' is only for $k \geq 1$. For the case $k = 0$, the restriction to Ω works, as seen in Theorem 1.3.20. The density of $C_c^\infty(\Omega)$ in $W^{k,p}(\Omega)$ may fail to generalise for an arbitrary *proper* subset $\Omega \subset \mathbb{R}^n$, because a "bad" derivative may be introduced at the boundary while extending by zero outside Ω.

Example 2.5. Let $\Omega = (0,1) \subset \mathbb{R}$ and $u \equiv 1$ on Ω. Then $u \in W^{1,p}(0,1)$. Setting $\tilde{u} = 0$ in $\mathbb{R} \setminus (0,1)$, we see that $\tilde{u} \in L^p(\mathbb{R})$ but not in $W^{1,p}(\mathbb{R})$. Because $D\tilde{u} = \delta_0 - \delta_1$ is not in $L^1_{\text{loc}}(\mathbb{R})$.

Recall the inclusion $C_c^\infty(\Omega) \subset C^\infty(\overline{\Omega}) \subset C^\infty(\Omega)$. The $C^\infty(\overline{\Omega})$ denotes all functions in $C^\infty(\Omega)$ such that all its derivatives can be extended continuously to $\overline{\Omega}$. For an arbitrary subset $\Omega \subseteq \mathbb{R}^n$ the best one can do is the following density result.

Theorem 2.4.3 (Meyers-Serrin)**.** *Let* $1 \leq p < \infty$ *and* $\Omega \subseteq \mathbb{R}^n$*. For* $u \in W^{k,p}(\Omega)$ *and any* $\varepsilon > 0$*, there is a* $\phi \in C^\infty(\Omega)$ *such that* $\|\phi\|_{k,p,\Omega} < \infty$ *and* $\|u - \phi\|_{k,p,\Omega} < \varepsilon$*.*

Proof. For each $m \in \mathbb{N}$, consider the sets

$$\omega_m := \{x \in \Omega \mid |x| < m \text{ and } \text{dist}(x, \partial\Omega) > \frac{1}{m}\}$$

and set $\omega_0 = \emptyset$. Define the collection of open sets $\{U_m\}$ as $U_m := \omega_{m+1} \cap (\overline{\omega}_{m-1})^c$. Note that $\Omega = \cup_m U_m$ is an open covering of Ω. Thus, we choose the C^∞ locally finite partition of unity $\{\phi_m\} \subset C_c^\infty(\Omega)$ such that $\text{supp}(\phi_m) \subset U_m$, $0 \leq \phi_m \leq 1$ and $\sum_m \phi_m = 1$. For the given $u \in W^{k,p}(\Omega)$, note that

$\phi_m u \in W^{k,p}(U_m)$ with $\text{supp}(\phi_m u) \subset U_m$. We extend $\phi_m u$ to all of $\mathbb{R}^n$ by zero outside U_m, i.e.,

$$\widetilde{\phi_m u}(x) = \begin{cases} \phi_m u(x) & x \in U_m \\ 0 & x \in \mathbb{R}^n \setminus U_m. \end{cases}$$

Observe that $\widetilde{\phi_m u} \in W^{k,p}(\mathbb{R}^n)$. Let ρ_δ be the sequence of mollifiers and consider the sequence $\rho_\delta * \widetilde{\phi_m u}$ in $C^\infty(\mathbb{R}^n)$. Support of $\rho_\delta * \widetilde{\phi_m u} \subset U_m + B(0;\delta)$. Note that for all $x \in U_m$, $1/(m+1) < \text{dist}(x, \partial\Omega) < 1/(m-1)$. Thus, for all $0 < \delta < 1/(m+1)(m+2)$[4], $\text{supp}(\rho_\delta * \widetilde{\phi_m u}) \subset \omega_{m+2} \cap (\overline{\omega}_{m-2})^c$, which is compactly contained in Ω. Since $\phi_m u \in W^{k,p}(\Omega)$, we can choose a subsequence $\{\delta_m\}$ going to zero in $(0, 1/(m+1)(m+2)$ such that

$$\|\rho_{\delta_m} * \phi_m u - \phi_m u\|_{k,p,\Omega} = \|\rho_{\delta_m} * \widetilde{\phi_m u} - \phi_m u\|_{k,p,\mathbb{R}^n} < \frac{\varepsilon}{2^m}.$$

Set $\phi = \sum_m \rho_{\delta_m} * \phi_m u$. Note that $\phi \in C^\infty(\Omega)$ and $\|\phi\|_{k,p,\Omega} < \infty$. Since every U_m intersects U_{m-1} and U_{m+1}, we at most have three non-zero terms in the sum for each $x \in U_m$, i.e.,

$$\phi(x) = \sum_{i=-1}^{1} (\rho_{\delta_{m+i}} * \phi_{m+i})(x) \quad x \in U_m.$$

Therefore,

$$\|u - \phi\|_{k,p,\Omega} = \|\sum_m (u\phi_m - \rho_{\delta_m} * \phi_m u)\|_{k,p,\Omega} \leq \sum_m \|\phi_m u - \rho_{\delta_m} * \phi_m u\|_{k,p,\Omega} < \varepsilon.$$

This proves the density of $C^\infty(\Omega)$ in $W^{k,p}(\Omega)$. □

Let $C^{k,p}(\Omega)$ denote the closure of $E := \{\phi \in C^k(\Omega) \mid \|\phi\|_{k,p,\Omega} < \infty\}$ w.r.t the Sobolev norm $\|\cdot\|_{k,p,\Omega}$. The space $C^{k,p}(\Omega)$ is a subspace of $W^{k,p}(\Omega)$ because classical derivative (continuous) and distributional derivative coincide (because integraion by parts is valid). Since $W^{k,p}(\Omega)$ is a Banach space, we have $C^{k,p}(\Omega)$ is a closed subspace of $W^{k,p}(\Omega)$. In fact, a consequence of above result is that the Sobolev space $W^{k,p}(\Omega) = C^{k,p}(\Omega)$, a result due to Meyers and Serrin (cf. [MS64]) proved in 1964.

[4]The choice of this range for δ is motivated from the fact that $\left(\frac{1}{m+1} - \frac{1}{(m+1)(m+2)}\right) = \frac{1}{(m+2)}$

Corollary 2.4.4 (Meyers-Serrin)**.** *Let* $1 \leq p < \infty$ *and* $\Omega \subseteq \mathbb{R}^n$*, then* $C^{k,p}(\Omega) = W^{k,p}(\Omega)$*.*

Proof. It is enough to show that E is dense in $W^{k,p}(\Omega)$ because, as a consequence, $W^{k,p}(\Omega) = \overline{E} = C^{k,p}(\Omega)$. For each given $\varepsilon > 0$ and $u \in W^{k,p}(\Omega)$, we need to show the existence of $\phi \in E$ such that $\|u - \phi\|_{k,p,\Omega} < \varepsilon$. By Theorem 2.4.3 there is a $\phi \in C^\infty(\Omega) \cap E$ such that $\|u - \phi\|_{k,p,\Omega} < \varepsilon$. Hence proved. □

The density of $C^\infty(\overline{\Omega})$ is not true, in general, and fails for some "bad" domains as seen in examples below. This, in turn, means that $C_c^\infty(\Omega)$ cannot, in general, be dense in $W^{k,p}(\Omega)$.

Example 2.6. Let $\Omega := \{(x, y) \in \mathbb{R}^2 \mid 0 < |x| < 1 \text{ and } 0 < y < 1\}$. Consider the function $u : \Omega \to \mathbb{R}$ defined as

$$u(x, y) = \begin{cases} 1 & x > 0 \\ 0 & x < 0. \end{cases}$$

Then for any fixed $\varepsilon > 0$ there exists no $\phi \in C^1(\overline{\Omega})$ such that $\|u - \phi\|_{1,p} < \varepsilon$.

Example 2.7. Let $\Omega; = \{(r, \theta) \in \mathbb{R}^2 \mid 1 < r^2 < 2 \text{ and } \theta \neq 0\}$. Consider $u(r, \theta) = \theta$. Then there exists no $\phi \in C^1(\overline{\Omega})$ such that $\|u - \phi\|_{1,1} < 2\pi$.

The trouble with domains in above examples is that they lie on both sides of the boundary $\partial\Omega$ which becomes the main handicap while trying to approximate $W^{k,p}(\Omega)$ by $C^\infty(\overline{\Omega})$ functions.

Definition 2.4.5. *A subset* $\Omega \subset \mathbb{R}^n$ *is said to satisfy the* segment property *if for every* $x \in \partial\Omega$*, there is a open set* B_x*, containing* x*, and a non-zero unit vector* e_x *such that* $z + te_x \in \Omega$*, for all* $0 < t < 1$ *whenever* $z \in \overline{\Omega} \cap B_x$*.*

Observe that a domain with segment property cannot lie on both sides of its boundary.

Theorem 2.4.6. *Let* $1 \leq p < \infty$ *and* $\Omega \subseteq \mathbb{R}^n$ *satisfy the segment property. Then, for every* $u \in W^{k,p}(\Omega)$ *there exists a sequence* $\{v_m\} \subset C_c^\infty(\mathbb{R}^n)$ *such that* $\|v_m - u\|_{k,p,\Omega} \to 0$*, for all* $k \geq 1$*, i.e.,* $C_c^\infty(\mathbb{R}^n)$ *is dense in* $W^{k,p}(\Omega)$*.*

Theorem 2.4.1 is a particular case of the above result, because $\mathbb{R}^n$ trivially satisfies the segment property as it has 'no boundary'. A stronger version of above result is Theorem 2.5.2 proved in next section.

Corollary 2.4.7. *Let $1 \leq p < \infty$ and $\Omega \subseteq \mathbb{R}^n$ satisfy the segment property. Then $C^\infty(\overline{\Omega})$ is dense in $W^{k,p}(\Omega)$.*

In general, the closure of $C_c^\infty(\Omega)$ is a proper subspace of $W^{k,p}(\Omega)$ w.r.t the $\|\cdot\|_{k,p,\Omega}$.

Definition 2.4.8. *Let $W_0^{k,p}(\Omega)$ be the closure of $C_c^\infty(\Omega)$ in $W^{k,p}(\Omega)$. For $p = 2$, we denote $W_0^{k,p}(\Omega)$ by $H_0^k(\Omega)$.*

Exercise 60. Show that $W_0^{k,p}(\Omega)$ is a closed subspace of $W^{k,p}(\Omega)$.

In general, $W_0^{k,p}(\Omega)$ is a strict subspace of $W^{k,p}(\Omega)$. However, as observed in Theorem 2.4.1, for $\Omega = \mathbb{R}^n$, $W^{k,p}(\mathbb{R}^n) = W_0^{k,p}(\mathbb{R}^n)$ for all $1 \leq p < \infty$ and $k \geq 0$. In fact, $W^{k,p}(\Omega) = W_0^{k,p}(\Omega)$ iff $\text{cap}_p(\mathbb{R}^n \setminus \Omega) = 0$[5].

The density results discussed in this section is not true for $p = \infty$ case as seen from the example below.

Example 2.8. Let $\Omega = (-1, 1)$ and

$$u(x) = \begin{cases} 0 & x \leq 0 \\ x & x \geq 0. \end{cases}$$

Then its distributional derivative is $u'(x) = 1_{(0,\infty)}$. Let $\phi \in C^\infty(\Omega)$ such that $\|\phi' - u'\|_\infty < \varepsilon$. Thus, if $x < 0$, $|\phi'(x)| < \varepsilon$ and if $x > 0$, $|\phi'(x) - 1| < \varepsilon$. In particular, $\phi'(x) > 1 - \varepsilon$. By continuity, $\phi'(0) < \varepsilon$ and $\phi'(0) > 1 - \varepsilon$ which is impossible if $\varepsilon < 1/2$. Hence, u cannot be approximated by smooth functions in $W^{1,\infty}(\Omega)$ norm.

2.5 Extension Operators

In general, many properties of $W^{k,p}(\Omega)$ can be inherited from $W^{k,p}(\mathbb{R}^n)$ provided the domain is "nice". For instance, we will see that if $W^{k,p}(\mathbb{R}^n)$ is continuously imbedded in $L^q(\mathbb{R}^n)$ then, for nice domains, $W^{k,p}(\Omega)$ is also continuously imbedded in $L^q(\Omega)$. In this section we classify such classes of "nice" domain.

Definition 2.5.1. *Let Ω be an open subset of $\mathbb{R}^n$. We say P is an (k,p)-*extension *operator for Ω, if $P : W^{k,p}(\Omega) \to W^{k,p}(\mathbb{R}^n)$ is a bounded linear operator, i.e., there is a constant $C > 0$ (depending on Ω, k and p) such that*

$$\|Pu\|_{k,p,\mathbb{R}^n} \leq C\|u\|_{k,p,\Omega} \quad \forall u \in W^{k,p}(\Omega)$$

[5] (k,p) polar sets

and $Pu\,|_\Omega = u$ *a.e. for every* $u \in W^{k,p}(\Omega)$. *If* P *is same for all* $1 \le p < \infty$ *and* $0 \le m \le k$, *then* P *is called* strong k-extension *operator. If* P *is a strong* k*-extension operator for all* k *then* P *is called* total extension *operator.*

Example 2.9. There is a natural extension operator $P : W_0^{k,p}(\Omega) \to W^{k,p}(\mathbb{R}^n)$ which is the extension by zero. Define

$$Pu := \tilde{u} = \begin{cases} u(x) & x \in \Omega \\ 0 & x \in \mathbb{R}^n \setminus \Omega. \end{cases}$$

Obviously, $\tilde{u}\,|_\Omega = u$ and $\|\tilde{u}\|_{0,p,\mathbb{R}^n} = \|u\|_{0,p,\Omega}$. We shall show that $\|\tilde{u}\|_{k,p,\mathbb{R}^n} = \|u\|_{k,p,\Omega}$. Since $u \in W_0^{k,p}(\Omega)$, there is a sequence $\{\phi_m\} \subset C_c^\infty(\Omega)$ converging to u in $\|\cdot\|_{k,p,\Omega}$. For any $\phi \in C_c^\infty(\mathbb{R}^n)$ and $|\alpha| \le k$, we have

$$\begin{aligned} D^\alpha \tilde{u}(\phi) &= (-1)^{|\alpha|} \int_{\mathbb{R}^n} \tilde{u} D^\alpha \phi \, dx = (-1)^{|\alpha|} \int_\Omega u D^\alpha \phi \, dx \\ &= (-1)^{|\alpha|} \lim_{m\to\infty} \int_\Omega \phi_m D^\alpha \phi \, dx = \lim_{m\to\infty} \int_\Omega D^\alpha \phi_m \phi \, dx \\ &= \int_\Omega D^\alpha u \phi \, dx = \widetilde{D^\alpha u}(\phi). \end{aligned}$$

Thus, $D^\alpha \tilde{u} = \widetilde{D^\alpha u}$ and therefore

$$\|\tilde{u}\|_{k,p,\mathbb{R}^n} = \sum_{|\alpha|=0}^{k} \|D^\alpha \tilde{u}\|_{p,\mathbb{R}^n} = \sum_{|\alpha|=0}^{k} \|\widetilde{D^\alpha u}\|_{p,\mathbb{R}^n} = \sum_{|\alpha|=0}^{k} \|D^\alpha u\|_{p,\Omega} = \|u\|_{k,p,\Omega}.$$

Theorem 2.5.2. *Let* $1 \le p < \infty$. *If* Ω *is an open subset of* $\mathbb{R}^n$ *such that there is a* (k,p)*-extension operator* $P : W^{k,p}(\Omega) \to W^{k,p}(\mathbb{R}^n)$, *then* $C_c^\infty(\mathbb{R}^n)$ *is dense in* $W^{k,p}(\Omega)$. *In particular,* $C^\infty(\overline{\Omega})$ *is dense in* $W^{k,p}(\Omega)$ *w.r.t* $\|\cdot\|_{k,p,\Omega}$.

Proof. For each $u \in W^{k,p}(\Omega)$, we choose the sequence $\phi_m(\rho_m * Pu)$ in $C_c^\infty(\mathbb{R}^n)$ which converges to Pu in $W^{k,p}(\mathbb{R}^n)$ and their restriction to Ω is in $C^\infty(\overline{\Omega})$ and converges in $W^{k,p}(\Omega)$. □

Above result is a particular case of Theorem 2.4.6. A natural question provoked by Theorem 2.5.2 is: For what classes of open sets Ω can one expect an extension operator P.

Definition 2.5.3. *For an open set $\Omega \subset \mathbb{R}^n$ we say that its boundary $\partial\Omega$ is C^k ($k \geq 1$), if for every point $x \in \partial\Omega$, there is a $r > 0$ and a C^k diffeomorphism $\gamma : B_r(x) \to B_1(0)$ (i.e. γ^{-1} exists and both γ and γ^{-1} are k-times continuously differentiable) such that*

1. $\gamma(\partial\Omega \cap B_r(x)) \subset B_1(0) \cap \{x \in \mathbb{R}^n \mid x_n = 0\}$ *and*
2. $\gamma(\Omega \cap B_r(x)) \subset B_1(0) \cap \{x \in \mathbb{R}^n \mid x_n > 0\}$

We say $\partial\Omega$ is C^∞ if $\partial\Omega$ is C^k for all $k = 1, 2, \ldots$ and $\partial\Omega$ is analytic if γ is analytic.

Equivalently, a workable definition of C^k boundary would be the following: if for every point $x \in \partial\Omega$, there exists a neighbourhood B_x of x and a C^k function $\gamma : \mathbb{R}^{n-1} \to \mathbb{R}$ such that

$$\Omega \cap B_x = \{x \in B_x \mid x_n > \gamma(x_1, x_2, \ldots, x_{n-1})\}.$$

To keep the illustration simple, we shall restrict ourselves to $k = 1$. For more general results we refer to [Ada75]. We begin by constructing an extension operator for the half-space and then use it along with partition of unity to construct an extension operator for C^1 boundary domains.

Theorem 2.5.4. *Let $\mathbb{R}^n_+ := \{x = (x', x_n) \in \mathbb{R}^n \mid x_n > 0\}$ where $x' = (x_1, x_2, \ldots, x_{n-1})$. Given $u \in W^{1,p}(\mathbb{R}^n_+)$, we define the extension to $\mathbb{R}^n$ as*

$$Pu = u^*(x) := \begin{cases} u(x', x_n) & x_n > 0 \\ u(x', -x_n) & x_n < 0. \end{cases}$$

Then $\|u^*\|_{1,p,\mathbb{R}^n} \leq 2\|u\|_{1,p,\mathbb{R}^n_+}$ and $u^* \in W^{1,p}(\mathbb{R}^n)$.

Proof. Observe that $u^* \in L^p(\mathbb{R}^n)$ because

$$\begin{aligned} \|u^*\|^p_{p,\mathbb{R}^n} &= \int_{\mathbb{R}^n_+} |u(x', x_n)|^p \, dx + \int_{\mathbb{R}^n_-} |u(x', -x_n)|^p \, dx \\ &= \int_{\mathbb{R}^n_+} |u(x', x_n)|^p \, dx + \int_{\mathbb{R}^{n-1}} \int_0^\infty |u(x', y_n)|^p \, dx' dy_n \\ &= 2 \int_{\mathbb{R}^n_+} |u(x)|^p \, dx = 2\|u\|^p_{p,\mathbb{R}^n_+}. \end{aligned}$$

Thus, $\|u^*\|_{p,\mathbb{R}^n} = 2^{1/p}\|u\|_{p,\mathbb{R}^n_+}$. We now show that for $\alpha = e_i$, $1 \le i \le n-1$, $D^\alpha u^* = (D^\alpha u)^*$. Consider, for $\phi \in \mathcal{D}(\mathbb{R}^n)$,

$$\begin{aligned} D^\alpha u^*(\phi) &= -\int_{\mathbb{R}^n} u^* D^\alpha \phi \, dx \\ &= -\int_{\mathbb{R}^n_+} u(x', x_n) D^\alpha \phi \, dx - \int_{\mathbb{R}^n_-} u(x', -x_n) D^\alpha \phi \, dx \\ &= -\int_{\mathbb{R}^n_+} u(x', x_n) D^\alpha \phi(x', x_n) \, dx - \int_{\mathbb{R}^n_+} u(x', x_n) D^\alpha \phi(x', -x_n) \, dx. \end{aligned}$$

Hence,

$$D^\alpha u^*(\phi) = -\int_{\mathbb{R}^n_+} u D^\alpha \psi(x) \, dx, \tag{2.5.1}$$

where $\psi(x', x_n) = \phi(x', x_n) + \phi(x', -x_n)$ when $x_n > 0$. In general, $\psi \notin \mathcal{D}(\mathbb{R}^n_+)$. Thus, we shall multiply ψ by a suitable cut-off function so that the product is in $\mathcal{D}(\mathbb{R}^n_+)$. Choose a $\{\zeta_m\} \in C^\infty(\mathbb{R})$ such that

$$\zeta_m(t) = \begin{cases} 0 & \text{if } t < 1/2m \\ 1 & \text{if } t > 1/m, \end{cases}$$

then $\zeta_m(x_n)\psi(x) \in \mathcal{D}(\mathbb{R}^n_+)$. Since ζ_m is independent of x_i, for $1 \le i \le n-1$, we have

$$D^\alpha u(\zeta_m \psi) = -\int_{\mathbb{R}^n_+} u(x) D^\alpha(\zeta_m(x_n)\psi(x)) \, dx = -\int_{\mathbb{R}^n_+} u(x)\zeta_m(x_n) D^\alpha \psi(x) \, dx.$$

Passing to limit, as $m \to \infty$ both sides, we get

$$\int_{\mathbb{R}^n_+} D^\alpha u(x)\psi(x) \, dx = -\int_{\mathbb{R}^n_+} u(x) D^\alpha \psi(x) \, dx.$$

The RHS in above equation is same as the RHS obtained in (2.5.1). Hence, we have $D^\alpha u^*(\phi) = \int_{\mathbb{R}^n_+} D^\alpha u(x)\psi(x) \, dx$. By setting

$$(D^\alpha u)^*(x', x_n) := \begin{cases} D^\alpha u(x', x_n) & x_n > 0 \\ D^\alpha u(x', -x_n) & x_n < 0, \end{cases}$$

we get $D^\alpha u^* = (D^\alpha u)^*$. We now show a similar result for $\alpha = e_n$. Consider, for $\phi \in \mathcal{D}(\mathbb{R}^n)$,

$$\begin{aligned} D^{e_n} u^*(\phi) &= -\int_{\mathbb{R}^n} u^* D^{e_n}\phi \, dx \\ &= -\int_{\mathbb{R}^n_-} u(x', -x_n) D^{e_n}\phi \, dx - \int_{\mathbb{R}^n_+} u(x', x_n) D^{e_n}\phi \, dx \\ &= \int_{\mathbb{R}^n_+} u(x', x_n) D^{e_n}\phi(x', -x_n) \, dx - \int_{\mathbb{R}^n_+} u(x', x_n) D^{e_n}\phi(x', x_n) \, dx. \end{aligned}$$

Hence,

$$D^{e_n} u^*(\phi) = -\int_{\mathbb{R}^n_+} u D^{e_n}\psi(x) \, dx, \tag{2.5.2}$$

where $\psi(x', x_n) = \phi(x', x_n) - \phi(x', -x_n)$ when $x_n > 0$. Note that $\psi(x', 0) = 0$. Thus, by Mean value theorem, $|\psi(x', x_n)| \leq C|x_n|$. As before, in general, $\psi \notin \mathcal{D}(\mathbb{R}^n_+)$, so $\zeta_m(x_n)\psi(x) \in \mathcal{D}(\mathbb{R}^n_+)$. One such choice of ζ_m is by choosing $\zeta_m(t) = \zeta(mt)$ where

$$\zeta(t) = \begin{cases} 0 & \text{if } t < 1/2 \\ 1 & \text{if } t > 1. \end{cases}$$

Consider,

$$\begin{aligned} D^{e_n} u(\zeta_m \psi) &= -\int_{\mathbb{R}^n_+} u(x) D^{e_n}(\zeta_m(x_n)\psi(x)) \, dx \\ &= -\int_{\mathbb{R}^n_+} u(x)\zeta_m'(x_n)\psi(x) \, dx - \int_{\mathbb{R}^n_+} u(x)\zeta_m(x_n) D^{e_n}\psi(x) \, dx. \end{aligned}$$

Passing to limit, as $m \to \infty$ both sides, we get

$$\int_{\mathbb{R}^n_+} D^{e_n} u(x)\psi(x) \, dx = -\lim_{m\to\infty} \int_{\mathbb{R}^n_+} u(x)\zeta_m'(x_n)\psi(x) \, dx - \int_{\mathbb{R}^n_+} u(x) D^{e_n}\psi(x) \, dx.$$

Let us handle the first term in RHS. Note that

$$\begin{aligned} \left| \int_{\mathbb{R}^n_+} u(x)\zeta_m'(x_n)\psi(x) \, dx \right| &= m \left| \int_{\mathbb{R}^n_+} u(x)\zeta'(mx_n)\psi(x) \, dx \right| \\ &\leq mC\|\zeta'\|_{\infty,[0,1]} \int_{\mathbb{R}^n_+} |u(x)||x_n| \, dx \\ &\leq C\|\zeta'\|_{\infty,[0,1]} \int_{\mathbb{R}^{n-1}} \int_{0<x_n<1/m} |u(x)| \, dx. \end{aligned}$$

Therefore, $\lim_{m\to\infty}\int_{\mathbb{R}^n_+} u(x)\zeta_m'(x_n)\psi(x)\,dx = 0$ and hence

$$\int_{\mathbb{R}^n_+} D^{e_n}u(x)\psi(x)\,dx = -\int_{\mathbb{R}^n_+} u(x)D^{e_n}\psi(x)\,dx.$$

The RHS in above equation is same as the RHS obtained in (2.5.2). Hence, we have $D^{e_n}u^*(\phi) = \int_{\mathbb{R}^n_+} D^{e_n}u(x)\psi(x)\,dx$. By setting

$$(D^{e_n}u)^\sharp(x',x_n) := \begin{cases} D^{e_n}u(x',x_n) & x_n > 0 \\ -D^{e_n}u(x',-x_n) & x_n < 0, \end{cases}$$

we get $D^{e_n}u^* = (D^{e_n}u)^\sharp$. Therefore, we have the estimate,

$$\begin{aligned} \|u^*\|_{1,p,\mathbb{R}^n} &= \|u^*\|_{p,\mathbb{R}^n} + \sum_{|\alpha|=1}\|D^\alpha u^*\|_{p,\mathbb{R}^n} \\ &= 2^{1/p}\|u\|_{p,\mathbb{R}^n_+} + \sum_{i=1}^{n-1}\|(D^{e_i}u)^*\|_{p,\mathbb{R}^n} + \|(D^{e_n}u)^\sharp\|_{p,\mathbb{R}^n} \\ &= 2^{1/p}\|u\|_{1,p,\mathbb{R}^n_+}. \end{aligned}$$

Hence, $u^* \in W^{1,p}(\mathbb{R}^n)$. □

Theorem 2.5.5. *Let Ω be a bounded open subset of $\mathbb{R}^n$ with C^1 boundary. Then there is an extension operator $P : W^{1,p}(\Omega) \to W^{1,p}(\mathbb{R}^n)$.*

Corollary 2.5.6. *For $1 \le p < \infty$ and Ω be bounded open set with C^1 boundary, then $C^\infty(\overline{\Omega})$ is dense in $W^{k,p}(\Omega)$, for all $k \ge 1$.*

Appendices

Appendix A

Cantor Set and Cantor Function

Let us construct the Cantor set which plays a special role in analysis.

Consider $C_0 = [0, 1]$ and trisect C_0 and remove the middle open interval to get C_1. Thus, $C_1 = [0, 1/3] \cup [2/3, 1]$. Repeat the procedure for each interval in C_1, we get

$$C_2 = [0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1].$$

Repeating this procedure at each stage, we get a sequence of subsets $C_i \subseteq [0, 1]$, for $i = 0, 1, 2, \ldots$. Note that each C_k is a compact subset, since it is a finite union of compact sets. Moreover,

$$C_0 \supset C_1 \supset C_2 \supset \ldots \supset C_i \supset C_{i+1} \supset \ldots.$$

The Cantor set C is the intersection of all the nested C_i's, $C = \cap_{i=0}^{\infty} C_i$.

Lemma A.0.7. *C is compact.*

Proof. C is countable intersection of closed sets and hence is closed. $C \subset [0, 1]$ and hence is bounded. Thus, C is compact. □

The Cantor set C is non-empty, because the end-points of the closed intervals in C_i, for each $i = 0, 1, 2, \ldots$, belong to C. In fact, the Cantor set cannot contain any interval of positive length.

Lemma A.0.8. *For any $x, y \in C$, there is a $z \notin C$ such that $x < z < y$. (Disconnected)*

Proof. If $x, y \in C$ are such that $z \in C$ for all $z \in (x, y)$, then we have the open interval $(x, y) \subset C$. It is always possible to find i, j such that

$$\left(\frac{j}{3^i}, \frac{j+1}{3^i}\right) \subseteq (x, y)$$

but does not belong $C_i \supset C$. □

We show in example **??**, that C has length zero. Since C is non-empty, how 'big' is C? The number of end-points sitting in C are countable. But C has points other than the end-points of the closed intervals C_i for all i. For instance, $1/4$ (not an end-point) will never belong to the the intervals being removed at every step i, hence is in C. There are more! $3/4$ and $1/13$ are all in C which are not end-points of removed intervals. It is easy to observe these by considering the ternary expansion characterisation of C. Consider the ternary expansion of every $x \in [0, 1]$,

$$x = \sum_{i=1}^{\infty} \frac{a_i}{3^i} = 0.a_1a_2a_3\ldots_3 \quad \text{where } a_i = 0, 1 \text{ or } 2.$$

The decomposition of x in ternary form is not unique[1]. For instance, $1/3 = 0.1_3 = 0.022222\ldots_3$, $2/3 = 0.2_3 = 0.1222\ldots_3$ and $1 = 0.222\ldots_3$. At the C_1 stage, while removing the open interval $(1/3, 2/3)$, we are removing all numbers whose first digit in ternary expansion (in all possible representations) is 1. Thus, C_1 has all those numbers in $[0, 1]$ whose first digit in ternary expansion is not 1. Carrying forward this argument, we see that for each i, C_i contains all those numbers in $[0, 1]$ with digits upto ith place, in ternary expansion, is not 1. Thus, for any $x \in C$,

$$x = \sum_{i=1}^{\infty} \frac{a_i}{3^i} = 0.a_1a_2a_3\ldots_3 \quad \text{where } a_i = 0, 2.$$

Lemma A.0.9. *C is uncountable.*

Proof. Use Cantor's diagonal argument to show that the set of all sequences containing 0 and 2 is uncountable. □

[1]This is true for any positional system. For instance, $1 = 0.99999\ldots$ in decimal system

Cantor Function

We shall now define the *Cantor function* $f_C : C \to [0,1]$ as,

$$f_C(x) = f_C\left(\sum_{i=1}^{\infty} \frac{a_i}{3^i}\right) = \sum_{i=1}^{\infty} \frac{a_i}{2} 2^{-i}.$$

Since $a_i = 0$ or 2, the function replaces all 2 occurrences with 1 in the ternary expansion and we interpret the resulting number in binary system. Note, however, that the Cantor function f_C is not injective. For instance, one of the representation of $1/3$ is $0.0222\ldots_3$ and $2/3$ is 0.2. Under f_C they are mapped to $0.0111\ldots_2$ and 0.1_2, respectively, which are different representations of the same point. Since f_C is same on the end-points of the removed interval, we can extend f_C to $[0,1]$ by making it constant along the removed intervals.

Alternately, one can construct the Cantor function step-by-step as we remove middle open intervals to get C_i. Consider f_1 to be a function which takes the constant value $1/2$ in the removed interval $(1/3, 2/3)$ and is linear on the remaining two intervals such that f_1 is continuous. In the second stage, the function f_2 coincides with f_1 in $1/3, 2/3$, takes the constant value $1/4$ and $3/4$ on the two removed intervals and is linear in the remaining four intervals such that f_2 is continuous. Proceeding this way we have a sequence of monotonically increasing continuous functions $f_k : [0,1] \to [0,1]$. Moreover, $|f_{k+1}(x) - f_k(x)| < 2^{-k}$ for all $x \in [0,1]$ and f_k converges uniformly to $f_C : [0,1] \to [0,1]$.

Exercise 61. The Cantor function $f_C : [0,1] \to [0,1]$ is uniformly continuous, monotonically increasing and is differentiable a.e. and $f_C' = 0$ a.e.

Exercise 62. The function f_C is not absolutely continuous.

Generalised Cantor Set

We generalise the idea behind the construction of Cantor sets to build Cantor-like subsets of $[0,1]$. Choose a sequence $\{a_k\}$ such that $a_k \in (0, 1/2)$ for all k. In the first step we remove the open interval $(a_1, 1-a_1)$ from $[0,1]$ to get C_1. Hence $C_1 = [0, a_1] \cup [1-a_1, 1]$. Let

$$C_1^1 := [0, a_1] \quad \text{and } C_1^2 := [1-a_1, 1].$$

Hence, $C_1 = C_1^1 \cup C_1^2$. Note that C_1^i are sets of length a_1 carved out from the end-points of C_0. We repeat step one for each of the end-points of C_1^i of length a_1a_2. Therefore, we get four sets

$$C_2^1 := [0, a_1a_2] \quad C_2^2 := [a_1 - a_1a_2, a_1],$$
$$C_2^3 := [1 - a_1, 1 - a_1 + a_1a_2] \quad C_2^4 := [1 - a_1a_2, 1].$$

Define $C_2 = \cup_{i=1}^4 C_2^i$. Each C_2^i is of length a_1a_2. Note that $a_1a_2 < a_1$. Repeating the procedure successively for each term in the sequence $\{a_k\}$, we get a sequence of sets $C_k \subset [0, 1]$ whose length is $2^k a_1a_2 \dots a_k$. The "generalised" Cantor set C is the intersection of all the nested C_k's, $C = \cap_{k=0}^\infty C_k$ and each $C_k = \cup_{i=1}^{2^k} C_k^i$. Note that by choosing the constant sequence $a_k = 1/3$ for all k gives the Cantor set defined in the beginning of this Appendix. Similar arguments show that the generalised Cantor set C is compact. Moreover, C is non-empty, because the end-points of the closed intervals in C_k, for each $k = 0, 1, 2, \dots$, belong to C.

Lemma A.0.10. *For any $x, y \in C$, there is a $z \notin C$ such that $x < z < y$.*

Lemma A.0.11. *C is uncountable.*

We show in example **??**, that C has length $2^k a_1a_2 \dots a_k$.

The interesting fact about generalised Cantor set is that it can have non-zero "length".

Proposition A.0.12. *For each $\alpha \in [0, 1)$ there is a sequence $\{a_k\} \subset (0, 1/2)$ such that*

$$\lim_k 2^k a_1a_2 \dots a_k = \alpha.$$

Proof. Choose $a_1 \in (0, 1/2)$ such that $0 < 2a_1 - \alpha < 1$. Use similar arguments to choose $a_k \in (0, 1/2)$ such that $0 < 2^k a_1a_2 \dots a_k - \alpha < 1/k$. □

Generalised Cantor Function

We shall define the generalised Cantor function f_C on the generalised Cantor set C. Define the function $f_0 : [0, 1] \to [0, 1]$ as $f_0(x) = x$. f_0 is continuous on $[0, 1]$. We define $f_1 : [0, 1] \to [0, 1]$ such that f is linear and continuous on C_1^i, and $1/2$ on $[a_1, 1 - a_1]$, the closure of removed open interval at first stage. We define $f_k : [0, 1] \to [0, 1]$ continuous $f_k(0) = 0, f_k(1) = 1$ such that $f_k(x) = i/2^k$ on the removed interval immediate right to C_k^i.

Theorem A.0.13. *Each f_k is continuous, monotonically non-decreasing and uniformly converges to some $f_C : [0, 1] \to [0, 1]$.*

Thus, f_C being uniform limit of continuous function is continuous and is the called the generalised Cantor function.

Appendix B

Partition of Unity

B.1 Zero Sets

Let X be any set. Any subset $E \subset X$ is said to be a *zero set* in X if there exists a real valued function $f : X \to \mathbb{R}$ such that $E = f^{-1}(0)$. Since we have no restriction on f, we can always write a function which takes the value zero only on E and this can be done for every subset of X.

Definition B.1.1. *Let X be a topological space X. Any subset $E \subset X$ is said to be a* zero set *in X if there exists a continuous real valued function $f : X \to \mathbb{R}$ such that $E = f^{-1}(0)$.*

Note that by definition every zero set in a topological space is a closed set (with the usual distance topology on $\mathbb{R}$). How about the converse? Is every closed subset of X a zero set in X. The answer is a "no", in general, and an attempt to answer this motivates the separation axioms concept in topology.

Definition B.1.2. *A topological space X is said to be* normal *if for any two given non-empty disjoint closed subsets E and F of X, there exists open neighbourhoods $U \supset E$ and $V \supset F$ such that $U \cap V = \emptyset$.*

Theorem B.1.3 (Urysohn Lemma)**.** *A topological space X is normal iff for any two disjoint closed subsets E and F of X there is a continuous function $f : X \to [0, 1]$ such that $f(E) = 0$ and $f(F) = 1$.*

Note that, in the Urysohn lemma above, though f is zero identically on E, E may not be the zero set of f. f may take the zero value outside E too.

Definition B.1.4. *A topological space X is said to be* perfectly normal *if X is normal and, in addition, every closed set is a G_δ set (countable intersection of open sets).*

Proposition B.1.5. *A topological space X is perfectly normal iff for any two given disjoint non-empty closed subsets E and F of X there is a continuous function $f : X \to [0, 1]$ such that $f^{-1}(0) = E$ and $f^{-1}(1) = F$.*

In contrast to Urysohn lemma, in the above theorem, the given set E is precisely the zero set of f.

Theorem B.1.6. *X is perfectly normal iff every closed subset of X is a zero set in X.*

Theorem B.1.7. *Every topological manifold is perfectly normal (?)*

While working on smooth manifolds, we seek to know if every closed set is a zero set of a smooth or C^∞ function.

Lemma B.1.8. *Let E be a closed subset of $\mathbb{R}$. Also, let $f : \mathbb{R} \to \mathbb{R}$ be a continuous such that $f(E) = 0$ and f is C^1 in E^c. Set*

$$F(x) = \begin{cases} f'(x) & x \in E^c \\ 0 & x \in E. \end{cases}$$

Then F is continuous.

Lemma B.1.9. *If $K \subset U \subset \mathbb{R}^n$ with K compact and U open, then there is a smooth function $f : \mathbb{R}^n \to [0, \infty)$ such that $f^{-1}(0) = K$ and $f(U^c) = 1$.*

Theorem B.1.10. *Let M be a smooth manifold M. For every closed subset $E \subset M$, there is a smooth function $f : M \to [0, \infty)$ such that $E = f^{-1}(0)$.*

B.2 Continuous Partition of Unity

The partition of unity is an important tool to bridge global and local analysis on a manifold. A manifold can be seen as obtained by gluing open sets of $\mathbb{R}^n$. Thus, many of our proofs in manifold needs to reverse this gluing operation. A useful tool for this reverse operation is partition of unity.

Definition B.2.1. *A collection of open subsets $\{U_i\}_{i\in I}$ of a topological space X is said to be an* open cover *of X if $X = \cup_{i\in I} U_i$. Any subcollection of $\{U_j\} \subset \{U_i\}$, where $j \in J \subset I$, is said to be a* sub-cover *if $X = \cup_{j\in J} U_j$.*

Lemma B.2.2 (σ-compactness)**.** *For a second countable, locally compact, Hausdorff topological space X, there exists a countable open cover $\{U_i\}_1^\infty$ of X such that $\overline{U}_i \subset U_{i+1}$, for all i, and $\overline{U}_i$ is compact in X.*

Proof. Let $\mathcal{U}$ be the countable basis of X, since X is second countable. Since X is locally compact, every $x \in X$ there is an open set U_x, containing x, whose closure is compact. Since $\mathcal{U}$ is a basis, for each $x \in X$, there is a $B_x \in \mathcal{U}$ such that $x \in B_x \subseteq U_x$. Thus, B_x has compact closure and $X = \cup_{x\in X} \overline{B}_x$, where is union is countable because the basis is countable. The $\{B_x\}$ is a sub-collection of the basis of X. Let us denote the countable collection as $\{B_i\}$ where $X = \cup_i B_i$ and $\overline{B}_i$ is compact.

Now, choose $U_1 = B_1$. Let j_2 denote the smallest number such that $\overline{U}_1 = \overline{B}_1 \subset \cup_{i=1}^{j_2} B_i$, this is possible due to the compactness of $\overline{B}_1$. Set $U_2 = \cup_{i=1}^{j_2} U_i$. Similarly, proceeding this way, we define

$$U_{k+1} = \cup_{i=1}^{j_{k+1}}$$

where j_k is the smallest number such that $\overline{U}_k \subset \cup_{i=1}^{j_{k+1}} B_i$. By construction, $U_k \subset U_{k+1}$. □

Definition B.2.3. *We say an open cover $\{V_k\}_{k\in K}$ is a* refinement *of the open cover $\{U_i\}_{i\in I}$ if, for all $k \in K$, there is a $i \in I$ such that $V_k \subset U_i$.*

Definition B.2.4. *A collection of subsets $\{U_i\}_{i\in I}$, of a topological space X is said to be* locally finite *if, for all $x \in X$, there is a neighbourhood Ω of x such that $U_j \cap \Omega \neq \emptyset$ for all $j \in J$ where J is a finite subset of I.*

In particular, note that $x \in X$ is contained in finitely many U_i. For instance, the collection of intervals $\{U_{r,i}\}$ defined as $U_{r,i} := (r - 1/i, r + 1/i)$ on $\mathbb{R}$ is not locally finite.

Definition B.2.5. *A topological space is said to be* paracompact *if every open cover admits a locally finite refinement.*

Theorem B.2.6. *Every metric space is paracompact.*

Theorem B.2.7. *Every topological manifold is paracompact.*

Proposition B.2.8. *A Hausdorff paracompact space is normal.*

Lemma B.2.9. *Every locally finite open cover of a paracompact space admits a open sub-cover.*

Definition B.2.10. *Let X be a topological space. A* partition of unity *on X is a collection of continuous functions $\{\phi_i\}_{i\in I}$ from X to $\mathbb{R}$ such that*

(i) The collection of subsets, supp(ϕ_i), is locally finite. Equivalently, on every neighbourhood of each $x \in X$, the functions ϕ_i are all identically zero except for some finite number of them.

(ii) $\phi_i(x) \geq 0$ for all $x \in X$ and $i \in I$.

(iii) $\sum_{i\in I} \phi_i(x) = 1$ for all $x \in X$ (the sum is finite due to (i)).

Given an open cover $\{U_j\}_{j\in J}$ of M, we say a partition of unity $\{\phi_i\}_{i\in I}$ is subordinate to the open cover *if, for all $i \in I$, there is a $j \in J$ such that supp(ϕ_i)$\subset U_j$.*

Theorem B.2.11. *A Hausdorff space is paracompact iff every open cover admits a subordinate partition of unity.*

B.3 Smooth Partition of Unity

Lemma B.3.1. *Let M be a smooth manifold and $K \subset U \subset M$ with K compact and U open. Then there is a smooth function $g : M \to [0, \infty)$ such that $g(x) > 0$ for all $x \in K$ and supp(g)$\subset U$.*

Proposition B.3.2. *Let M be a manifold with basis $\{B_i\}_{i\in I}$ such that $\overline{B}_i$ is compact for all $i \in I$. For any open cover $\{U_j\}_{j\in J}$ of M there exists a countable locally finite refinement of $\{U_j\}$.*

Lemma B.3.3. *For any manifold M there exists a basis $\{U_i\}_{i\in I}$ such that $\overline{U}_i$ is compact and for each $i \in I$, there is a smooth function $\phi_i : M \to \mathbb{R}$ such that*

$$\phi_i(x) \begin{cases} = 0 & x \notin U_i \\ > 0 & x \in U_i. \end{cases}$$

Definition B.3.4. *A* partition of unity *on a manifold M is a collection of smooth functions $\{\phi_i\}_{i\in I}$ from M to $\mathbb{R}$ such that*

(i) The collection of subsets, supp(ϕ_i), is locally finite.

(ii) $\phi_i(p) \geq 0$ *for all* $p \in M$ *and* $i \in I$.

(iii) $\sum_{i \in I} \phi_i(p) = 1$ *for all* $p \in M$.

Given an open cover $\{U_j\}_{j \in J}$ *of* M*, we say a partition of unity* $\{\phi_i\}_{i \in I}$ *is* subordinate to the open cover *if forf all* $i \in I$ *there is a* $j \in J$ *such that supp*(ϕ_i)$\subset U_j$.

Theorem B.3.5. *For any open cover of a manifold* M *there is a countable partition of unity subordinate to the open cover and the support of each function is compact.*

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Index

Dirac, 1, 2
    delta function, 2
    measure, 2
    sequence, 11
Dirac sequence, 36

function
    Dirac delta, 2
    Heaviside, 1
    test, 16

gradient, 12

Hessian matrix, 12
hypoelliptic operator, 70

inequality
    Young's, 18

Laplace operator, 12
Lebesgue, 2

Schwartz, 2
sequence
    mollifiers, 10
solution
    fundamental, 70